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DEPARTMENT OF DEFENSE HANDBOOK

GUIDELINES FOR INSPECTION OF AIRCRAFT ELECTRICAL WIRING INTERCONNECT SYSTEMS

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AMSC N/A FSC 6145

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FOREWORD

1. This handbook is approved for use by all Departments and Agencies of the Department of Defense (DoD).

2. This handbook is a guide for aircraft Electrical Wiring Interconnect System (EWIS) inspection. In case of conflict between the aircraft/platform-specific maintenance manual and this handbook, the platform-specific manual will take precedence. Not all possible wire systems factors have been addressed. This handbook captures in one document, under a suitable subject heading, fundamental installation criteria guidelines. The opportunity to focus on a single document results in substantial savings to the Government.

3. All aircraft are filled with miles of wiring and hundreds of wiring devices that connect and transfer power and signals to and from electrical components. Virtually all aircraft systems rely heavily on some type of wiring for safe operation. Much like the structural components of an aircraft, the health and integrity of the EWIS can be significantly compromised due to premature aging, damage, and failure of wiring insulation. It is integral to the overall maintenance and sustainment of all aircraft that the EWIS be treated as a system and as a system is afforded the same level of importance as the aircraft structure and other critical flight control systems.

4. The majority of aircraft wiring in military service is of a thin-walled construction, and by its very nature, is susceptible to mechanical damage. However, there are several factors that may contribute to premature aging, damage, and failure of wiring insulation, including but not limited to:

a. Wire/Bundle location (Severe Wind and Moisture Problem (SWAMP) Areas)
b. Temperature cycling
c. Contamination
d. Improper installation
e. Mishandling
f. Poor maintenance practices
g. Lack of effective inspection and maintenance training
h. Battle damage

5. This handbook was prepared by, and is regularly updated through, the cooperative efforts of Government and industry. Applicable documents are listed in each individual guideline. Additional information regarding aerospace vehicle wiring requirements is found in SAE AS50881 “Aerospace Vehicle Wiring.” Copies of SAE AS50881 are available from the Society of Automotive Engineers at the website specified in section 2 of this handbook.

6. When a Joint Service manual is referenced throughout this document, the Navy equivalent is listed in each guideline. For other services’ technical manual numbers, refer to table I. See 2.5 for information on how to obtain copies of maintenance manuals referenced in this handbook.

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TABLE I. Topic and tri-service maintenance manual cross-reference.

Manual Topic	Navy	Air Force	Army	USMC
General Wiring	01-1A-505-1	1-1A-14	1-1500-323-24-1	01-1A-505-1
Circular Connectors	01-1A-505-2	1-1A-14-2	1-1500-323-24-2	01-1A-505-2
Rectangular Connectors	01-1A-505-3	1-1A-14-3	1-1500-323-24-3	01-1A-505-3
Fiber Optics	01-1A-505-4	1-1A-14-4	1-1500-323-24-4	01-1A-505-4
Soldering/ESD	01-1A-23	00-25-259	5895-45/1C	5895-45/1E
Corrosion Program and Corrosion Theory	01-1A-509-1	1-1-689-1	1-1500-344-23-1	01-1A-509-1
Avionics and Electronics Cleaning and Corrosion Prevention/Control	01-1A-509-3	1-1-689-3	1-1500-344-23-3	01-1A-509-3

7. Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, System Standardization and PHS&T Branch, Code BL32600, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to LKE-NAVAIRDocComments@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil. The technical content contained in this handbook has been provided by the NAVAIR Wiring Systems Branch (48298 Shaw Road Bldg. 1461, Patuxent River, MD 20670-1161).

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1. SCOPE

1.1 Guidelines applicable to aircraft Electrical Wiring Interconnect Systems (EWIS). This handbook provides guidance and lessons learned in the inspection of EWIS.

1.2 Method of reference. Guidelines contained herein should be referenced by citing this handbook and the guideline number. The information given in each guideline is intended for guidance only.

1.3 Interrelationship of guidelines. Each guideline is intended to cover some discipline in the inspection of EWIS, such as a procedure, a process, or the selection and application of parts and materials. Many of these disciplines, however, cannot retain a clear-cut separation or isolation from others so that when guidelines of MIL-HDBK-522 are referenced in a specification, some guidelines will undoubtedly have a direct interrelationship with other guidelines. This interrelationship should be taken into consideration when referencing these guidelines.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed below are not necessarily all of the documents referenced herein but are those needed to understand the information provided by this handbook.

2.2 Government document.

2.2.1 Specification. The following specification forms a part of this document to the extent specified herein.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-DTL-81381 - Wire, Electric, Polyimide-Insulated, Copper or Copper Alloy

(Copies of military documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publication. The following document forms a part of this document to the extent specified herein.

SAE INTERNATIONAL

SAE AS23053 - Insulation Sleeving, Electrical, Heat Shrinkable, General Specification For

(Copies of this document are available online at http://www.sae.org.)

2.4 Individual guidelines. See section 2 of each individual guideline for a listing of applicable documents. Documents referenced in the individual documents apply to the extent that they are cited herein. (Copies of military documents are available online at https://quicksearch.dla.mil.)

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2.5 Copies of maintenance manuals. Copies of maintenance manuals NAVAIR 01-1A-505-1 through -4, may be obtained by DoD employees and uniformed users by going to the NATEC site https://mynatec.navair.navy.mil/ using their Common Access Card (CAC) to access the documents. Commercial entities requesting copies of these documents should reference this site for maintenance manual access via the Freedom of Information Act requests.

2.6 Industry addresses. Addresses for obtaining documents referenced in the guidelines but not obtainable from the Government are as follows:

National Electrical Manufacturers Association (NEMA)
1300 North 17th Street Suite 1752
Roslyn, Virginia 22209
Online: http://www.nema.org

Society of Automotive Engineers (SAE)
400 Commonwealth Drive
Warrendale, PA 15096-0001 USA
Online: http://www.sae.org

3. ACRONYMS AND DEFINITIONS

3.1 Acronyms. The following acronyms are applicable to this handbook.

AIR	Denotes aviation for terminal lugs
AS	Aerospace Standard
AWG	American Wire Gauge
°C	Degrees Celsius
CAC	Common Access Card
CAGE	Commercial and Government Entity
CEA	Cognizant Engineering Authority
CLF	Cable Lacing Fastener
CPC	Corrosion Prevention (or Preventive) Compound
DoD	Department of Defense
DVI	Detailed Visual Inspection
EI	Engineering Investigation
EMI	Electromagnetic Interference
EN	Electroless Nickel
ESD	Electrostatic Discharge
EWIS	Electrical Wiring Interconnect (or Interconnection) System
°F	Degrees Fahrenheit
FO	Fiber Optic
FOD	Foreign Object Damage
FST	Fleet Support Team
GVI	General Visual Inspection
HMR	Hazardous Material Report
JDRS	Joint Deficiency Reporting System

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JSWAG	Joint Services Wiring Action Group
MDL	McGuire-Dix-Lakehurst
MIL-DTL	Military Detail Specification
MIL-HDBK	Military Handbook
MIL-PRF	Military Performance Specification
NASM	National Aerospace Standard (Metric)
NATEC	Naval Air Technical Data and Engineering Service Center
NAVAIR	Naval Air Systems Command
NEMA	National Electrical Manufacturers Association
NSN	National Stock Number
QPL	Qualified Products List
PTFE	Polytetrafluoroethylene
RF	Radio Frequency
RFI	Radio Frequency Interference.
RoHS	Restriction of Hazardous Substances
SAE	Society of Automotive Engineers
SM&R	Source, Maintenance and Recovery
SWAMP	Severe Wind and Moisture Problem
USMC	United States Marine Corps
WP	Work Package
WRA	Weapons Replaceable Assembly

3.2 Definitions.

3.2.1 Airborne, space, aerospace. "Airborne" denotes those applications peculiar to aircraft and missile or other systems designed for operation primarily within the earth's atmosphere; "space" denotes application peculiar to spacecraft and systems designed for operation near or beyond the upper reaches of the earth's atmosphere; and "aerospace" includes both airborne and space applications.

3.2.2 Bend radius. The maximum amount a wire, cable, fiber, or fiber cable can be bent without causing damage. Usually called minimum safe bending radius.

3.2.3 Bird cage. Defect in stranded wire where the strands in the stripped portion between the covering of an insulated wire and a soldered connection (or an end-tinned lead) have separated from the normal lay of the strands.

3.2.4 Bonded assembly. Connector assembly in which the components are bonded together using an electrically appropriate adhesive in a sandwich-type structure. Provides sealing against moisture and other environmental conditions that weaken electrical insulating properties.

3.2.5 Bundle. Number of wires and/or cables, groups, or harnesses routed together

3.2.6 Chamfer. Funnel type angle on the inside edge of the barrel entrance of a connecter insert and/or socket contact, which permits easier insertion of a pin contact into the barrel.

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3.2.7 Coaxial cable. Cable consisting of two cylindrical conductors with a common axis. The two conductors are separated by a dielectric. The outer conductor or shield, normally at ground potential, acts as a return path for current flowing through the center conductor and prevents energy radiation from the cable. The outer conductor is commonly used to prevent external radiation from affecting the current flowing in the inner conductor. The outer conductor consists of woven strands of wire or is a metal sheath.

3.2.8 Conduit. Tube or trough in which insulated wires and cables are run.

3.2.9 Connector. Describes all interface devices, either plug or receptacle, used to provide rapid connect/disconnect service for electrical cable and wire interconnections. A fixed connector is used for attachment to a rigid surface, while a free connector mates with the wire or cable. Connectors used in military applications generally fall into three broad categories: single contact coaxial connectors, circular multi-contact connectors, and rectangular multi-contact connectors.

3.2.10 Discrepancy. A clearly identifiable deviation from the original design of the system as identified in the source data.

3.2.11 Dust cover. An item that is specifically designed to cover the mating end of a connector for mechanical and/or environmental protection.

3.2.12 Electrical Wiring Interconnect System (EWIS). Known as aircraft wiring, EWIS is defined as any wire, fiber optic link, wiring or fiber device, or a combination of these items (including terminations) installed in any area of the aircraft for the purpose of transmitting electrical energy, signals, or data between two or more electrical end points.

3.2.13 Electromagnetic Interference (EMI). The disruption of the operation of an electronic device when it is in the vicinity of an electromagnetic field that is caused by another electrical or electronic device. There is a wide range of frequencies at which EMI can occur. Shielding materials for the entire EMI spectrum are not readily available.

3.2.14 Environmentally sealed. A device that is provided with gaskets, seals, grommets, potting, or other means to keep out moisture, dirt, air, or dust which might reduce performance. Does not include non-physical environments such as Radio Frequency (RF) and radiation.

3.2.15 Grommet. A rubber ring or tube through which wires pass. Grommets can be used to provide sealing around wires where they enter a connector or to provide mechanical protection for wires where they pass through a hole in a structure.

3.2.16 Harness. An assembly of any number of wires, electrical/optical cables, and/or groups and their terminations designed and fabricated to allow for installation and removal as a unit. A harness may be an open harness or a protected harness.

3.2.17 Heat shrinkable. A term describing tubes, sleeves, caps, boots, films, or other forms of plastic which shrink to encapsulate, protect, or insulate connections, splices, terminations, and

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other configurations with the application of heat. Heat shrinkable sleeves are typically defined in SAE AS23053.

3.2.18 Lacing tape. Flexible, flat fabric tape for tying harnesses and wire bundles, securing of sleeves and other items, and general lacing and tying applications. Lacing tape is available in various materials and impregnates. When installed, it may be called “spot tie.”

3.2.19 Lay. The Lay of any helical element of a cable is the axial length of a turn of the helix of that element. A cable's helical elements may be each strand in a concentric-lay cable or each insulated conductor in a multi-conductor cable. Lay is often referred to as pitch.

3.2.20 Lay, direction of. The direction in which the strands or a conductor or components in a cable pass over the top of the bundle as they recede from an observer looking along the axis of the conductor or cable. Termed right hand or left hand (see lay).

3.2.21 Marker tape. Tape laid parallel to the conductors under the sheath in a cable, imprinted with the manufacturer’s name and the specification to which the cable is made.

3.2.22 Plating. Overlaying of a thin coating of metal on metallic components to improve conductivity, provide for easy soldering, or prevent rusting or corrosion.

3.2.23 Polyimide. A trademark of the DuPont Company for their polyimide resin film used as wire insulation. Polyimide (Kapton®) was a popular aircraft wiring insulation from the 1960s through the early 1990s because of its excellent mechanical properties at high temperatures, light weight, low smoke generation, and low flammability. However, weathered polyimide was found to have a high risk of arcing due to hydrolysis (degradation caused by water absorption) and carbon arc tracking (self-propagation of arc faults along the surface of the insulation). Polyimide is now used almost exclusively in hybrid wiring insulation, where it is sandwiched between layers of Teflon® for safety. Kapton® polyimide film is transparent and is amber in color. An opaque top coat is applied to provide different colors of wire and a surface for wire printing. Kapton® wire configurations are defined in Revision B of MIL-DTL-81381. Kapton® wire has poor life characteristics and is no longer recommended for Navy aircraft.

3.2.24 Potting. Process of completely enclosing an article in an envelope of liquid dielectric material, which then changes to a solid. Potting is performed to improve and protect the electrical functions of the unit. The compound acts as a dielectric and provides strain relief and protection to the unit from the environment.

3.2.25 Routing. The path followed by a cable or conductor.

3.2.26 Safety wire. Securing wire used to prevent the loosening or vibrating free of the attached part. Also named lockwire.

3.2.27 Severe Wind and Moisture Problem (SWAMP) areas. Areas such as wheel wells, wing folds, areas near wing flaps, and areas directly exposed to extended weather conditions are considered SWAMP areas on aerospace vehicles.

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3.2.28 Spiral wrap. Term used to describe the helical wrap of a tape or thread over a core.

3.2.29 Strand. One of the wires or groups of wires of any stranded conductor.

3.2.30 Strand lay. Distance of advance of one strand of a spirally stranded conductor, in one turn, measured axially (see lay).

3.2.31 Stranded conductor. Conductor composed of a group of wires or of any combination of groups of wires. The wires in a stranded conductor are usually twisted or braided together.

3.2.32 Stress relief chamfer. The tapering or rounding of an interior or exterior corner of a mechanical part, incorporated in the design to eliminate areas of high-stress concentration.

3.2.33 Stripping wire. Removal of a predetermined portion of insulation without affecting the mechanical or electrical characteristics of the conductor or the remaining insulation.

3.2.34 Tape, pressure-sensitive. Pressure-sensitive tapes contain an adhesive coating applied to the backing material, which allows the backing to be positioned with the application of pressure only. The use of an activator such as heat, solvent, or water is not required. The two primary functions of electrical pressure-sensitive tapes are holding and insulating. These tapes are commonly adhered to conductors or other insulating devices and serve the purpose of holding or anchoring them in the desired manner.

3.2.35 Tape wrap. Term denoting a spirally or longitudinally applied tape material wrapped around the wire, either insulated or uninsulated, and used as an insulation or mechanical barrier.

3.2.36 Terminal. Metal wire termination devices designed to handle one or more conductors to be attached to a board, bus, or block with mechanical fasteners, or clipped on. Types include ring, tongue, spade, flag, hook, blade, quick-connect, off-set, and flanged. Special types include taper pins, taper tab, and others, either insulated or non-insulated.

3.2.37 Terminal lug. A device designed to be affixed, usually at one end, to a post, stud, chassis, or similar device, and with provision for attachment of an electrical conductor(s) in order to establish an electrical connection.

4. GENERAL GUIDELINES

4.1 Application. The guidelines contained herein are intended to provide guidance applicable to the inspection and sustainment of EWIS.

4.2 Use of selection and application standards. When a selection and application standard is referenced in a guideline, the devices or parts selected should conform to the applicable military or commercial specifications referenced in the standard.

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5. DETAIL GUIDELINES

5.1 Individual guidelines for EWIS inspection. The individual guidelines for EWIS are located after section 6.

6. NOTES

6.1 Intended use. The guidelines in this document are intended to provide information on the proper inspection techniques and tools for sustaining EWIS.

6.2 Subject term (key word) listing.

Circular connectors
Corrosion prevention
Electronics cleaning
Fiber Optics
Maintenance procedures
Pressure sensitive tape
Rectangular connectors
Severe Wind and Moisture Problem Areas
Soldering
Stress relief chamfer

6.3 Changes from the previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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GUIDELINE 1

INSPECTION PREPARATION, TECHNIQUES AND DOCUMENTATION

1. Purpose. This guideline demonstrates proper inspection techniques used when working with the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

MIL-HDBK-525	Electrical Wiring Interconnect System (EWIS) Integrity Program
NA 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 004 01	Aircraft Wiring System Inspection
Work Package 026 00	Connector Cleaning and Preservation
Work Package 028 00	Protective Devices Installation

3. General inspection guidelines. The following steps should be used when performing EWIS inspections:

WARNING

Ensure all circuits under inspection or repair, or that are disconnected to facilitate the inspection or repair are not powered before proceeding with any of the following instructions or routine maintenance.
Failure to do so can result in severe injury, or death to personnel, and/or damage to equipment.

a. Prior to the EWIS inspection, the aircraft, or component should be safe for maintenance and not powered, using the platform-specific maintenance manual. This may include removal of external power and/or battery power disconnection, such that EWIS is de-energized and safe for handling, inspection, or maintenance. Lockout and tag applicable circuits as required (see WP 028 00 for authorized lockout devices).

b. The components to be inspected should be clean enough to minimize the possibility of collected dirt, grease, or other contaminants, which may potentially hide unsatisfactory conditions that would otherwise be obvious.

NOTE

A bright white light (not red or night vision green) with an incandescent (conventional) or LED bulb has been shown to be the most effective in identifying discrepancies.

c. EWIS Inspection should be performed using a bright light (incandescent or LED) and mirror, focusing on all the accessible wire system components (EWIS) in the applicable zone/area (see figure 1-1). For difficult to access locations, a borescope may be employed.

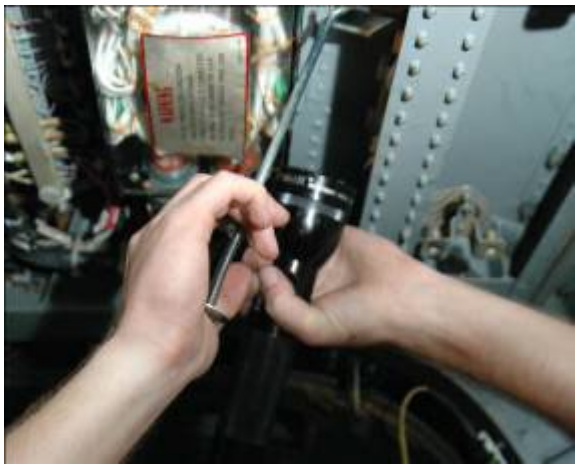
d. Use a consistent approach for each zone/area (e.g., left to right, top to bottom) and repeat for subsequent zones/areas to be inspected. Focus on all the accessible EWIS components in the applicable zone/area. Inspect the entire wiring system within the required zone, panel, or opening.

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e. Document any deficiencies using the procedures in the applicable aircraft/system maintenance manual and WP 004 01. Based on severity, some discrepancies may need to be corrected immediately while access to the area is opened. Deficiencies that are not corrected are required to retain the appropriate documentation in applicable aircraft/end item records/forms. These discrepancies are required to be worked on at the next available major maintenance action. To aid in capturing the inspection findings, see the inspection worksheets given in Appendix D of MIL-HDBK-525 for an optional format of an EWIS Inspection Worksheet/Aircraft Physical Inspection Log.



Acceptable
Proper tools for inspecting EWIS: Bright light and mirror.



Acceptable
Proper use of a mirror and flashlight inspecting a wiring harness.

FIGURE 1-1. EWIS inspection tools.

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GUIDELINE 2

INCOMING/RECEIVING WIRE INSPECTION

1. Purpose. This guideline provides incoming wire inspection criteria for wire used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 004 00	Wire Characteristics, Replacement, and Inspection Techniques
Work Package 007 00	Connectors, Wiring and Harness Stowage

3. General inspection guidelines. The following steps should be followed when inspecting incoming wire received from the supply system:

a. Ensure that wire/cable is received from the supply system on a spool (see figure 2-1) and wire is legibly marked with part number and CAGE code at intervals of 6 to 60 inches. Typically, markings are every 12 inches. Marking colors are black or green. Wire 26 gauge and smaller do not require marking (see figure 2-2).

b. Only wire/cable from qualified sources may be installed on aircraft. Refer to the applicable Qualified Products List (QPL) for approved manufacturing sources. Copies of QPLs are available from NAVAIR QPL at https://www.navair.navy.mil/qpl/ and from the ASSIST database at https://quicksearch.dla.mil.

c. Visually inspect exposed wire while wrapped on the spool for physical damage, nicks, cuts, burrs, abrasion, etc. (see figure 2-3). Unwind the first 3-4 feet and check for general condition, insulation smoothness, kinks, insulation discontinuity, discoloration, or bleaching. Remove damaged sections. If damage is throughout the length of the wire, it is unserviceable.

CAUTION

Silver-plated conductors are prone to cuprous oxide (red plague) corrosion due to improper processing during manufacture (see figure 2-3). Thoroughly inspect the stripped wire for evidence of red corrosion on the conductor. This is an unserviceable condition; submit a deficiency report.

Fluoropolymer-insulated wire and cable (e.g., ETFE, XL-ETFE, or composite, multi-layer insulation) are prone to HF outgassing (white plague) corrosion of the conductor or shield due to improper processing during manufacture (see figure 2-4). Thoroughly inspect the stripped wire for evidence of greenish or white corrosion on the conductor. This is an unserviceable condition; submit a deficiency report.

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NOTE

If the wire end is not capped, trim off one to two inches of the wire from the loose end. This ensures that any corroded/contaminated section is removed.

d. Visually inspect both ends of the wire by stripping insulation from the last one inch of wire, untwist strands and closely examine the conductor for any signs of corrosion. For silver-plated conductors, thoroughly inspect the conductor for evidence of red/reddish-brown discoloration (corrosion). Any sign of corrosion is cause for rejection (see figure 2-3). For Fluoropolymer-insulated wire and cable, inspect the conductor for evidence of white/green discoloration (corrosion). Any sign of corrosion is cause for rejection (see figure 2-4).

e. Gain access to both ends of the spooled wire and check wire for continuity. Continuity value should match the value on the label of the spool. If there is no label or missing value, refer to NAVAIR 01-1A-505-1 WP 004 00, table 1. Upon completion of continuity check, cap the exposed ends of the wire or cable to prevent wicking of moisture and potential corrosion.



Acceptable
The wire is received from supply on a spool.



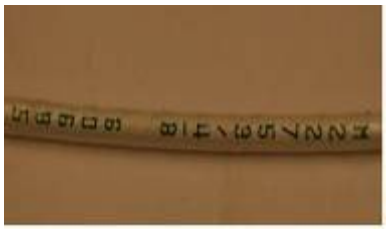
Unacceptable
Wire received from supply in an unusable condition. The wire is not on a spool.



Unacceptable
Wire received from supply in an unusable condition. The wire is in several small pieces.

FIGURE 2-1. Examples of acceptable and unacceptable wire received from the supply system.

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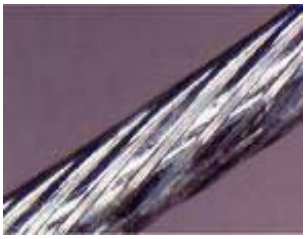


Acceptable
QPL wire with proper CAGE and part number markings from the manufacturer.



Unacceptable
No wire marking on the wire.

FIGURE 2-2. Examples of acceptable and unacceptable wire marking.



Acceptable
Typical wire conductor without any corrosion.



Unacceptable
Broken strands and signs of corrosion on conductor/shielding.

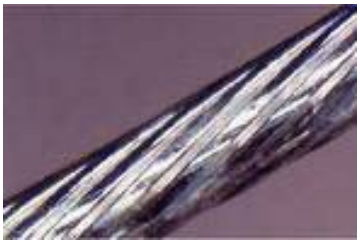


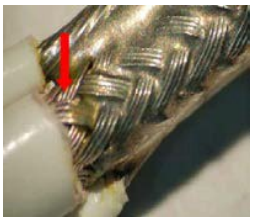
Unacceptable
Severe corrosion on wire conductor/shielding.

FIGURE 2-3. Examples of acceptable and unacceptable corroded conductors.

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Acceptable	**Unacceptable**	**Unacceptable**
Typical wire conductor without any corrosion.	Signs of severe corrosion on the conductor (white/ green).	Severe corrosion on wire conductor/shielding.

FIGURE 2-4. Examples of acceptable and unacceptable conductors with and without corrosion.

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GUIDELINE 3

WIRE INSULATION INSPECTION (EXCLUDING POLYIMIDE/KAPTON® WIRE)

1. Purpose. This guideline demonstrates inspection criteria for wire or cable insulation (other than polyimide type (see guideline 4)) used in the EWIS (such as, but not limited to, M22759/XX-XX).

NOTE

The "XX-XX" part number extension is for reference only, showing there are various configurations. Substitute for the actual part number extension as applicable in place of the "XX-XX."

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 004 01	Aircraft Wiring System Inspection
Work Package 004 00	Wire Characteristics, Replacement
Work Package 029 01	Basic Fault Isolation Methods
NAVAIR 01-1A-505-4	Aircraft Fiber Optic Cabling

3. General inspection guidelines. The following steps detail the inspection of wiring and reporting requirements for EWIS damage.

a. Examine insulation for physical damage to cable assemblies/bundles/wires such as:

(1) Evidence of arcing is present, such as burned insulation, burned/melted conductor strands, flash over marks on structure (see figure 3-1).

(2) Wire damaged, including cut, torn or separated braid/shield/insulation material, abraded or severed conductors, and abrasion against fluid carrying lines, or direct physical contact of size eight and larger power cables against a grounded structure.

(3) Damage to cable assembly/bundle/wire exterior (no damaged or abraded internal conductor), consisting of chafed or torn protective braid or insulation material.

(4) Wire damage to multi-layer insulated wire (such as composite) applying to M22759/80-92, M22759/180-192, M27500-XXDB-DR, and M27500-XXWB-WR wire/cable types. Damage to composite wire/cable outer layer consists of damage to only the PTFE (white outer tape exposing inner insulator layer).

(5) The following are examples of insulation damage and/or a configuration which may lead to wiring damage and should be documented as discrepancies requiring repair:

(i) Cracking.

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(ii) Chafing.

(iii) Flaking, or peeling of the insulation of any wire or cable (excluding polyimide topcoat, see guideline 4) is a discrepancy.

(6) Radial cracking (circumferential to the axis of wire or cable) is a discrepancy (see figure 3-2). Some insulation may be more susceptible to cracking within 1/2 inch of clamps.

(7) Heat damage: overheat/pronounced darkening wire insulation discoloration, melting, scorching, charring, and blistering is a discrepancy (see figure 3-3).

(8) Fluid/moisture effects: swelling, blistering or cracking is a discrepancy.

(9) Mechanical damage that is caused by the installation or removal of equipment, crew movement, shifting cargo, etc., is a discrepancy.

(10) Wiring that bears evidence of having been crushed is a discrepancy.



Acceptable
Insulation is not damaged

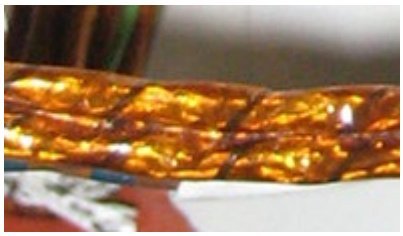


Unacceptable
Wire bundle with more than six wires showing insulation overheat damage

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FIGURE 3-1. Examples of wire in acceptable condition and with severe wire damage.

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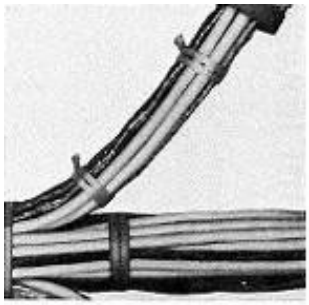


Acceptable
Insulation is not damaged, chafed, or flaking.



Unacceptable
Radial crack in the insulation.

FIGURE 3-2. Examples of insulation in acceptable and unacceptable condition.



Acceptable
Serviceable wire harness with no signs of damage.



Unacceptable
Arc damage



Unacceptable
Overheating is seen as a discoloration of the insulation (brown) compared to the white insulated wire.



Unacceptable
Overheating of the wire insulation is seen at the splice termination.

FIGURE 3-3. Examples of an undamaged wiring harness and wire damaged due to arc tracking and overheating.

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GUIDELINE 4

POLYIMIDE (KAPTON®) WIRE INSULATION INSPECTION

1. Purpose. This guideline demonstrates EWIS inspection criteria for wire or cable, which employs polyimide type insulation, outer jacket, or topcoat. The guidance given in this guideline applies to any M81381/XX, or M81381/7, M81381/9, M27500-XXMR, and M27500- XXMT wire/cable types.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 004 01	Aircraft Wiring System Inspection

3. General inspection guidelines. The following steps detail the inspection of polyimide insulated wire for EWIS damage.

a. Inspect polyimide insulated wires for the following types of damage:

(1) Wire damage to cable assemblies/bundles/wires which include wire flash-over/arc tracking, burned/brittle insulation, or burned/broken conductor strands (see figure 4-1).

(2) Wire damage to cable which include cut, torn or separated braid/shield/insulation material, abraded or severed conductors, abrasion against fluid carrying lines, or abrasion of size eight and larger power cables against grounded structure, or wire flash-over/arc tracking, burned/brittle insulation (see figure 4-2).

(3) Damage to cable assembly/bundle/wire exterior (no damaged or abraded internal conductor), consisting of chafed or torn protective braid or insulation material (see figure 4-3).

(4) Superficial damage to Kapton® insulated wire (such as flaking) applies to all M81381/7, M81381/9, M27500-XXMR, and M27500- XXMT wire/cable types. Superficial damage to Kapton®, wire/cable exterior, consists of damage only to the outer jacket (known as topcoat flaking). It is usually found at connector/boot area or termination points and is due to repetitive bending or flexing of cable assembly/bundle/wire where the conductor is not exposed, and there is no insulation breaches/cracks (see figure 4-4).

(5) Radial cracking (circumferential to the axis of wire or cable) is a discrepancy (see figure 4-5). Wire or cable with insulation that has this type of damage is susceptible to cracking within ½ inch of clamps or at areas of relative motion.

(6) Wiring that bears evidence of having been crushed is a discrepancy. Classification of damage is dependent on severity.

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Acceptable
Insulation is not damaged, chafed, or flaking.



Unacceptable
Wire and insulation is damaged, overheated, on more than six wires in the bundle.

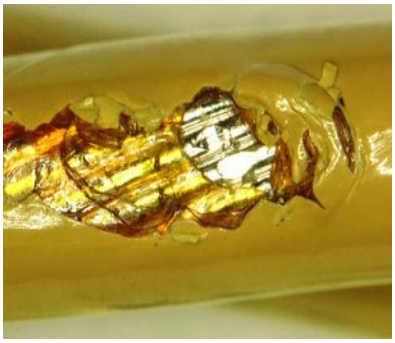
FIGURE 4-1. Examples of polyimide insulated wire in an acceptable condition and polyimide insulated wire with severe damage.



Acceptable



Unacceptable



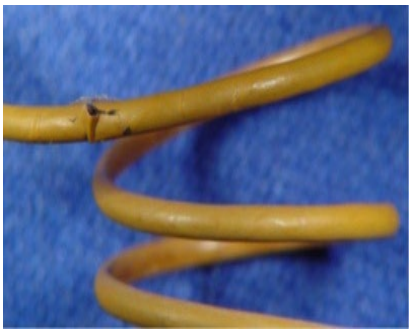
Unacceptable

FIGURE 4-2. Examples of polyimide insulated wire in acceptable condition and polyimide insulated wire damage with unserviceable, cracked, and severe insulation damage.

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Acceptable
No wire insulation damage.



Acceptable
Only outer topcoat damage. No inner polyimide insulation (brown) damage; conductor not exposed.

FIGURE 4-3. Examples of polyimide insulated wire in acceptable condition and polyimide insulated wire with serviceable, superficial topcoat insulation damage.



Acceptable

Polyimide insulation topcoat flaking; no damaged polyimide insulation (brown) no repair required.



Acceptable

Polyimide insulation topcoat flaking; no damaged polyimide insulation (brown); no repair required.



Unacceptable

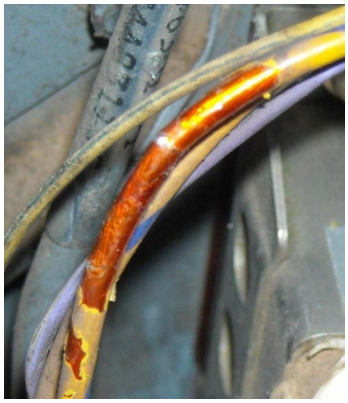
One wire shows exposed conductor; repair required.

FIGURE 4-4. Examples of polyimide insulated wire in acceptable condition and polyimide insulated wire with unserviceable, cracked and with severe insulation damage.

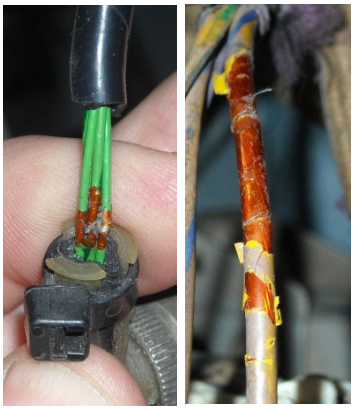
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Acceptable
No wire insulation damage.



Unacceptable
Polyimide insulation cracked; conductor exposed.



Unacceptable
Polyimide insulation cracked; conductor exposed.

FIGURE 4-5. Examples of polyimide insulated wire in acceptable condition and polyimide with radial cracking damage.

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GUIDELINE 5

CONNECTOR MATING INSPECTION

1. Purpose. This guideline gives criteria for inspecting connectors used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

MIL-DTL-38999	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded, and Breech Coupling), Environment Resistant with, Crimp Removable Contacts or Hermetically Sealed with Fixed, Solderable Contacts, General Specification for
MS27488	Plug, End Seal, Electrical Connector
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
NAVAIR 01-1A-505-2	Aircraft Circular Electrical Connectors and Accessories

3. General inspection guidelines. The following steps should be followed when inspecting connectors used in the EWIS:

a. Before coupling the connector, examine the mating halves for properly seated contacts and:

(1) Confirm that all cavities are filled with contacts (except unused cavities for coaxial contacts).

(2) Verify that there are no bent contacts.

(3) Confirm that all unwired cavities are fitted with the proper sealing plugs (e.g., MS27488).

(4) Verify that the applicable backshell, if required, is tightened on the connector, and where cable clamp is used, saddle bars are tightened.

(5) Verify that grommet or silicone pressure tape is positioned under saddle bars. See figure 36-2.

WARNING

Unless otherwise required by specific equipment technical data, power should be removed from the affected circuit to avoid shock hazard and possible arcing of connector.

b. After examination and mating, verify that connector has locked or is tight, depending on the type.

c. If the threaded connector has no mating indicator line/band, ensure that it has been sufficiently tightened. With moderate pressure applied to the back of the connector body, verify

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that the connector is not loose. If the connector body is loose, tighten the coupling nut as required. If the coupling nut is fully tightened and the connector body is still loose, de-mate the connector, inspect, and/or replace the faulty connector.

d. If the connector has a mating indicator line/band, check for the proper location of the red, locked indicator band. For MIL-DTL-38999, series 3 connectors, if the red band is properly located and the connector is fully mated, the red band will not be visible. If the connector is fully mated and the red band is visible, two conditions may apply (see figure 5-1):

(1) The red band has been mislocated on the receptacle, but the connection system otherwise functions as designed.

(2) The red band is correctly located on the receptacle or plug, but the red band is mislocated on the mating connector half due to allowable tolerances defined in MIL-DTL-38999, Revision K. This condition has been corrected in Revision L of MIL-DTL-38999, but these connectors may be installed in various systems/aircraft. If this condition exists, use the steps given in this guideline to confirm that the connector is properly mated and mark the connector as shown in figure 5-2.

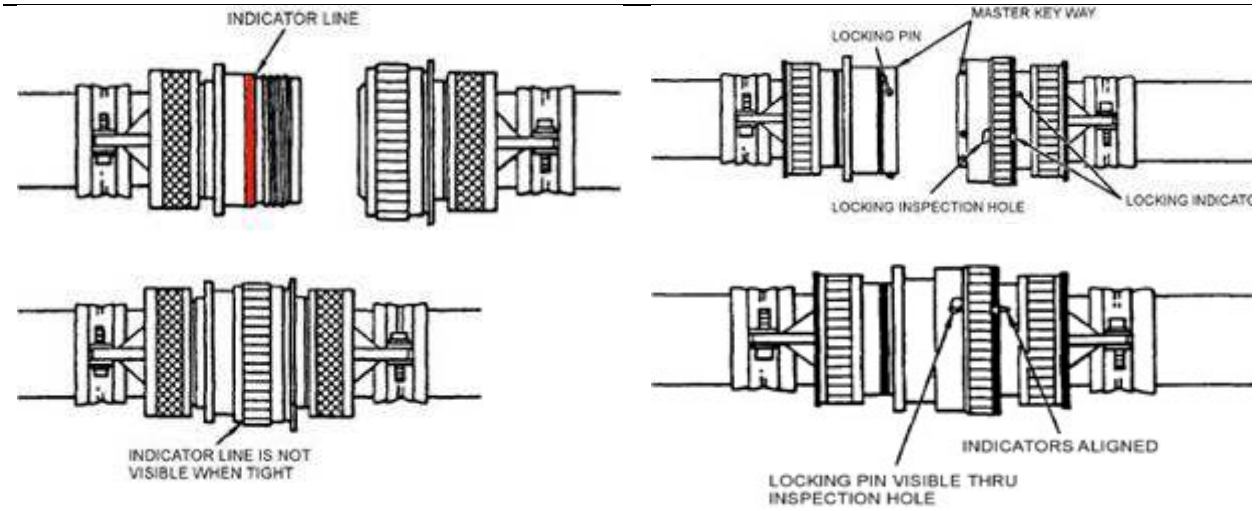
e. If the connector has a bayonet system, then the following procedure is followed:

(1) The bayonet system employs three locking pins spaced 120 degrees apart on the outside perimeter of the receptacle.

(2) Make sure that all locking pins of the coupling are engaged.

(3) When connected, verify that the locking pins are visible through the inspection holes. The locking pins are usually colored white or blue.

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Acceptable **Acceptable**

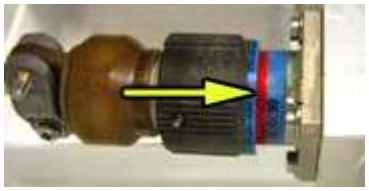
FIGURE 5-1. Example of correctly mated connectors using an indicator line and locking pin with inspection hole.



Acceptable
Upon successful system functional test (all applicable systems pass), a yellow stripe is marked on the receptacle.



Acceptable
Plug indicating a known mislocated red band on an otherwise functional receptacle.



Unacceptable
The red band is visible and the connector has no yellow marking stripe.

FIGURE 5-2. Examples of acceptable and unacceptable marking of yellow stripes on connectors.

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GUIDELINE 6

COAXIAL CABLE INSTALLATION INSPECTION

1. Purpose. This guideline gives criteria for the inspection of coaxial cable installed in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

A-A-52083	Tape, Lacing and Tying, Glass
A-A-52084	Tape, Lacing and Tying, Aramid
MIL-DTL-32554	Straps, Tie-Down, Adjustable, Non-Metallic, Cable Bundling, General Specification for
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 006 00	Radio Frequency (RF) Cable Characteristics and Replacements

3. General inspection guidelines. When inspecting cable harnesses that include coaxial cables, the following items should be checked:

a. Cable clamps and spot ties used on coaxial cables should be examined to confirm that they are not excessively tight. Only MIL-DTL-32554 Cable Lacing Fastener (CLF), A-A-52083, or A-A-52084 lacing tape/tie string should be used for tying wire and cable bundles containing coaxial cables. Figure 6-1 shows the proper use of lacing tape/tie string. See guideline 51 for MIL-DTL-32554 CLF inspection.

b. Confirm that coaxial cable is routed as directly as possible.

c. Verify that the minimum bend radius requirements are not violated; if none noted, use ten times the diameter of the largest cable in the harness (see figures 6-2, 6-3, and 6-4).

d. Confirm that unnecessary or sharp bends are avoided to preserve the cable's dielectric integrity.

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Acceptable
Proper installation of lacing tape installed on a harness. A-A-52083 or A-A-52084 lacing tape, ties are neat and properly spaced.



Acceptable
A-A-52083 or A-A-52084 lacing tape installed tight and secure on the harness.



Unacceptable
Continuous lacing tape installation is not approved for securing wires in harnesses.

FIGURE 6-1. Examples of acceptable and unacceptable use of lacing tape.

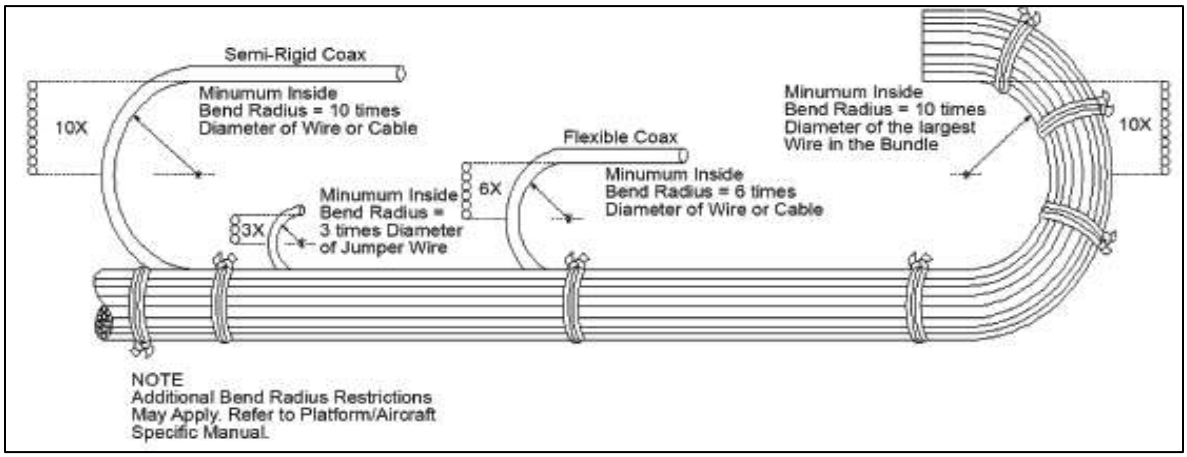
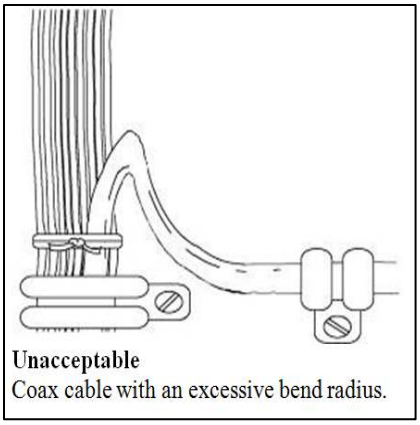


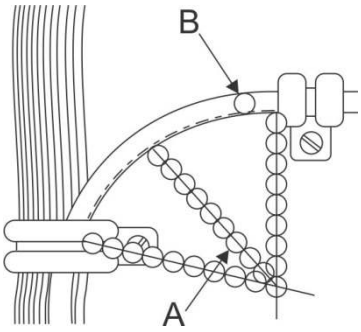
FIGURE 6-2. Acceptable bend radius.

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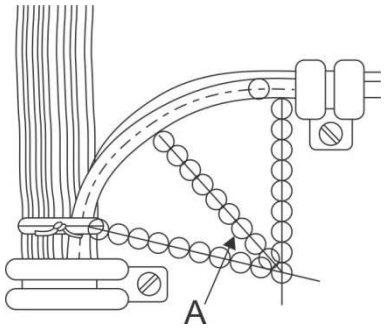


Unacceptable
Coax cable with an excessive bend radius.

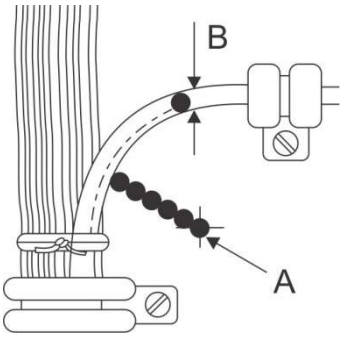
FIGURE 6-3. Example of unacceptable coaxial bend radius.



Acceptable
Wires and cables should have a minimum bend radius (A) of ten times the diameter of the largest wire insulation (B) contained in the bundle.



Acceptable
The bend radius is shown (A) measures slightly more than ten times the diameter of the largest wire in the harness breakout.



Unacceptable
The radius (A) of the wire breakout measures only six times the wire diameter (B). Excessive stress is exerted on wires installed with sharp bends.

FIGURE 6-4. Acceptable and unacceptable bend radius.

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GUIDELINE 7

PROPER MARKING OF WIRE/FIBER OPTIC AND CABLE HARNESSES

1. Purpose. This guideline gives inspection criteria on the proper marking of wire and cable used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

A-A-52083	Tape, Lacing and Tying, Glass
A-A-52084	Tape, Lacing and Tying, Aramid
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 008 00	Wire, Cable, and Harness Marking
NAVAIR 01-1A-505-4	Aircraft Fiber Optic Cabling
Work Package 003 03	Marking Methods and Labels

3. General inspection guidelines. The following procedures should be followed when inspecting wire and fiber optic cables and harnesses.

a. There are two methods of marking wire and cables: direct and indirect marking. Inspect marked wire/cable as follows:

WARNING

Hot stamp marking directly on the wire or cable is not authorized for any aircraft EWIS application. Hot stamping degrades wire insulation, resulting in system failure and potential injury to personnel.

(1) Direct marking is printed on the wire or cable outer covering. Inspect cable assembly/harness marker as follows:

(i) Ensure that the marking can be identified at intervals not longer than 3 inches along the entire length of wire or cable.

(ii) Verify that direct marking is legible and contains the applicable identifier for the system, sub-system, reference designator, or item identification.

(iii) Hot stamp marking of wires is not approved for aerospace applications (see figure 7-3).

(2) Indirect wire or cable marking is performed using tags or labels attached to the wire or cable. Inspect cable assembly/harness marker as follows:

(i) Verify that the indirect marked wire or cable is identified with printed heat shrinkable sleeves or labels after the last clamp and within 12 inches of the cable termination and at intervals of 3 feet throughout the length of wire or cable or harness (see figure 7-5).

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(ii) Verify that the marking labels are installed after the last clamp and within 12 inches of the termination point.

(iii) Individual wire not in a cable should be identified with a marking label in the same fashion as if it were a harness.

(iv) Individual wires less than 6 inches in length need not be marked.

(v) Ensure that the heat-shrinkable marking labels have been shrunk/recovered onto the harness (see figure 7-5).

(vi) If, after examining the harness, labels are missing, print and affix new marking labels.

(vii) Verify that marking is legible and contains the applicable identifier for the system, sub-system, reference designator, or item identification (see figure 7-4).

(viii) If non-adhesive marking labels are used, ensure that they are secured with A-A-52083, A-A-52084, or equivalent lacing tape in accordance with NAVAIR 01-1A-505-1, WP 010 00 (see figure 7-1).

CAUTION

Continuous/direct printing methods such as Inkjet, Laser, and Hot stamp marking are not authorized for any fiber optic application as it may cause system degradation and/or failure.

b. Marking of fiber optic cables can only be performed using indirect marking methods. Indirectly marked cable bundle or cable harness should be identified with printed sleeves at the following locations unless otherwise superseded in the platform maintenance manual. Inspect fiber optic cables as follows:

(1) After the last clamp.

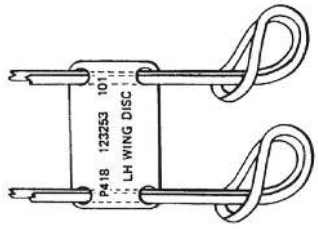
(2) Within 12 inches of the cable termination.

(3) At intervals of 3 feet throughout the length of the individual cables, cable bundle, or cable harness.

(4) Within six inches of the cable assembly entering or exiting conduit for routing through a fuel tank or used as protection in a severe environment.

(5) The recommended fiber optic cable marking should contain applicable text and border. The border or the band should be violet, and the background color yellow (see figure 7-2).

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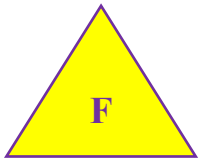






Acceptable	**Acceptable**	**Unacceptable**
Marking label secured to wire bundle using lacing tape/tie string.	Wrap around wire marking label secured with lacing tape/tie string.	Wrap-around, non-adhesive wire marking label is not secured with lacing tape/tie string.

FIGURE 7-1. Examples of acceptable and unacceptable wire marking with and without lacing tape.





Acceptable
Weapon Repairable Assembly (WRA) Label. This label is usually found within 6 inches from the fiber optic connection port. The text and border color are violet, and the background color is yellow.

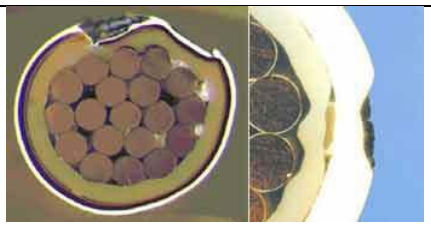
Acceptable
The cable is marked within 12 inches of the termination point.

FIGURE 7-2. Examples of acceptable fiber optic labels and cable marking.

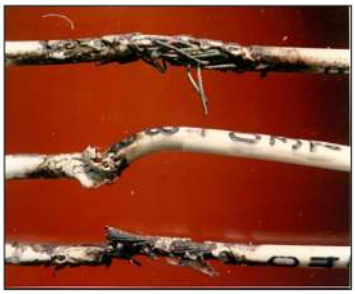
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Acceptable
UV laser wire marking on the cable is legible and does not damage the insulation.



Unacceptable
Hot-stamp marking of wire damages the insulation. This marking method is not for aerospace applications.



Unacceptable
Hot-stamp marking of wire damages the insulation. This marking method is not for aerospace applications.

FIGURE 7-3. Examples of direct wire/cable marking acceptable and unacceptable condition.



Acceptable

Adhesive marking tape is secure and legibly identifies the wire harness.



Unacceptable

Adhesive marking tape is not secure and is peeling off.

FIGURE 7-4. Examples of indirect wire harness marking using tape; in acceptable and unacceptable condition.

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Acceptable
Heat shrink tubing marker label sleeve is secure within 12 inches of the connector and legibly identifies the wire harness.



Unacceptable
Heat shrink tubing marker label sleeve is secure, but harness id information is missing, or worn off and fails to identify the wire harness.

FIGURE 7-5. Examples of indirect wire harness marking using heat shrink tubing sleeve; in acceptable and unacceptable condition.

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GUIDELINE 8

CAPPING AND STOWAGE OF ON-AIRCRAFT CONNECTORS INSPECTION

1. Purpose. This guideline gives inspection criteria and procedures for the proper capping and stowing of connectors used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

A-A-52083	Tape, Lacing and Tying, Glass
A-A-52084	Tape, Lacing and Tying, Aramid
A-A-59163	Insulation Tape, Electrical, Self-Adhering Unsupported Silicone Rubber
ANSI/ESD S20.20	ESD Association Standard for the Development of an Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices)
Air Force Technical Order 00-25-234	General Shop Practice Requirements for the Repair, Maintenance, and Test of Electrical Equipment
SAE AS21919	Clamp, Loop Type, Cushioned Support
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 007 00	Connectors, Wiring and Harness Stowage for operational and Non-operational Aircraft.
Work Package 026 00	EWIS Cleaning and Preservation
NAVAIR 01-1A-505-4	Aircraft Fiber Optic Cabling
Work Package 012 02	Dust Cover Preparation and Capping Methods

(ANSI/ESD S20.20 is available from the Electrostatic Discharge Association www.esda.org.)

3. General inspection guidelines. When fiber optic and electrical connectors are capped or stowed in the EWIS, the following procedures and details apply:

a. If fiber optic termini are installed in the connectors used in the EWIS, refer to NAVAIR 01-1A-505-4 for identification and follow-on maintenance procedures.

b. If electrical connectors are being examined for proper capping or covering, the following actions should be taken:

(1) Ensure that the connector is clean and the exterior (external metal surfaces) has corrosion preventative compound sparingly applied. See NAVAIR 01-1A-505-1, Work Package 026 00 for further information.

(2) Ensure that an approved capping method is employed based on application (operational/non-operational) and duration (short/long term).

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(i) Metal cap/plug. When inspecting a connector using a metal cap or plug, verify that the connector is clean and that a military standard protective metal cover is used (see figure 8-1).

CAUTION

Plastic caps are not authorized for use at any time on ready-for-flight aircraft, as they are a FOD hazard. Only military standard metal covers or pressure-sensitive tape are authorized.

(ii) Plastic connector covers are only to be in aircraft undergoing depot maintenance, off aircraft maintenance, and for shipping (see figure 8-1).

WARNING

Do not use Conductive Shielding Tape (ESD Grid Tape, NSN 5999-01-378-8454) to cap aircraft connectors. It is a FOD hazard; when removed, it can leave a conductive residue. Application of this tape to on-aircraft connectors can cause damage to aircraft and injury to personnel. For Air Force Applications, refer to ANSI/ESD S20.20 and Technical Order 00-25-234, Section VII.

NOTE

Plastic bags are not authorized for use to cap and stow connectors.

(iii) When examining connectors wrapped with pressure-sensitive, self-fusing silicone tape, verify that the connectors have been cleaned and preserved. Also, confirm that A-A-59163 type II tape is being used, and it is secured with lacing tape/tie string A-A-52083 or A-A-52084 in accordance with NAVAIR 01-1A-505-1, WP 010 00 (see figure 8-1).

(3) When inspecting connectors secured to adjacent structure or harnesses, confirm that they are secured using cushion clamps (e.g., SAE AS21919) or tie string/lacing tape (e.g., A-A-52083 or A-A-52084). See NAVAIR 01-1A-505-1, WP 007 00, paragraph 10 with examples shown on figures 3 through 6.

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Acceptable
Typical metal or composite protective cover.
Typical dummy receptacle for stowing connectors. Cap is secured with a chain or cable lanyard.



Acceptable
Pressure-sensitive tape wrapped connector.





Unacceptable
Plastic ESD bag or ESD Grid Tape used to cap and stow connector is not suitable.





Unacceptable
Plastic Dust Cap; not for operational aircraft; only authorized for aircraft in depot maintenance or off aircraft for shipping and storage. Plastic bag and plastic tie are also not suitable.

FIGURE 8-1. Examples of acceptable and unacceptable connector capping and stowage methods.

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GUIDELINE 9

MECHANICAL STRIPPING WIRE INSPECTION

1. Purpose. This guideline gives criteria for inspecting wire used in the EWIS after using a mechanical stripping device.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 009 00	Wire and Cable Stripping

3. General inspection guidelines. When inspecting or repairing wire stripped using a mechanical stripping device, the following procedures should be followed:

a. Visually inspect the wire and determine if any of the following unacceptable conditions exist (see figures 9-1 and 9-2). Unacceptable wire strip may be caused by incorrect strip tool choice, worn strip blade/gripper pad, or improper tool use. Inspect wire for:

(1) Nicked or cut strands

(2) Frayed insulation

(3) Broken wire strands (see table 9-I)

WARNING:
Care should be exercised when smoothing insulation or twisting conductors as nicked, frayed, or broken strands can cause injury.

(4) Unstranded, splayed, or bird cage strands (see figure 9-1). If un-twisting or bird-caging occurs, correct and reshape conductor strands by twisting the strands in the same direction as the normal lay of the wire. The conductor is recommended only to be twisted by hand. The conductor should not be over-twisted.

b. Visually inspect the wire insulation to determine if it has been damaged during wire stripping. When examined, the wire insulation (see figure 9-3):

(1) Should not be punctured, crushed, or cut by the tool.

(2) Should not have deformation exceeding 20 percent of the insulation thickness.

(3) Should not have gouges, ragged edges, be loose, nor frayed.

(4) The end of the insulation should be cut as squarely and cleanly as required to meet any soldering or crimping requirements.

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(5) Insulation damage should not exceed 1/32 inch, or 50 percent of the insulation's outside diameter, whichever is greater.



Acceptable
Wire insulation removed without disturbing the normal lay of the wire, as shown. Wire strands are free of nicks or cuts.



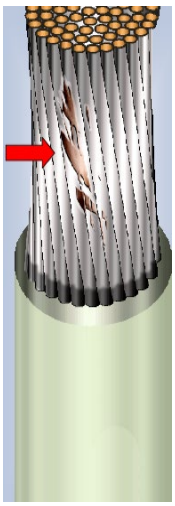
Unacceptable
Strands, re-twisted and overlapping each other, as shown, will result in increased stress and difficulty in forming a mechanical joint.



Unacceptable
Wire strands that are re-twisted in excess of their normal lay exert increased stress and may break.

FIGURE 9-1. Examples of acceptable and unacceptable wire stripping.

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Unacceptable
Wire strands are nicked due to misalignment of wire and stripping blades. Nicked strands reveal base metal and may break.



Unacceptable
Wire strands show evidence of a ringed condition (arrow) caused by stripper blades. Outer strands are weakened and may break.



Unacceptable
Several wire strands are cut (arrow). This condition may be the result of placing the wire in the wrong hole size of the mechanical stripper.



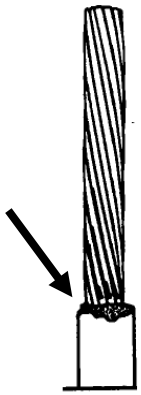
Unacceptable
Wire strands are bird-caged due to misalignment of wire and stripping blades.

FIGURE 9-2. Examples of wire strands in an unacceptable condition after wire stripping.

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Acceptable
Wire stripped with mechanical strippers that show no evidence of damage to either the insulation or the wire strands.



Acceptable
Although wire insulation appears to be cut or split, close examination shows that only the outer coating is damaged and does not exceed 1/32 inch, or 50 percent of insulation's thickness (whichever is greater).



Unacceptable
The insulation has been pinched by the mechanical strippers. This condition was caused by using too small of a hole/setting of the mechanical stripper.

FIGURE 9-3. Examples of wire insulation in acceptable and unacceptable condition after wire stripping.

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TABLE 9-I. Allowable nicked or broken strands.

Number of Strands per Conductor*	Total Allowable Nicked or Broken Strands 1/
7	None Nicked, Broken, or Severed
19	2 Nicked, None Broken or Severed
37	4 Nicked, None Broken or Severed
More than 37	6 Nicked, None Broken or Severed
1/ No nicked or broken strands are permitted for aluminum conductor regardless of the number of conductor strands.	

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GUIDELINE 10

MECHANICAL STRIPPING/SHIELDING REMOVAL INSPECTION

1. Purpose. This guideline gives criteria for inspecting cables with shielding after they have been stripped using a mechanical stripping device.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 009 00	Wire and Cable Stripping

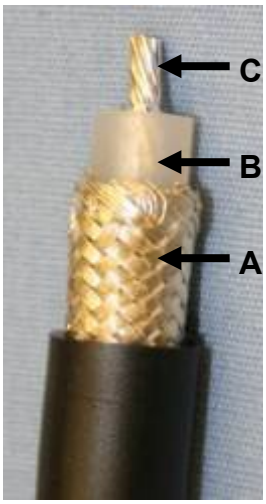
3. General inspection guidelines. When mechanically stripped shielded cable is inspected, the following procedure should be followed (see figure 10-1):

a. Verify that there are no nicks or cuts on the shielding or inner wire insulation, or the conductor.

b. Verify that the shielding is uniformly trimmed and shows no evidence of unraveling.

c. Verify the dielectric insulator is uniformly cut (flush/squared), and no nicks are present.

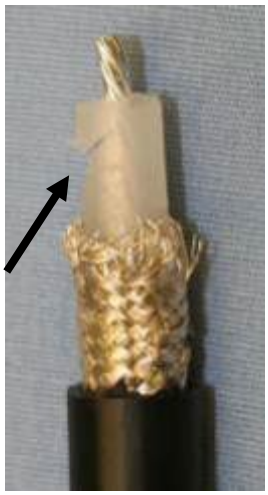
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Acceptable
After stripping, there should be no nicks or cuts on the shielding (A) or inner wire insulation (B), or the conductor (C).



Unacceptable
Cut shielding strands, as shown, are caused during outer insulation removal.



Unacceptable
The nicked dielectric insulator is caused during shielding removal.

FIGURE 10-1. Examples of acceptable and unacceptable mechanical stripping of shielded cable.

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GUIDELINE 11

THERMAL/LASER STRIPPING CABLE JACKET INSPECTION

1. Purpose. This guideline gives criteria for inspecting wire after using a thermal/laser stripping device.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 009 00	Wire and Cable Stripping

3. General inspection guidelines. Figure 11-1 shows cable in acceptable condition after stripping using a thermal or laser stripper. When inspecting cable jackets that have been stripped using a thermal or laser stripper, the following procedure should be followed:

WARNING

Thermal strippers can cause serious burns. Keep flammables away from thermal strippers. Do not leave thermal strippers unattended during operating or cool down. Thermal strippers are not for use on fueled aircraft as they are not explosion-proof.

a. Allow the wire to cool before handling, then verify that there are no nicks or cuts on the shielding or inner wire insulation or the conductor.

b. Verify that there has been no contact made by the wire with the thermal stripper beyond the stripping area.

c. Verify that the insulation is not charred or blistered.

d. Verify that the insulation is not pulled in strings adhering to the conductor.

e. Verify that all insulation is removed from the conductor.

f. Verify that the conductor is not damaged.

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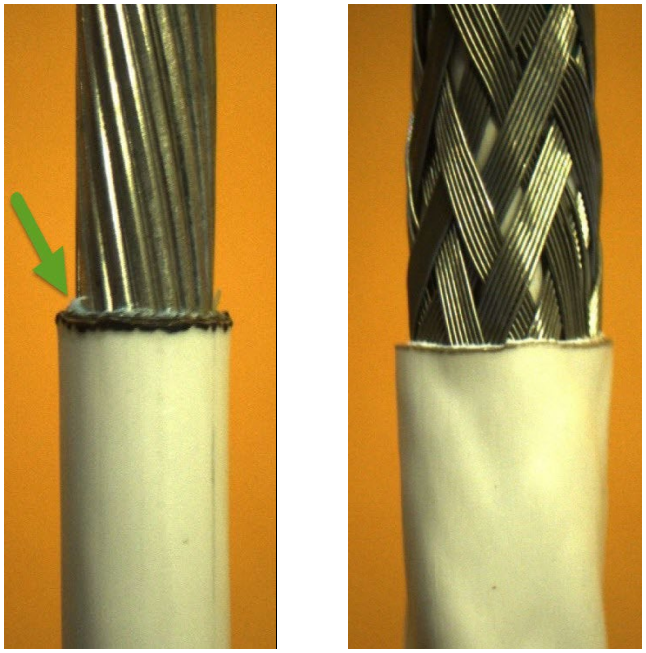


FIGURE 11-1. Examples of acceptable stripping of wire and shielded cable by thermal/laser strippers.

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GUIDELINE 12

HARNESS ROUTING INSPECTION

1. Purpose. This guideline gives criteria for inspecting proper harness routing of wires and cables that make up the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

A-A-52083	Tape, Lacing and Tying, Glass
A-A-52084	Tape, Lacing and Tying, Aramid
MIL-DTL-32554	Straps, Tie-Down, Adjustable, Non-Metallic, Cable Bundling, General Specification for
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 010 00	Harness Installation
NAVAIR 01-1A-505-4	Aircraft Fiber Optic Cabling
Work Package 012 01	Heating Tools for on Aircraft Maintenance

3. General inspection guidelines. When inspecting harness routing in aircraft, the following procedures should be followed:

a. Confirm that individual wires in a bundle are uniformly arranged and of the same length (see figure 12-1).

b. Confirm that the wire length is short enough so that there are no wire loops where individual wires can be easily damaged (see figure 12-2).

c. Ensure that cabling is routed so that relative motion does not result in abrasion between wires with dissimilar insulation.

d. Confirm that any wiring of dissimilar insulation that crosses over or under other wires is secured using lacing tape (e.g., A-A-52083 or A-A-52084) or MIL-DTL-32554 Cable Lacing Fastener (for information on CLF, see guideline 51) to prevent chafing before and after the cross over, but not on the cross over point. This is especially important in the case of polyimide insulation when it is in contact with other insulation types.

e. Ensure that sufficient slack exists for full extension of shock mounts or vibration isolators on cabling affixed or connected to shock/vibration protected equipment.

f. Confirm that sufficient slack exists to permit maintenance access.

g. Ensure that bundles are secured to exhibit a smooth appearance, without protruding wires which can be snagged or damaged.

h. Confirm that there are no twisted wires under the lacing string (see figure 12-3).

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i. When fiber optic cabling is routed with electrical cable/wire:

(1) Ensure that fiber optical cable does not support electrical cables.

(2) Verify that fiber optic cables are located on top of electrical cables.

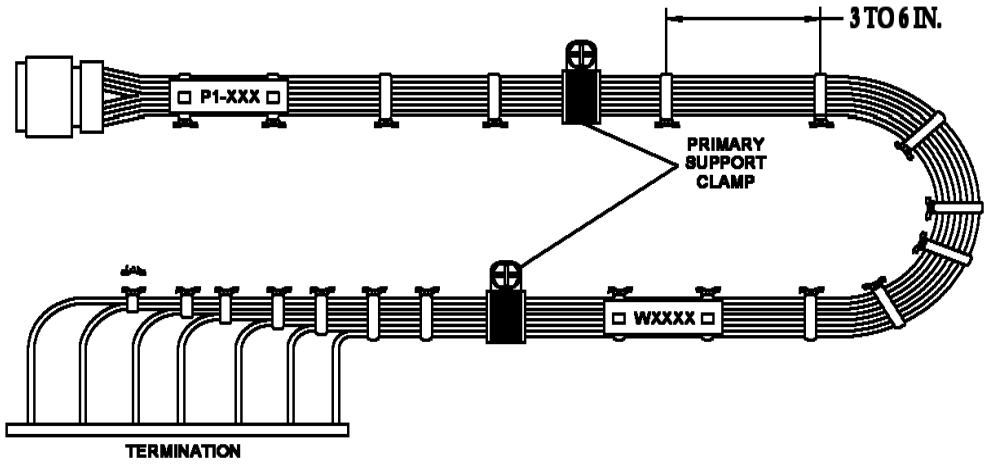
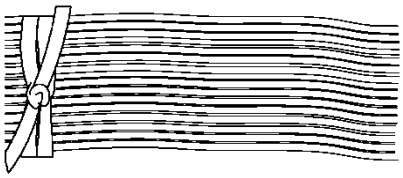
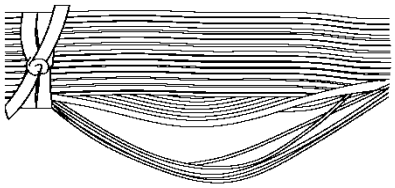


FIGURE 12-1. Typical wire harness assembly.



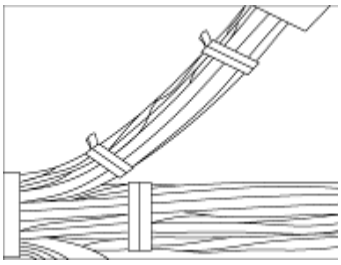
Acceptable
Individual wires in a wire bundle should be uniformly arranged and of the same length. Varying lengths of wire will increase the bundle diameter.



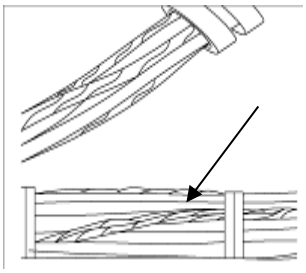
Unacceptable
Excessive wire length has formed loops that can easily be damaged and have increased the bundle diameter.

FIGURE 12-2. Examples of acceptable and unacceptable wires in a wire bundle.

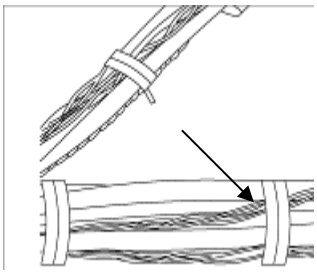
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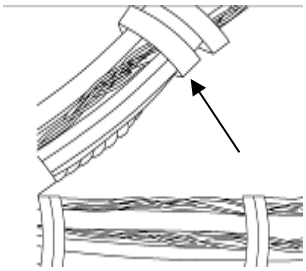
Acceptable
Wires in a harness should run straight and parallel to each other, as shown. Although some minor wire crossing is permitted, wires should never cross each other under a spot tie or clamp.



Acceptable
Wires (arrow) are shown crossing over the top of other wires. However, the crossover does not occur under a spot tie or clamp.



Unacceptable
The twisted wires (arrow) are shown crossing another wire underneath a spot tie. Damage to insulation may result.



Unacceptable
The twisted wires (arrows) have crossed over the top of the other wires under the clamp. Damage to individual wires may occur.

FIGURE 12-3. Examples of acceptable and unacceptable wires in a wiring harness that are straight and twisted.

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GUIDELINE 13

CABLE HARNESS COVERING OR PROTECTION INSPECTION

1. Purpose. This guideline gives criteria for inspecting cable harness coverings and abrasion protection.

2. Applicable documents. The documents listed below are those applicable to this guideline.

A-A-52083	Tape, Lacing and Tying, Glass
A-A-52084	Tape, Lacing and Tying, Aramid
SAE AS21919	Clamp, Loop Type, Cushioned Support
SAE AS23190	Wiring, Positioning, and Support Accessories
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 010 00	Harness Installation

3. General inspection guidelines. When inspecting and repairing cable harness covering and protection, the following procedures should be followed:

a. Examine cable harness covering (convoluted tubing or cable wrap) for:

(1) Loosening or fraying.

(2) Proper support using rigid conduit or SAE AS21919/AS23190 cable clamps (see figure 13-1).

(3) Proper installation of spot tie on convoluted tube wrapped harness and spiral wrap (see figures 13-1 and 13-2).

(4) Secured wrap ends are spot tied using lacing tape (A-A-52083 or A-A-52084) (see figure 13-2).

(5) Twisting of harness covering, a minimum of one revolution per foot. This accommodates for uniform coverage around bends or twists around the harness being protected (see figure 13-2).

b. Examine cable harness for chafing where wire or cable is routed near structural members and equipment (see figure 13-3).

(1) Crosses over/under other wiring.

(2) Passes through lightening holes. If found, cover all feed-throughs with an edge grommet (see guideline 45) and wire bundle or harness in a lightening hole (see figure 13-4).

(3) Moves/flexes when the door(s) are opened/closed.

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(4) Passes over or near hinged areas.

(5) Turns or bends near components and at connector backshells and is flexed during removal and installation of components.

(6) Around generator power wiring routing areas.

(7) At conduit exit points (see figure 13-1).

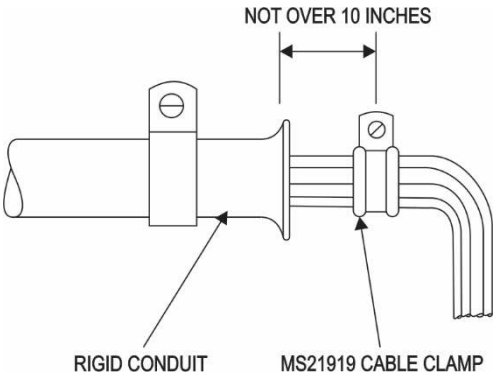
c. Clearance from structure, surfaces, and equipment should be a minimum of ½ inch. Where physical separation of at least ½ inch cannot be maintained, the edges should be covered with suitable protection, or the harness should have secondary protection.

d. For bundles containing polyimide (Kapton®) insulated wire (e.g., M81381), if the separation is ½ inch or less, use secondary protection over the wire bundle. Examples of secondary protection for wiring harnesses are heat shrink, spiral chafe wrap, chafe pad, Expando sleeving, or wrap-around sleeving. All harness wraps and sleeving should be secured with lacing tape/tie string within an inch of the ends. Refer to NAVAIR 01-1A-505-1, WP 010 00 for additional requirements.

e. Examine places where wire bundles come into contact with sharp edges. Re-route/reposition and secure wire bundle to achieve minimum ½ inch clearance. Employ chafe protection material only if there is limited space or bundle length prevents repositioning. Proximate, sharp edges should be covered using a Teflon sheet (see figure 13-5).

f. Examine places where wiring harnesses come into contact with abrasive surfaces. If found, Teflon tape with a minimum of 50 percent overlap should be used on these wiring harnesses. Spot ties should be used on each end of the tape to prevent unraveling. Teflon sheet or other chafe protection may also be used (see figures 13-4 and 13-5).

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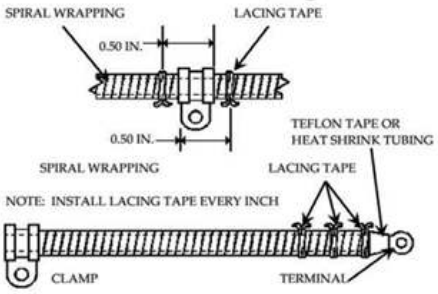


Acceptable
This is an example of how a wire harness should be supported after exiting a rigid conduit.

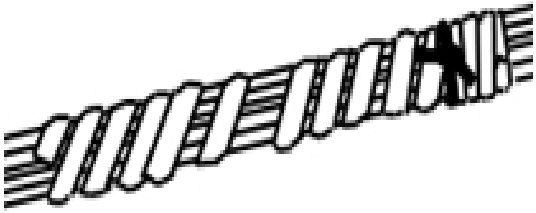


Acceptable
Correct installation of spot tie on a convoluted tube wrapped harness with a clamp.

FIGURE 13-1. Examples of acceptable cable harness support and protection.



Acceptable
Correct installation of spot tie on a spiral-wrapped harness with a clamp.



Unacceptable
Spot tie not installed on a spiral wrap. The spiral wrap is loosened and unraveling.

FIGURE 13-2. Examples of acceptable and unacceptable spiral-wrapped cable harness.

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Acceptable
Proper installation of harness covering.

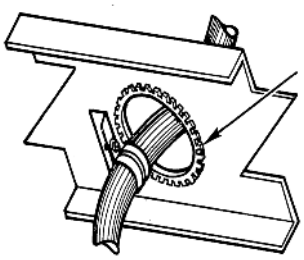


Unacceptable
Improper twisting of harness covering.

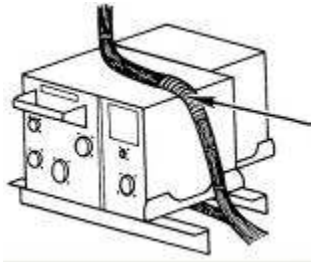


Unacceptable
No spot tie on harness covering.

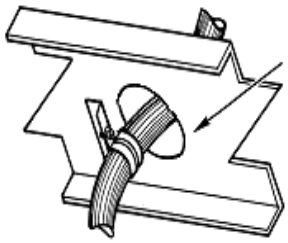
FIGURE 13-3. Examples of acceptable and unacceptable wires in a wiring harness that are straight and twisted.



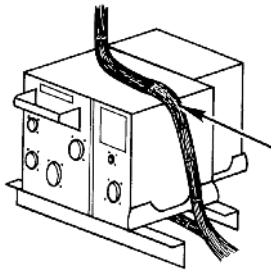
Acceptable
Feedthrough with edge grommet (arrow). Center bundle/harness in lightening hole.



Acceptable
Wire harnesses wrapped with Teflon spiral chafe wrap or chafe pad and secured with lacing tape/tie string.



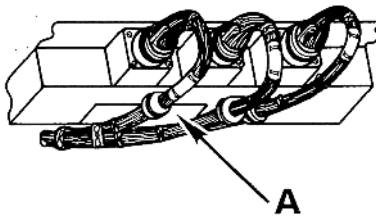
Unacceptable
Missing edge grommet.



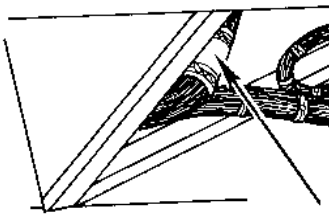
Unacceptable
A harness is routed on a sharp edge of a structure. Cable insulation will be damaged due to aircraft vibration.

FIGURE 13-4. Examples of acceptable and unacceptable routing of cables in feed-throughs and aircraft structures.

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Acceptable
(A) Positive clearance maintained and sharp edges covered with Teflon sheet.



Acceptable
(B) Teflon sheet with a minimum 50 percent overlap.

FIGURE 13-5. Examples of acceptable use of Teflon sheet.

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GUIDELINE 14

CRITICAL CLAMP MARKER INSPECTION

1. Purpose. This guideline gives inspection criteria on critical clamp markers used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 008 00	Wire, Cable, and Harness Marking
Work Package 010 00	Harness Installation

3. General guidelines. The following criteria for critical clamp marker installation are as follows:

WARNING

Critical clamp marker labels may not be moved from their design drawing-required location. If moved beyond allowable limits, the harness may be damaged, or cut, resulting in aircraft system failure or injury to personnel.

NOTE

Critical clamp markers installed on designated wire harnesses provide a means of ensuring installation clearance requirements are met. Typically, critical clamp markers are installed on very long/branched harnesses (landing gear, flight controls) subjected to motion.

a. Ensure that markers are permanent and cannot be moved to facilitate maintenance.

b. Verify that critical routing points are indicated by colored markers (typically one to two inches wide, white or yellow, marked by paint, tape, or label) on the bundle, which are located under cable clamp such that the colored marker is exposed on both sides of the clamp (see figure 14-1).

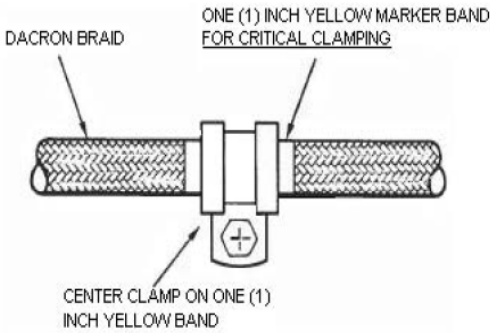
c. Verify that the cable assembly routing and clamping of specific cables or harnesses is correct. Refer to the applicable aircraft electrical wiring installation drawings or Illustrated Parts Breakdown. If not available, contact the Cognizant Engineering Authority for direction.

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Acceptable

Correct marker type; centered on the clamp, with both sides shown



Acceptable

Ideal clamp marker installation



Unacceptable

Markers are not under the clamps or only partially

FIGURE 14-1. Acceptable and unacceptable critical clamp marker installation.

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GUIDELINE 15

WIRE/HARNESS CLEARANCE INSPECTION

1. Purpose. This guideline gives criteria for inspecting wire or harness clearances.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 010 00	Harness Installation
Work Package 014 02	Heat Shrink and Repair
Work Package 029 01	EWIS Test Methods
NAVAIR 01-1A-505-4	Aircraft Fiber Optic Cabling
Work Package 012 01	General Practices for Cable Harness Installation

3. General inspection guidelines. When inspecting installed wire/harness and fiber optic cable clearance, the following guidance should be followed:

a. Clearance from structure, surfaces, and equipment should be a minimum of 1/2 inch. Re-route/reposition wire bundle to achieve minimum 1/2 inch clearance. Secure wire bundle using cushioned clamps. Employ chafe protection material only if there is limited space or bundle length prevents repositioning. Where physical separation of at least 1/2 inch cannot be maintained, the edges should be covered with suitable protection, or the harness should have secondary protection (e.g., heat shrink, chafe wrap, etc.). Examples of where these minimum clearances can exist include the distance between:

(1) Wire/harness and linkages, throttle controls, boxes, covers, structures, control cables, and component mounting hardware (see figure 15-1).

(2) Terminal lugs attached to other posts, adjacent components, and nearby structures at contactors, circuit breakers, relays, power control relays, and terminal boards.

b. Clearance from a flammable fluid carrying lines and tubes. The following should be considered when inspecting wire harness routed near fluid carrying lines. Request CEA/FSR (Field Service Representative) assistance if minimum clearance cannot be achieved.

(1) Wire/harness should be routed level with or above all fluid lines.

NOTE

Separation between wire/harness and fluid carrying lines is not required when a conduit, bulkhead, or other continuous structure separates the wire bundle from fluid lines.

(2) When six-inch clearance cannot be maintained, the wire bundle and flammable line should be clamped, so there is no relative motion between them (see figure 15-2).

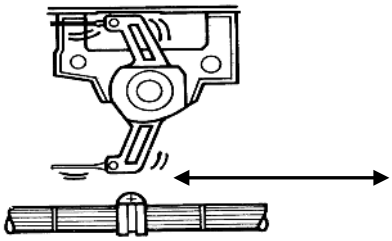
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(3) When a 2-inch clearance between the wire bundle and fluid carrying lines cannot be maintained, the wire bundle is required to be clamped (see figure 15-3, center detail).

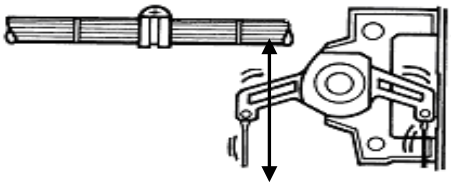
(4) The minimum clearance between the wire bundle and fluid carrying lines should be 1/2 inch. For all wire type applications, use clamp-to-structure or two cable clamps back-to-back (butterflied) to maintain a rigid separation from flammable fluid carrying lines (see figure 15-3, left detail).

(5) For wire bundles containing polyimide (Kapton®) insulated wire (e.g., M81381), if the separation is between 1/2 inch and 2 inches, use protective sleeve over the wire bundle (i.e., heat shrink or wraparound sleeve (see guideline 13). Limit protective sleeve application to maximum 2 inches beyond either side of the affected area.

(6) Wire harnesses are only clamped to fluid-carrying lines to maintain rigid separation and are not to be used as primary support. Primary supports should be located within every 24 inches.



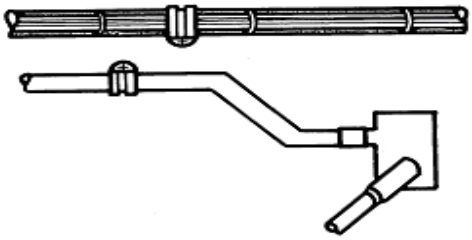
Acceptable
Secure wire harness to avoid moving parts and high-temperature ducts or equipment.



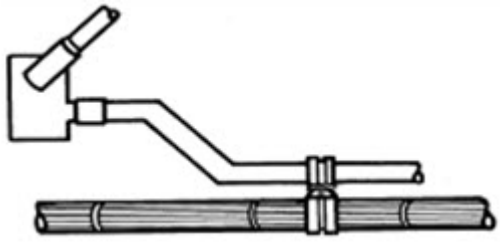
Unacceptable
Reposition wire harness to prevent the harness from being damaged by mechanical linkage during travel.

FIGURE 15-1. Examples of acceptable and unacceptable placement of wire harnesses near equipment and linkage.

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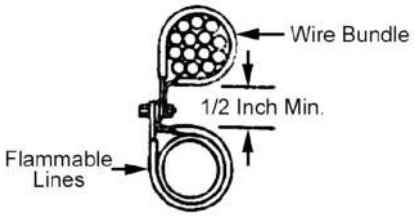
Acceptable
Cables, wires, and harnesses should maintain a separation of not less than 2 inches from lines carrying flammable fluids (fuels, hydraulic fluid, and coolant). Where design requirements are less than 2 inches, the cable should be rigidly supported and covered with suitable secondary protection material.



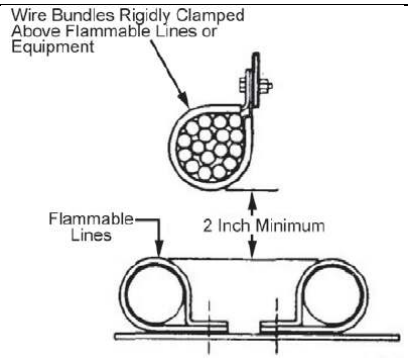
Unacceptable
Cables, wires, and harnesses should not be clamped or tied below fluid lines. The cable harness should be routed above fluid lines.

FIGURE 15-2. Examples of acceptable and unacceptable placement of wires, cables, and harnesses near fuel lines.

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Acceptable
Cables, wires, and harnesses should maintain a separation of not less than 2 inches from lines carrying flammable fluids (fuels, hydraulic fluid, and coolant). Where design requirements are less than 2 inches, the cable should be rigidly supported, no closer than ½ inch, and covered with suitable secondary protection material.



Acceptable
Cables, wires, and harnesses maintain a separation of more than 2 inches from lines carrying flammable fluids. The cable or harness is rigidly supported above the flammable fluid line.



Unacceptable
Cables, wires, and harnesses do not maintain a minimum separation of ½ inch from lines carrying flammable fluids or from airframe structure. The harness is not rigidly supported.

FIGURE 15-3. Examples of acceptable and unacceptable routing of wire harnesses near flammable fluid lines.

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GUIDELINE 16

CABLE HARNESS BEND RADIUS INSPECTION

1. Purpose. This guideline gives criteria for inspecting cable harness bend radius in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 010 00	Harness Installation
NAVAIR 01-1A-505-4	Aircraft Fiber Optic Cabling
Work Package 006 01	General Handling Practices for Fiber Optic Cable Harnesses

3. General inspection guidelines. The following procedures should be followed when inspecting the bend radius of electrical cables and harnesses. The maintenance of minimum bend radius requirements ensures optimum system performance.

a. For wiring system applications:

(1) Confirm that wiring groups, bundles or harnesses, and cables that are individually routed and supported meet the minimum bend radius. At the point individual wiring breaks out from a group, harness, or bundle, the minimum bend radius should be no less than ten times the outside diameter of the largest included wire or cable, provided the wire is suitably supported at the breakout point (see figure 16-1).

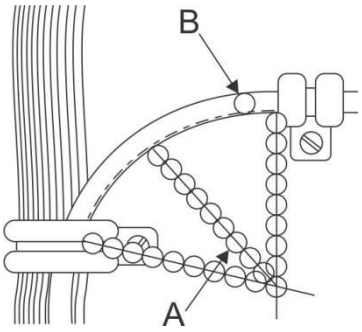
(2) Verify that wires used as shield terminators or jumpers, when required to reverse direction, have no less than a minimum bend radius three times the wire diameter at the reversal point, provided the wire is adequately supported.

(3) Flexible coaxial cables should have a minimum bend radius of six times the cable diameter, and semi-rigid coaxial cable should have no less than a minimum bend radius of 10 times the cable diameter (see figure 16-2).

(4) Ensure that the minimum bend radius, as measured on the inside radius of the harness/cable is no less than ten times the outside diameter of the largest wire/cable in the harness (see figures 16-1 and 16-2).

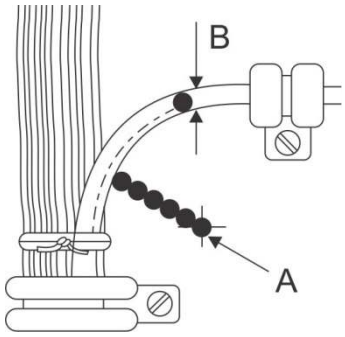
(5) For Fiber Optic (FO) cable applications, ensure that the minimum bend radius, as measured on the inside radius of the harness/cable is no less than 16 or 20 times the outside diameter of the FO cable/conduit/harness (see figure 16-3).

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Acceptable

Wires and cables should have a minimum bend radius (A) of ten times the diameter of the largest wire insulation (B) contained in the bundle.



Unacceptable

The radius (A) of the wire breakout measures only six times the wire diameter (B). Excessive stress is exerted on wires installed with sharp bends.

FIGURE 16-1. Examples of wires and cables with acceptable and unacceptable bend radius.

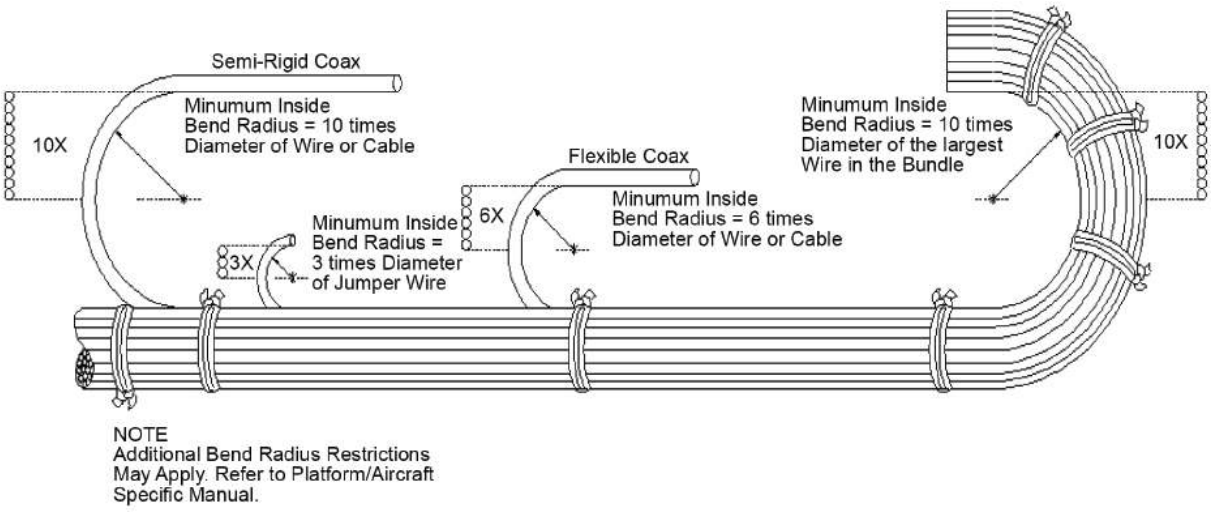


FIGURE 16-2. EWIS minimum bend radius.

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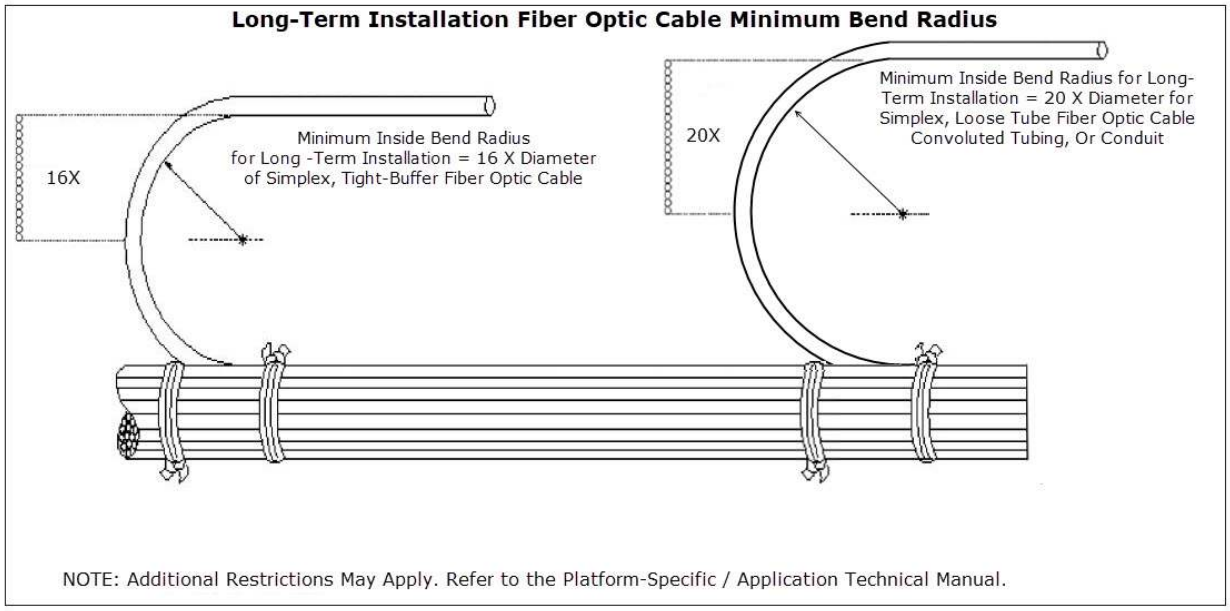


FIGURE 16-3. Long-term minimum bend radius for fiber optic cable.

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GUIDELINE 17

LACING TAPE INSPECTION

1. Purpose. This guideline gives criteria for using or inspecting secondary support spot tie/lacing tape used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

A-A-52083	Tape, Lacing and Tying, Glass
A-A-52084	Tape, Lacing and Tying, Aramid
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 010 00	Harness Installation
NAVAIR 01-1A-505-4	Aircraft Fiber Optic Cabling
Work Package 012 01	Heating Tools for on Aircraft Maintenance

3. General inspection guidelines. Use the following installation procedures when inspecting lacing tape or tie string:

a. Confirm that the wire bundle is secured using lacing tape/tie string (part number A-A-52083 or A-A-52084), where A-A-52084 is used for all general applications and A-A-52083 is only used for high-temperature applications. Ensure that only A-A-52083, finish C, size 2 or 3 is employed (e.g., A-A-52083-C-2). For applications where black color is required, use A-A-52084-C-2-BLK or A-A-52084-C-3-BLK.

b. Verify that a clove hitch and square knot has been used and trim off excess squarely to 3/8″ ±1/8″ in length (see figures 17-1 and 17-2).

c. Verify that wire/cable bundles have been tied tightly enough to prevent slipping, but not so tightly that the lacing tape/tie string cuts into or deforms the insulation. Care must be taken when lacing or tying a coaxial cable, which has a soft dielectric insulation. If the dielectric is deformed, signal integrity will be degraded.

d. Lacing tape/tie string should be uniformly spaced and located every 3″ to 6″ throughout the length of the harness. The spacing of spot ties used should be as indicated in table 17-I (see figure 17-3).

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TABLE 17-I. Lacing tape/tie string spacing on aircraft harnesses (secondary support).

Wire bundle diameter	Spot Tie Spacing	
	Max	Not less than
up to ½″	4″	3″
½″ to 1″	5″	3″
1″ & larger	6″	3″

e. Lacing tape/tie string installed inside electronic assembly/panel wiring/circuit breaker panel should be uniformly spaced in accordance with table 17-II throughout the length of the harness (see figure 17-4).

TABLE 17-II. Lacing tape/tie string spacing for inside electronic assembly and panel harnesses.

Bundle Diameter	Preferred Mimimum Spot Tie Spacing	Maximum Spot Tie Spacing
1/2 inch and less	Approx. 1/2 inch	1 - 1/2 inch
Over 1/2 inch up thru 1 inch	Approx. bundle diameter	1 - 1/2 inch
Over 1 inch	Approx. bundle diameter	Bundle diameter + 1/2 inch

f. Lacing string/tie string should not be closer to the clamp than a distance equal to the width of the clamp (see figure 17-5).

g. When lacing string/tie string is used at breakouts, they should be located as shown on figure 17-6.

CAUTION

Applying lacing ties too tightly can lead to degraded optical performance or fiber breakage. Use extreme caution when applying lacing tape to ensure that the ties do not deform the cable, cable bundles, or cable harness.

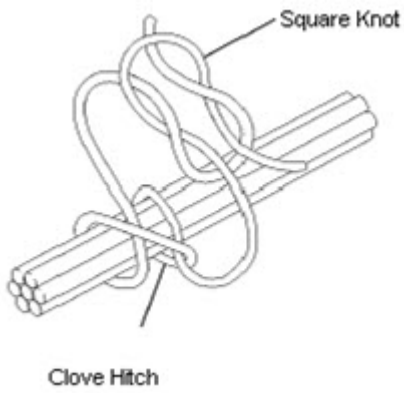
h. When inspecting lacing string/tie string installed on wiring groups, bundles, cables and wiring, and fiber optic harnesses, the following should be considered:

(1) Ties should not be used on the part of a cable group or bundle located inside a conduit.

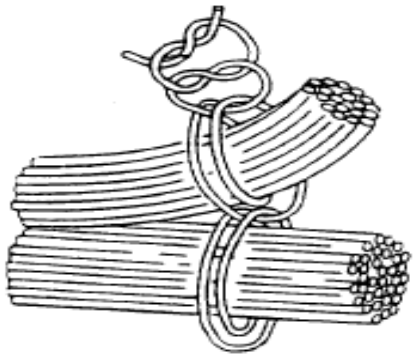
(2) Continuous lacing may not be used for secondary support of fiber optic cable, cable bundles, or cable harness.

(3) Plastic cable straps (zip ties) are prohibited for use as secondary support. Refer to guideline 21 for information regarding plastic cable straps.

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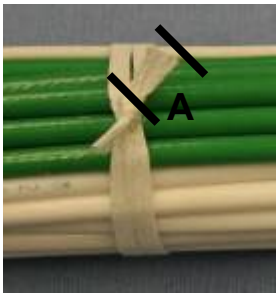


Acceptable
Clove hitch, followed by a square knot.



Acceptable
Breakout of the bundle into two separate bundles. Wrap cord around wire group or bundle, as shown. Make a clove hitch, followed by a square knot with an extra loop. Trim free ends of cord 3/8″ +/- 1/8″.

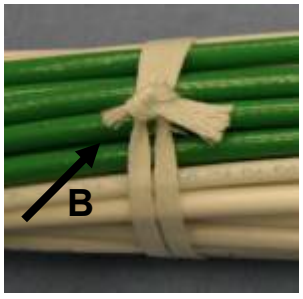
FIGURE 17-1. Examples of acceptable use of lacing tape.



Acceptable
The ends of the tying material should be approximately even in length, cut square at the ends, and measure 3/8″ +/-1/8″ from the knot as shown (A).



Acceptable
The cut ends of the knot are uneven in length, indicating poor trimming technique. However, the length of both ends are within the allowable limits.



Unacceptable
The tying material is trimmed short of the required 1/4″ (B) minimum length.



Unacceptable
The loose ends of the spot tie knot have been trimmed closer than the 1/4″ minimum requirement.

FIGURE 17-2. Examples of acceptable and unacceptable tying of lacing tape.

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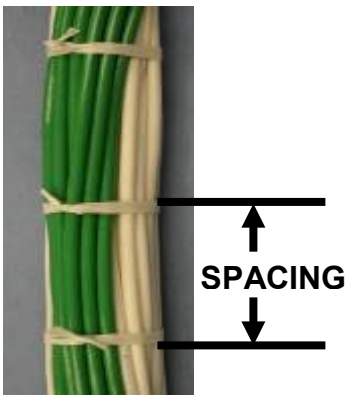


Illustration of spacing of spot ties (see table 17-1).



Acceptable



Unacceptable

The maximum allowable spot tie spacing has been exceeded. Wires are not contained neatly. Loose wires may be damaged.



Unacceptable

FIGURE 17-3. Examples of acceptable and unacceptable lacing tape/tie string spacing.

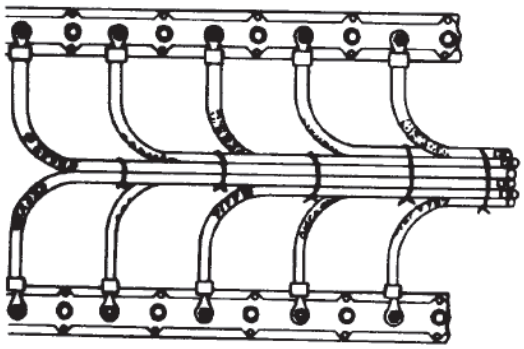
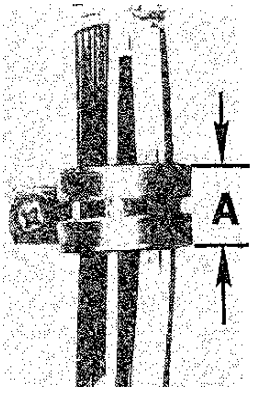
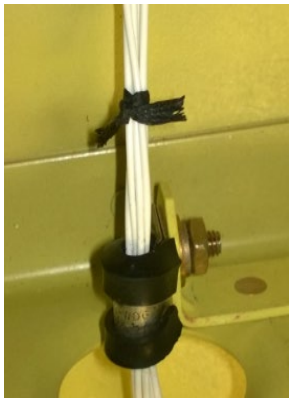


FIGURE 17-4. Lacing tape/tie string at termination inside electronic assembly and panel harnesses.

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Acceptable
Spot ties are no closer to the clamp than a distance equal to the width of the clamp (A). This will allow the wire bundle to be supported by the clamp, as shown.



Acceptable
Spot tie wrap is located at a distance from the clamp, equal to the clamp width. The bundle is able to conform to the shape of the clamp.



Unacceptable
Spot tie has been positioned too close to clamp, resulting in the cut end of the spot tie being pinched by the clamp.



Unacceptable
The clamp has been placed over the spot tie. When tightened, the clamp will press the tie-down into the wire insulation, causing serious damage.

FIGURE 17-5. Examples of lacing tape/tie string correctly located and located too close to a cable clamp.

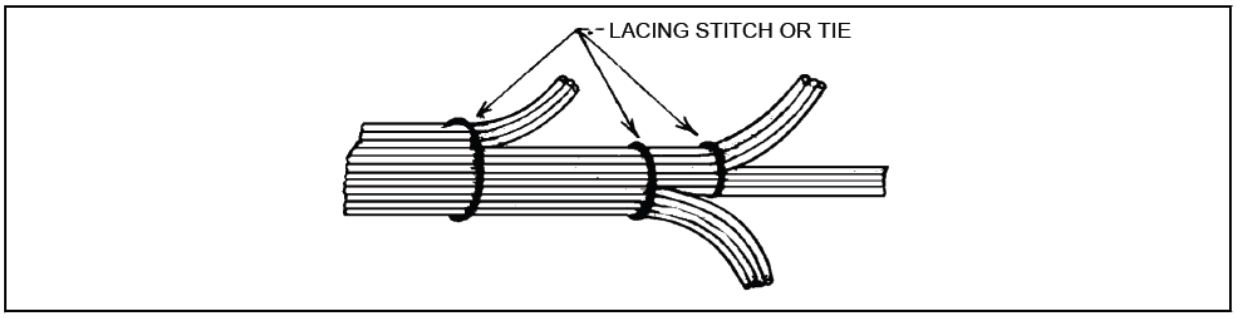


FIGURE 17-6. Spot ties at breakouts should be located as shown.

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GUIDELINE 18

PRIMARY SUPPORT CABLE CLAMP INSPECTION

1. Purpose. This guideline gives criteria for inspecting cable harness primary support cable clamps used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

A-A-52083	Tape, Lacing and Tying Glass
A-A-52084	Tape, Lacing and Tying, Aramid
A-A-59163	Insulation Tape, Electrical, Self-Adhering Unsupported Silicone Rubber
MIL-DTL-85052/7	Clamp, Loop, Aluminum 2024-T42, 212 Degrees F, Chloroprene
MIL-DTL-85052/10	Clamp, Loop, Aluminum 2024-T42, 275 Degrees F, Nitrile Butadiene Rubber Fuel and Petroleum Based Hydraulic Fluid Resistant
MIL-DTL-85052/12	Clamp, Loop, Stainless Steel, 400 Degrees F, Fluorosilicone
SAE AS21919	Clamp, Loop Type, Cushioned Support
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 010 00	Harness Installation
Work Package 012 01	Heating Tools for on Aircraft Maintenance
NAVAIR 01-1A-505-4	Aircraft Fiber Optic Cabling

3. General inspection guidelines. When inspecting clamps used in the EWIS, the following procedure should be used:

a. Primary support clamps:

(1) Confirm that the SAE AS21919 primary support clamp part number includes a “W” in the part number if it is a wedge type clamp; this does not apply to plastic or specialty clamp configurations (see figure 18-1). The substitute MIL-DTL-85052/7, /10 or /12 cushioned clamps do not use a "W" in the part number, but do incorporate the wedge-type clamp cushion.

(2) Inspect the base or wedge of cushion material to ensure that wires are not pinched in the metal band (see figure 18-1).

(3) Ensure that clamps are not too large or too small for the wire bundle on which they are installed (see figure 18-2). If the wire bundle is smaller than the nearest clamp size, or if a clamp of the proper size is not available, wrap the wire bundle with the necessary number of turns of A-A-59163 Type 2, non-adhesive insulating tape, so that the bundle will be held securely in the clamp.

(4) Confirm no plastic clamps are used where ambient temperature may exceed

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185 °F. Plastic cable clamps are not permitted for use as primary support of fiber optic cable. Where plastic clamps are used to support electrical wiring, at least every fourth clamp should be a rubber cushion type clamp.

(5) Confirm there are no loose, broken, or deteriorated cushion clamps.

(6) Verify that all clamps used are able to withstand the environment to which they are exposed.

(7) Confirm there are no deformed clamps and that there are no cracks in the metal portion, particularly at the bolt location (see figure 18-3).

(8) Verify that metal cushion clamps are used as the primary means to support fiber optic cabling.

(9) Ensure that lacing tape/tie string (A-A-52083 or A-A-52084) is used only for secondary support.

(10) Verify that wire harnesses are held firmly and fill the clamp completely (see figure 18-4).

(11) Confirm that clamps are secure enough to prevent harness movement and chafing.

(12) Verify that the clamp does not compress the wire while maintaining continuous contact throughout the clamp (see figure 18-4).

(13) Confirm proper clamp tightness by following all wire/harness runs and lightly shaking at all clamps or support devices. Inspect for proper torque by attempting to rotate clamp around bolt/screw axis. If not tight, the length may be improper or bolt bottomed out.

(14) Examine clamp attaching hardware for proper thread protrusion from the back of the clamp. Three to five threads are optimum (2 minimum).

(15) Ensure that airframe clips, nut plates, and brackets do not have loose rivets or fasteners.

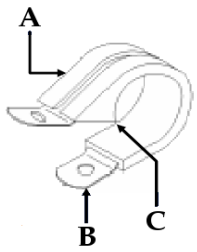
(16) Verify that the space between clamps is not greater than 24 inches (see figure 18-5).

(17) Confirm that fuel lines have not been used to support wire harnesses.

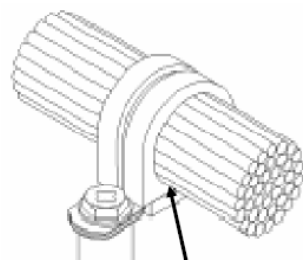
b. Support clamp installation options. Figure 18-6 illustrates the different parts of clamps used for single, double harness clamping, and multi-harness wiring.

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c. For the MIL-DTL-85052/7, /10 and /12 locking hinge, cushioned clamps (see figure 18-7), ensure the locking feature is engaged securely. All other inspection criteria from the SAE AS21919 clamp also apply to this type clamp.



Acceptable
(A) Rubber cushion. (B) Clamp tabs. (C) Wedge.



Acceptable
Verify that there are no pinched wires at the wedge.

FIGURE 18-1. Examples of acceptable clamps and clamp installation.

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Acceptable Wire harness is held firmly and fills the clamp completely. The clamp is not distorted and the clamp does not crush the wires. The clamp is secure enough so that the harness does not move and there is no chafing.	**Unacceptable** Wires are compressed tightly into the clamp, decreasing bundle below normal diameter. Visible evidence of tight clamping may be a curved contour or compressed rubber cushion (A). Spreading of the metal band and wedge not following the contour of the bundle is also visible (B).	**Unacceptable** The clamp shown with wire pinched under wedge.

FIGURE 18-2. Examples of clamps that are properly and improperly installed on wire bundles.

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Acceptable
The rubber cushion should be free from gouges that expose the metal clamp and from cuts or cracks.



Acceptable
Slight abrasions and minor imperfections are acceptable provided there is no exposed metal or sharp indentation, which may develop into cracks.





Unacceptable
The cushion is weakened by a cut. A cut will tend to propagate and increase cushion separation.



Unacceptable
The metal clamp has been exposed by a gouge in the cushion.

FIGURE 18-3. Examples of clamps in acceptable and unacceptable condition.

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Acceptable
The wire harness should be held firmly within the clamp to prevent excessive wire movement.



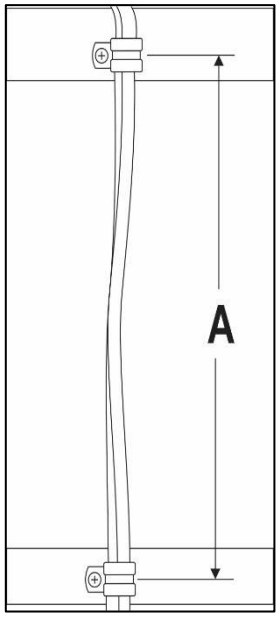
Acceptable
Although the wire clamp is not completely tight, it contacts enough of the wire bundle to assure firm grip and prevent harness movement.



Unacceptable
The wire bundle is not held tightly by the clamp. Too large a clamp, used as shown, will not provide a snug grip. Chafing of the wires may occur.

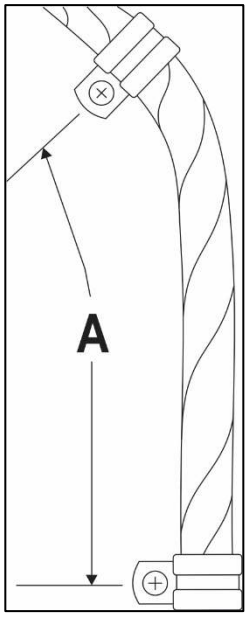
FIGURE 18-4. Examples of acceptable and unacceptable installation of clamps to prevent wire chafing.

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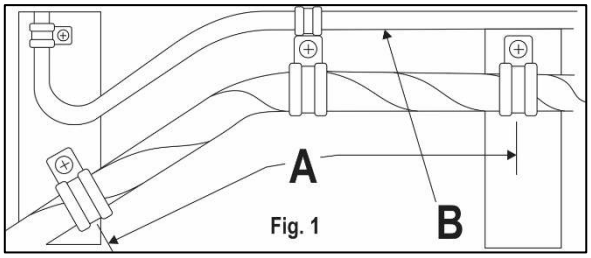
Acceptable

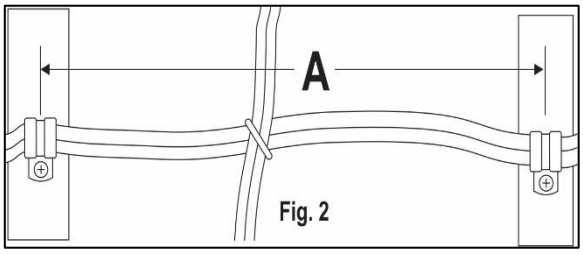
Where dictated by design, cable support clamps may be a maximum of 24 inches apart (A). Wire harnesses should not be supported by other harness assemblies or by fuel lines, or other nonstructural members.



Acceptable

Cable clamps are within the maximum spacing. Clamp spacing (A) is measured along the contour of the harness, as shown.



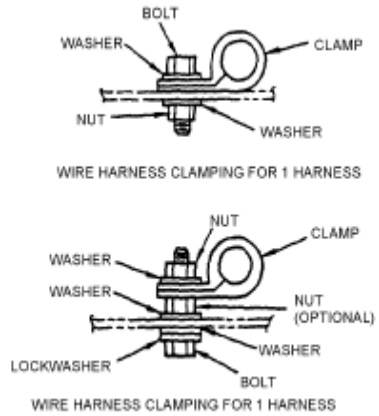


Unacceptable

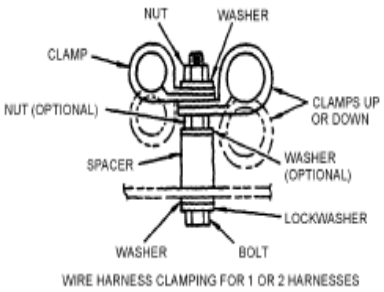
Fig. 1 - Harness is supported by fuel lines (B). Structurally supported clamps are located more than 24-inch maximum clamp spacing allowable (A).
Fig. 2 - Harness is spot tied to another harness (arrow) for support because of the excess spacing (A) between clamps. This is not adequate support.

FIGURE 18-5. Examples of acceptable clamp spacing and unacceptable use of fuel lines and spot ties as a means of support.

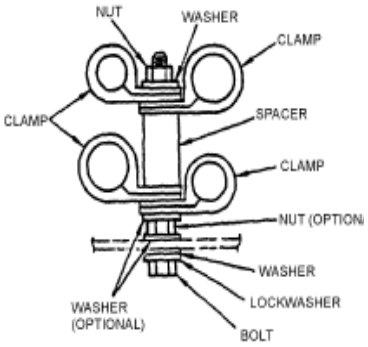
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Typical single harness clamping

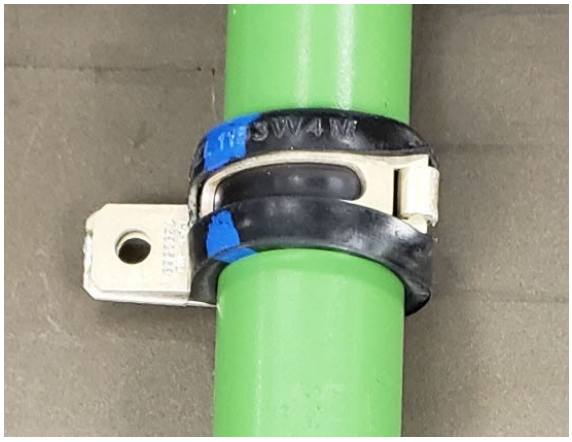


Typical two harness clamping
(Top and bottom clamping is commonly referred to as marriage clamping.)



Typical multi-harness wiring
(Side by side clamping commonly referred to as butterfly clamping.)

FIGURE 18-6. Acceptable hardware mounting configurations for clamps.



Acceptable

Typical M85052/7 cushioned hinge clamp installed.



Acceptable

Detailed locking feature of the hinge clamp.

FIGURE 18-7. Acceptable installation and feature of the M85052/7, /10, and /12 hinge locking clamp.

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GUIDELINE 19

HARNESS DRIP LOOP INSPECTION

1. Purpose. This guideline gives criteria for inspecting cable harness drip loop.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 010 00	Harness Installation

3. General inspection guidelines. When inspecting drip loops used in EWIS wiring, the following should be examined:

a. Wiring should be examined for proper drip loop installation. Wiring dressed down to a connector should have a drip loop/trap to prevent fluids or condensation from running down the wiring into the connector. A drip loop should also be installed between the connector and the first primary support. Figure 19-1 gives examples of acceptable and unacceptable drip loops. If a drip loop is present, it should be inspected as follows:

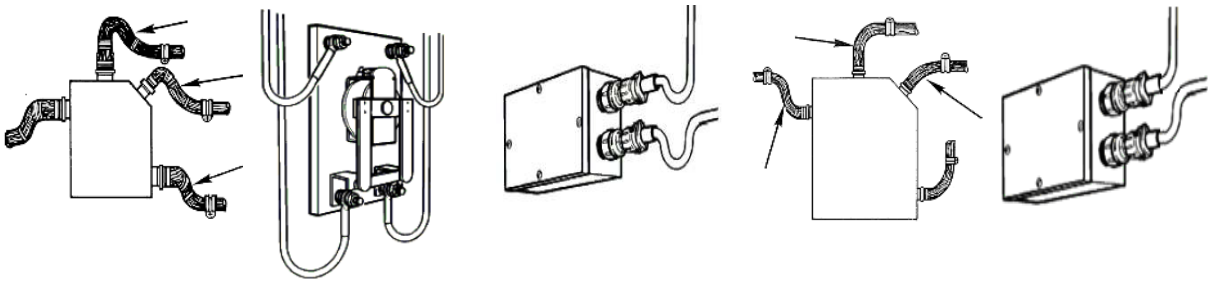
(1) Verify that the drainage hole in tape or tubing (if installed) is at the lowest point. If none exists, create a drainage hole.

(2) Ensure that the hole is not clogged or covered and that no fluids are present.

NOTE

Potted connectors do not require a drip loop.

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Acceptable **Acceptable** **Acceptable** **Unacceptable** **Unacceptable**

Draining, in both on-ground and in-flight conditions, should be provided in wires and cables at electrical and electronic components.

No drip loop provided. Any moisture collected will run straight into the connector and/or components.

FIGURE 19-1. Examples of acceptable and unacceptable drip loops.

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GUIDELINE 20

SHIELD TERMINATION FERRULE INSPECTION

1. Purpose. This guideline gives criteria for inspecting a shield crimp ring after installation in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 015 00	Shield Terminations
SAE AS21980	Ferrule, Outer, Uninsulated, Shield Terminating, Type I, Two-Piece, Class 1, for Shielded Cables
SAE AS21981	Ferrule, Inner, Uninsulated, Shield Terminating, Type I Two-Piece, Class 1, for Shielded Cables

3. General inspection guidelines. Crimped on shield termination ferrule is designed to contain and terminate the cable shielding, allowing up to 1/16 inch of shielding material to protrude past the outer ring. The SAE AS21980 or SAE AS21981 shield ferrule assembly uses an inner and outer ring (see figure 20-1). The outer terminates the shield over the inner ring. Figure 20-2 shows the proper and improper installation of shield termination ferrules. When inspecting shield termination ferrules used in EWIS wiring, the following should be examined:

a. Ensure that inner and outer crimp rings overlap and align on top of each other. No more than 1/32 of an inch of the inner crimp ring should be visible

b. Ensure that the cable shield is trimmed to 1/32 to 1/16 of an inch past the outer ring around cable circumference.

c. If the ground wire is installed under the outer crimp ring, ensure that the metal conductor is visible next to the outer crimp ring. This validates that no wire insulation is under the crimp ring, and optimal continuity is achieved.

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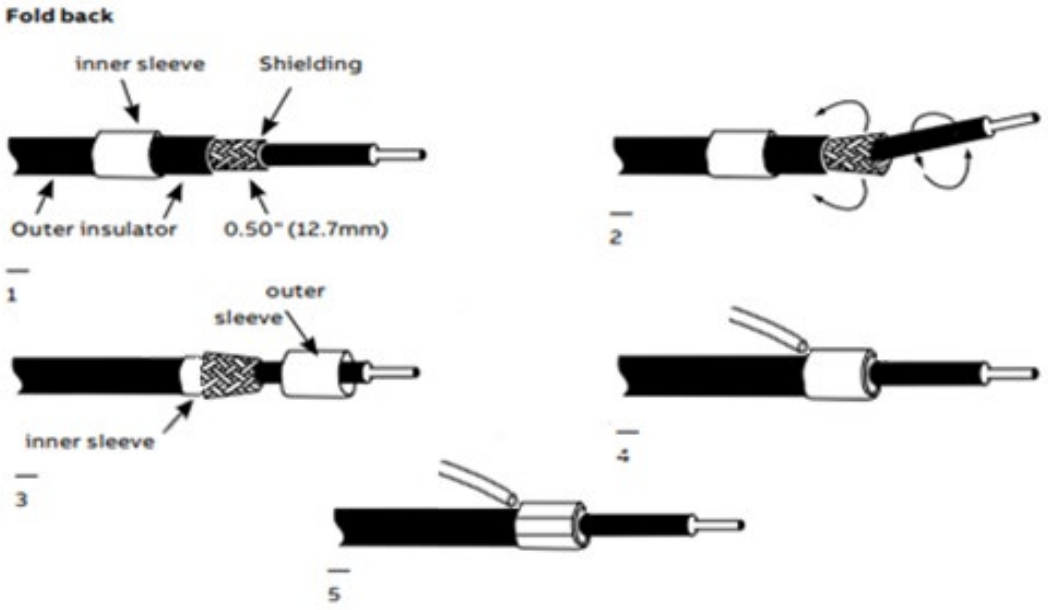


FIGURE 20-1. Typical two-piece shield termination crimp ferrule assembly.

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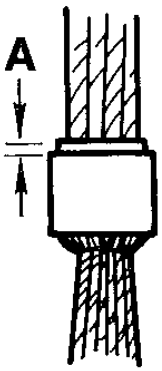
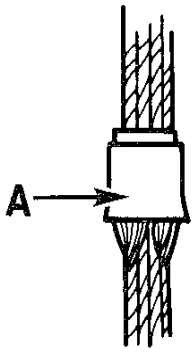
			
Acceptable (A) Inner ring aligned with the outer ring or may extend past outer ring by 1/32 inch. Shielding protrudes 1/16 to 1/32 inch past the outer ring.	**Unacceptable** No shielding protrudes beyond the outer ring. There is not enough shielding in the contact surface between the inner and outer ring.	**Unacceptable** Inner ring extends beyond the maximum allowed, resulting in reduction of the crimp contact surface and insufficient crimp.	**Unacceptable** (A) Outer ring is not completely formed because of improper positioning in the crimp tool.

FIGURE 20-2. Examples of shield termination ferrules in acceptable and unacceptable condition.

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GUIDELINE 21

SECONDARY SUPPORT DEVICES INSPECTION

1. Purpose. This guideline gives criteria for inspecting secondary support devices used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

A-A-52083	Tape, Lacing and Tying, Glass
A-A-52084	Tape, Lacing and Tying, Aramid
MIL-DTL-32554	Straps, Tie-Down, Adjustable, Non-Metallic, Cable Bundling, General Specification for
SAE AS50881	Wiring Aerospace Vehicle
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 010 00	Harness Installation

3. General inspection guidelines. When inspecting secondary support devices used in the EWIS, they should be examined for the following:

CAUTION

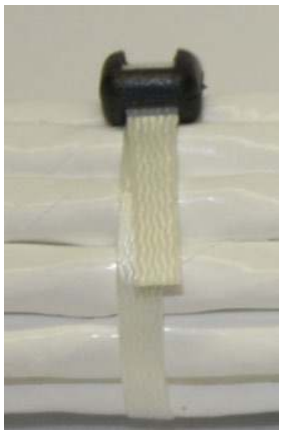
The use of plastic cable straps is strictly prohibited in all instances. Damage to EWIS and surrounding components may result from their use.

a. Installation: The use of plastic cable straps is strictly prohibited in all instances. When maintenance is to be performed on a wire bundle secured with plastic cable straps, only remove enough straps to affect the applicable repair. Upon restoration, install lacing tape/tie string (e.g., A-A-52083, A-A-52084, or MIL-DTL-32554) (see figure 21-1). Refer to guideline 17 for additional inspection of lacing tape and to guideline 51 for inspection of Cable Lacing Fasteners.

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Acceptable
Proper installation of lacing tape installed on a harness.



Acceptable
The M32554 CLF Strap installed on a harness.



Unacceptable
A plastic strap is installed after the repair is performed. Only remove enough straps to affect the applicable repair. Lacing tape should be installed after restoration.

FIGURE 21-1. Examples of acceptable and unacceptable secondary support installation.

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GUIDELINE 22

EMI SHIELDED WRAP-AROUND REPAIR INSPECTION

1. Purpose. This guideline gives criteria for inspecting EMI shield wrap-around repair when this repair is present in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

A-A-52083	Tape, Lacing and Tying, Glass
A-A-52084	Tape, Lacing and Tying, Aramid
A-A-59163	Insulation Tape, Electrical, Self-Fusing, Unsupported Silicone Rubber
SAE AS85049/93	Connector Accessories, Electrical, Termination, Shield Split Support Ring, Composite, Nonenvironmental, Straight, Category 7
SAE AS85049/128	Connector Accessories, Electrical Backshell, Shield Band, Category 7 (For AS85049/82-/90, /93, /109-/117 Accessories)
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 011 01	Protected Wire Harness Fabrication and Repair

3. General inspection guidelines. When inspecting EMI shield wrap used in the EWIS, the following should be considered:

a. Round-it 2000NXEMI-B is the EMI shielded wrap configuration. It is identified by a visible blue tracer, stitched longitudinally on the wrap. This blue tracer will be continuously visible.

NOTE

All wrap-around protective sleeving material incorporates a white-colored tracer as a means to ensure that there is minimum coverage/overlap. The white tracer is applied longitudinally along the length of the wrap and should not be visible if properly installed.

b. Ensure that a minimum 90° overlap exists in the EMI two-layer sleeving material (see figure 22-1). The white tracer line on the overlap direction should not be visible if the correct size sleeving was employed with the required overlap.

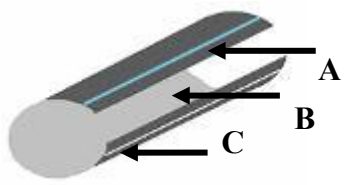
c. Ensure that a minimum twist of 1 to 2 turns per yard of the wire bundle is maintained (see figure 22-1).

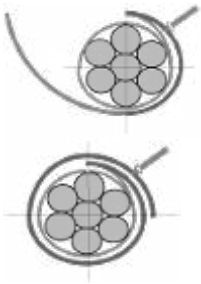
d. Ensure that the Round-it 2000NXEMI-B is terminated with metallic EMI shield termination bands (SAE AS85049/128) at each end. This will ensure shield continuity and EMI coverage for the harness.

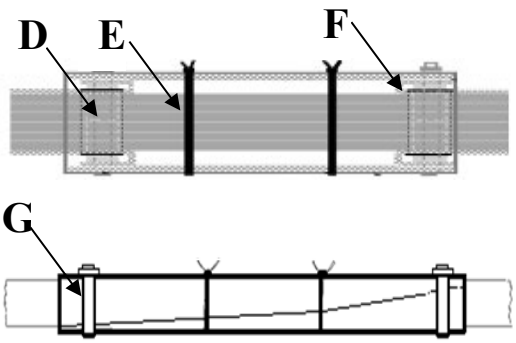
e. Verify that the sleeving on the bundle is tight. There should be no folds or gaps.

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f. Ensure that sleeving has been secured with lacing and tying tape (A-A-52084 or A-A-52083) every 2 +/- 0.5 inches.







(A) External layer: Nomex for mechanical Protection.
(B) Internal layer: Flat braid in nickel-plated copper for EMI Protection.
(C) White tracer indicating the maximum operating diameter. If the required overlap is applied, the tracer should not be exposed.

Ensure that there is a 90° overlap of both the metal braid layer and the Nomex layer.

Wrap around repair using EMI metal bands with the required sleeve twist and overlap requirements.
(D) Split Ring (part number M85049/93-XX)
(E) Spot Tie (A-A-52084).
(F) Self-Fusing Silicone tape A-A-59163, Type II.
(G) EMI metal band (part number M85049/128-1 or -2).

FIGURE 22-1. Illustration of EMI wrap-around braid and examples of proper installation.

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GUIDELINE 23

CONTACT CRIMP INSPECTION

1. Purpose. This guideline gives criteria for inspecting contact crimps.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 013 00	Contacts and Terminals
SAE AS39029	Contacts, Electrical Connector, General Specification For

3. General inspection guidelines. Electrical crimp contact configurations vary, but the typical design incorporates features resembling those on figure 23-1. When inspecting crimped contacts used in the EWIS, confirm that the following minimum requirements are satisfied:

a. Ensure that the metal conductor is visible in the contact inspection hole. It is possible to have the crimp indents deform the inspection hole of a crimp contact (see figure 23-2).

b. Verify that the four double (8 total) indents are visible on the crimped contact (see figure 23-3).

c. Confirm that the conductor is visible around the contact crimp barrel. Ensure that the metallic conductor is visible (insulation gap) and not greater than 1/32″. Re-terminate the contact as required to meet the insulation gap limit.

d. Confirm that the contact crimp barrel is not cracked next to/parallel to the crimp indents. Replace contact if cracked (see figure 23-3).

e. Gently pull/tug on contact in a direction parallel to the wire length to ensure an adequate crimp. Ensure that there is no movement or slippage of the contact from the conductor. Replace contact if any movement is observed.

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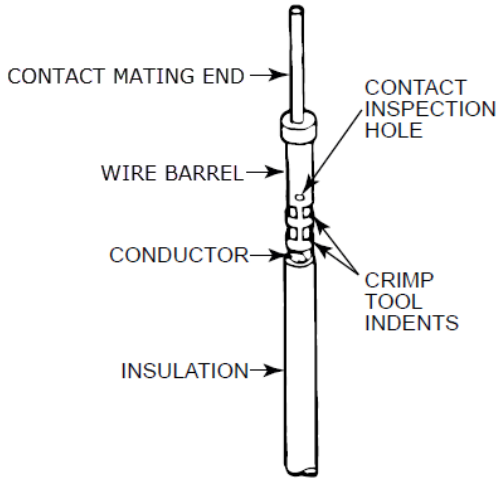


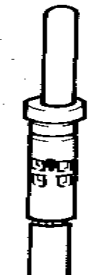
FIGURE 23-1. Example of typical crimp contact configuration (SAE AS39029).



Acceptable
Crimp indents deform the inspection hole boundary but do not inhibit the visibility of the conductor.



Acceptable
Crimp indent enters the inspection hole. Contact is not cracked, split, or has exposed base metal (Copper) in or around the inspection hole. The conductor is visible through the deformed hole.



Unacceptable
Crimp indents have deformed the inspection hole such that the conductor is not visible. Contact is cracked, split, or has exposed base metal (Copper),

FIGURE 23-2. Examples of acceptable and unacceptable crimp indents and their location with the inspection hole.

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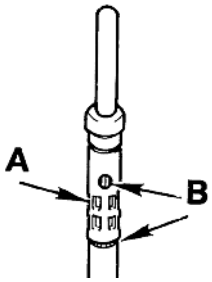
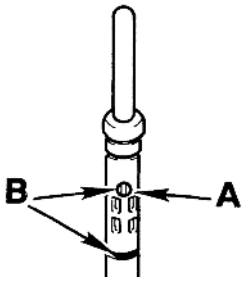
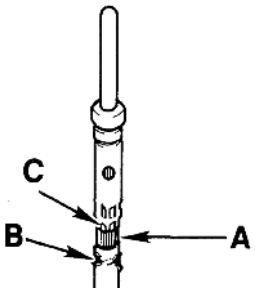
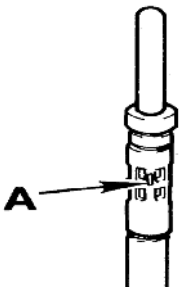
			
Acceptable (A) Proper contact crimp. Crimp indents are uniform. (B) The conductor is just visible around the wire barrel and in the inspection hole with an inspection gap of 1/32″.	**Acceptable** (A) The crimp borders the inspection hole but does not intrude into the hole. (B) The conductor is visible in the inspection hole and around the end of the wire barrel.	**Unacceptable** (A) Gap between the wire insulation and the crimp barrel exceeds 1/32″ maximum allowable. Strip length is too long. (B) Wire insulation is damaged from the stripping tool. (C) The contact was improperly positioned in the tool. The crimp has occurred on the end of the barrel.	**Unacceptable** (A) The contact was improperly positioned in the tool. The crimp has occurred over the inspection hole. The conductor is not visible in the inspection hole.

FIGURE 23-3. Examples of acceptable and unacceptable contact crimps.

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GUIDELINE 24

CONTACT FRETTING CORROSION INSPECTION

1. Purpose. This guideline provides the criteria for inspecting Contact Fretting Corrosion in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 020 00	Military Standard Circular Connectors
Work Package 013 00	Contacts and Terminals
Work Package 029 01	Basic Fault Isolation Methods
NAVAIR 01-1A-509-1	Corrosion Program and Corrosion Theory

3. General inspection guidelines. Contact fretting corrosion should be examined as follows:

NOTE

Fretting is a condition where slight movement between mated surfaces occurs, which continually exposes fresh metal to corrosion.

a. Perform a detailed inspection to verify that that connector plugs and receptacles do not have evidence of gold flaking on interfaces. Evidence of gold flakes in the bottom of connector plug/receptacle requires cleaning and re-inspecting (see figure 24-1).

b. Ensure that the connector pin does not show any signs of cracking or surface plating damage (see figure 24-2).

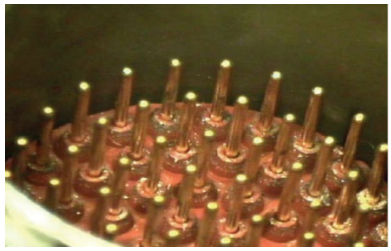
c. Examine connector for damaged threads (mating or accessory threads), bayonet pins, and keyways.

d. Repair and replace any contact that shows evidence of fretting corrosion. See NAVAIR 01-1A-505-1, Work package 013 00 for Contacts and Terminals.

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Acceptable
Connector is clean, no signs of corrosion or gold flaking on interfacial seal or o-ring.



Unacceptable
Gold flaking on interfacial seal; evidence of fretting.

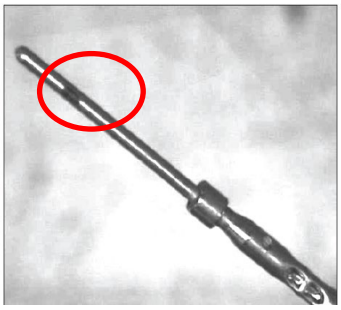


Unacceptable
Excessive plating on interfacial seal; bare copper is showing on contacts.

FIGURE 24-1. Evidence of fretting corrosion on connector.



Acceptable
Contact is clean; no signs of corrosion, and gold plating intact.



Unacceptable
Gold plating is chipped off;
Copper is exposed and corroded;
evidence of fretting corrosion.

FIGURE 24-2. Evidence of fretting corrosion on a typical SAE AS39029 contact.

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GUIDELINE 25

COPPER TERMINAL LUG INSPECTION

1. Purpose. This guideline gives criteria for inspecting copper terminal lugs (SAE AS7928, SAE AS7928/4, and SAE AS20659) used in the EWIS. For cold-applied terminal lugs (SAE AS7928/14), see guideline 26.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 013 00	Contacts and Terminals
SAE AS7928	Terminals, Lug: Splices, Conductor: Crimp Style, Copper, General Specification for
SAE AS7928/4	Terminals, Lug and Splices, Conductor, Crimp Style, Copper Terminal, Lug, Insulated, Ring Tongue, Bell-Mouthed, Type II, Class 1 (For 150 °C Total Conductor Temperature)
SAE AS7928/14	Terminal, Electric, Permanent, Crimp Style, Tin-Coated Copper, Insulated, Environment Resistant, Class 1, 150°C, Heatless Sealing
SAE AS20659	Terminal Lugs, Crimp Style, Copper Uninsulated, Ring Tongue, Type I, Class 1, for 175 °C or 260 °C Total Conductor Temperature

3. General inspection guidelines. Terminal lug configurations vary, but the typical design incorporates features resembling those on figure 25-1. In order to verify that that the terminal lugs are crimped properly when used in the EWIS, they should be examined as follows:

a. For insulated terminals lugs, check that the wire insulation is inserted in the support area of the terminal barrel (see figure 25-2).

b. Confirm that the conductor is extended through the terminal barrel.

c. Verify that stripped wire ends are flush to terminal stop with not more than 1/32 inch protruding (see figure 25-3).

d. Confirm that the crimp indent is centered and applied on the surface of the terminal wire barrel.

CAUTION

In cases where the wire insulation diameter is small enough to enter the conductor crimp barrel area of the terminal lug, care should be used to prevent crimping over insulation.

e. Verify that no wire insulation is present in the wire crimp barrel.

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f. Confirm that the conductor insulation has been inserted into the insulation support area of the terminal lug barrel (see figures 25-2 and 25-3, detail A).

g. Verify that the amount of insulation stripped from the wire is not excessive, such that conductors extend into the hardware mounting area (see detail A on figure 25-2 and figure 25-3).

h. Ensure that the terminal lug insulation sleeve is not cracked.

i. Verify that the terminal lug wire crimp barrel is not cracked (see figure 25-3).

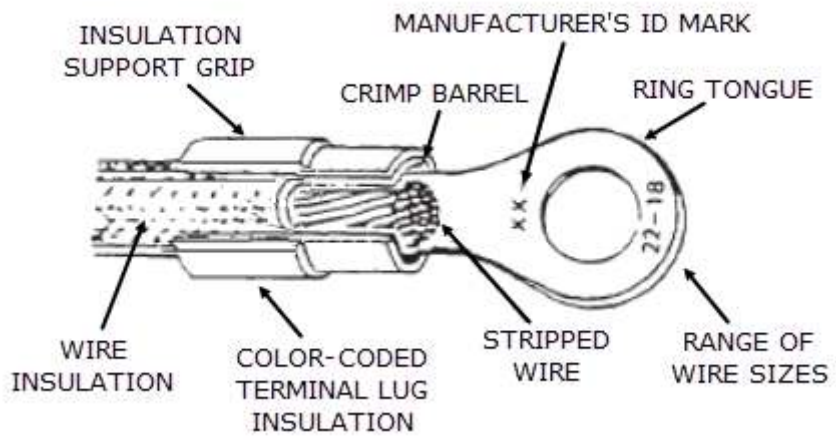
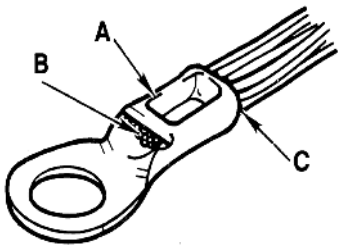
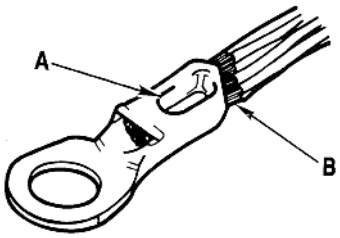


FIGURE 25-1. Terminal lug features (typical SAE AS7928/4 insulated terminal lug shown).

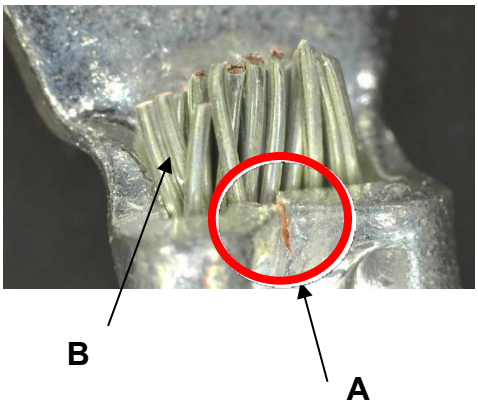
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Acceptable
(A) Lug has been properly positioned in the tool. (B) Stripped wire ends are flush to terminal stop (Not more than 1/32″ protruding). (C) No insulation is in the barrel.



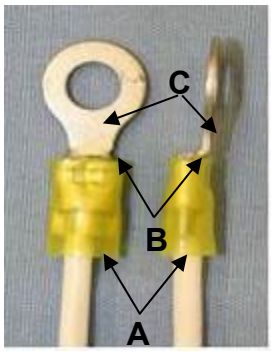
Unacceptable
(A) Lug is improperly positioned in tool resulting in deformation of the terminal with crimp indent not centered on barrel. (B) Excessive wire strip length.



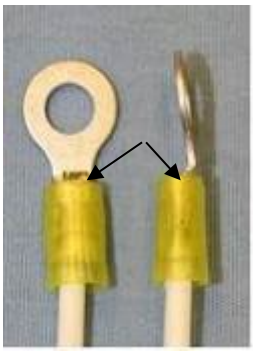
Unacceptable
(A) Lug is cracked with exposed bare metal on crimp barrel.
(B) Excessive wire strip length protruding out of lug.

FIGURE 25-2. Examples of acceptable and unacceptable crimping of terminal lugs.

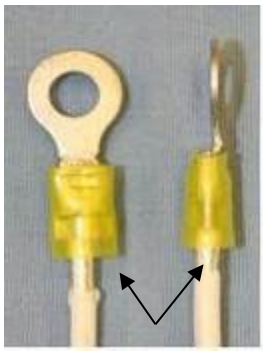
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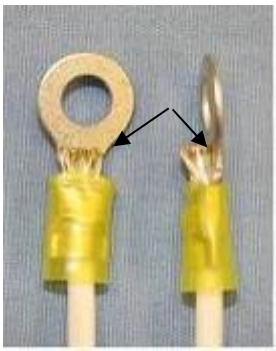
Acceptable
(A) Conductor insulation should be inserted within the support area of the terminal barrel.
(B) Stripped conductor extends fully through the terminal barrel, as shown, with no more than 1/32″ protruding.
(C) No conductors extend into the hardware mounting area.



Unacceptable
Terminal lug and wire is not completely inserted into the crimping tool. The front end of the barrel has been pinched (arrows), causing insufficient crimp on the conductor.



Unacceptable
Conductor insulation (arrows) has not been inserted into the insulation support area of the terminal lug barrel. Excessive strip length.



Unacceptable
The conductor (arrows) extends into the hardware mounting area. Excessive strip length.

FIGURE 25-3. Examples of acceptable and unacceptable wire installation in terminal lugs.

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GUIDELINE 26

COLD-APPLIED TERMINAL LUG INSPECTION

1. Purpose. This guideline provides the criteria for inspecting cold-applied terminal lugs used in the EWIS. For conventional, copper terminal lugs (i.e., SAE AS7928, SAE AS7928/4, and SAE AS20659), see guideline 25.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 013 00	Contacts and Terminals
SAE AS7928	Terminals, Lug: Splices, Conductor: Crimp Style, Copper, General Specification for
SAE AS7928/4	Terminals, Lug and Splices, Conductor, Crimp Style, Copper Terminal, Lug, Insulated, Ring Tongue, Bell-Mouthed, Type II, Class 1 (For 150 °C Total Conductor Temperature)
SAE AS7928/14	Terminal, Electric, Permanent, Crimp Style, Tin-coated Copper, Insulated, environment Resistant, Class 1, 150 Degrees C, Heatless Sealing
SAE AS20659	Terminal Lugs, Crimp Style, Copper Uninsulated, Ring Tongue, Type I, Class 1, for 175 °C or 260 °C Total Conductor Temperature

3. General inspection guidelines. In order to verify that the cold-applied terminal lugs (SAE AS7928/14) are crimped properly when used in the EWIS, they should be examined as follows (see figures 26-1 and 26-2):

a. Verify that the crimp indent of the barrel is visible and parallel to the terminal ring.

b. Ensure that wire strands are visible at the ring end of the lug.

c. Ensure that the crimp barrel and insulation sleeve are not cracked.

d. Check for adequate crimp of terminal lug by tugging on wire end. Verify that there is no movement or slippage of the wire from the terminal lug with average hand pull force. Replace terminal lug if any movement is observed at the crimp.

e. Ensure that all conductor strands are inserted in the crimp barrel.

f. Verify that there are no bubbles are visible in the gel.

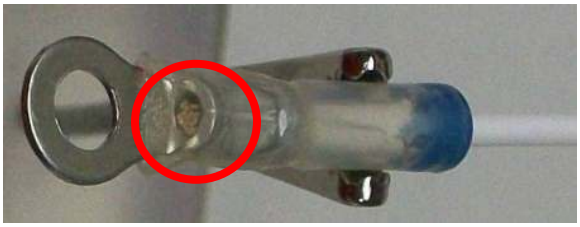
g. Ensure that the gel overlaps wire insulation by at least 3/16″.

h. Verify that there is a visible gap of 1/32″ exists between crimp barrel and wire insulation.

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Acceptable
Crimp placement is centered and parallel to the terminal tongue; no air bubbles in gel; no cracks in the sleeve, adequate gel coverage on the wire in excess of 3/16″.

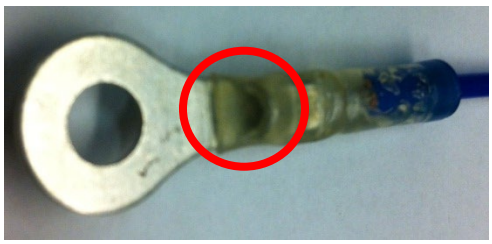


Acceptable
Stripped wire is visible in the inspection window.

FIGURE 26-1. Examples of acceptable cold-applied SAE AS7928/14 terminal lugs.



Unacceptable
Air bubbles visible inside the insulating gel. Excess of 1/32″ insulation exposed.



Unacceptable
Wire strands not visible at the end of the terminal lug crimp barrel.

FIGURE 26-2. Examples of unacceptable cold-applied SAE AS7928/14 terminal lugs.

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GUIDELINE 27

SOLDER SLEEVE/SHIELDING TERMINATION INSPECTION

1. Purpose. This guideline gives the criteria for inspecting solder sleeve/shielding terminations (SAE AS83519/1 through /6) used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

SAE AS83519	Shield Termination, Solder Style, Insulated, Heat-Shrinkable, Environment Resistant General Specification for
SAE AS83519/1	Shield Termination, Solder Style, Insulated, Heat-Shrinkable, Environment Resistant For Cables Having Tin or Silver Plated Shields (Class I)
SAE AS83519/2	Shield Termination, Solder Style, Insulated, Heat-Shrinkable, Environment Resistant with Pre Installed Leads for Cables Having Tin Or Silver Plated Shields (Class I)
SAE AS83519/3	Shield Termination, Solder Style, Insulated, Heat Shrinkable, Environment Resistant with Preinstalled Braid, Class 1, Non-ROHS
SAE AS83519/4	Termination, Solder Style, Insulated, Heat-Shrinkable, Environment Resistant with Multiple Opening Sealant, 150 °C and 200 °C
SAE AS83519/5	Termination, Solder Style, Insulated, Heat-Shrinkable, Environment Resistant with Preinstalled Braid, NI Plated, Class 1, ROHS
SAE AS83519/6	Shield Termination, Solder Style, Insulated, Heat-Shrinkable, Environment Resistant for Cables Having NI Plated Shields, 200 °C, ROHS, Class I
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 015 00	Shield Terminations

3. General inspection guidelines.

a. There are several configurations of SAE AS83519 solder sleeves that meet application temperature requirements. There is low-temperature solder, rated to 150 °C, such as SAE AS83519/1 without a ground lead, SAE AS83519/2 with a ground wire, and SAE AS83519/3 with a flat braid ground lead. The SAE AS83519/4 is a multi-wire configuration that can accommodate sealing for up to six wires for either low or high-temperature applications. The high-temperature solder configurations are rated to 200 °C and include SAE AS83519/5 with a flat braid ground lead and the SAE AS83519/6 with a ground wire lead.

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NOTE

The low-temperature solder sleeves employ a red-colored thermochromatic indicator that melts and dissipates/turns clear when the solder melting temperature is reached (see figure 27-1).

b. Low-temperature solder sleeves (SAE AS83519/1 through /4). When inspecting solder sleeves, the minimum and maximum solder flow should be verified (see figure 27-2):

(1) For acceptable minimum solder flow, the following should be present:

(i) Slight traces of dull red color (thermal indicator).

(ii) Solder has lost all original shape.

(iii) Sealant inserts have melted and flowed along wires.

(iv) Shield and lead contours are visible.

(v) A definite fillet is visible between lead and shield.

(2) For acceptable maximum solder flow, the following should be present:

(i) The dull red color has disappeared.

(ii) Slight traces of dull red color (thermal indicator) in sealant insert area are acceptable.

(iii) Sealant inserts have melted and flowed out along wires.

(iv) A definite fillet is clearly visible between lead and shield.

(v) Joint area is visible despite browning of the sleeve.

(3) When there is insufficient heat during the soldering process, unacceptable solder flow results. In this situation, the following can be observed when inspecting the solder flow:

(i) The dull red color (thermal indicator) is clearly visible.

(ii) Original shape of the solder preform is clearly visible.

(iii) Melt-able sealing inserts have not flowed.

(iv) Contour of braid and/or lead is blocked by solder.

(4) When there is too much heat used during the soldering process, unacceptable solder flow results. In this situation, the following can be observed when inspecting the solder flow:

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(i) Joint area is not visible because of severe darkening of the outer sleeve.

(ii) Solder fillet is not visible along lead and shield interface.

(iii) Wire insulation damaged outside of the sleeve.

c. High-temperature solder sleeves (SAE AS83519/5, and /6). When inspecting solder sleeves and shielding terminals, the minimum and maximum solder flow should be verified (see figure 27-4):

NOTE

The high-temperature solder sleeves employ a solder wire as the solder melt indicator. This indicator doesn't change color, but when melting temperature is reached, both the solder ring and the wire melt (see figure 27-3 for un-melted and figure 27-4 for melted example).

(1) For acceptable minimum solder flow, the following should be present:

(i) Solder has lost all appearance of ring shape.

(ii) Inserts have melted and flowed along wires.

(iii) Shield and lead contours are visible.

(iv) There is a definite fillet visible along the lead and shield interface.

(2) For acceptable maximum solder flow, the following should be present.

(i) Fillet is clearly visible between the lead and shield.

(ii) Joint area is visible despite browning of the sleeve.

(3) When there is insufficient heat during the soldering process, unacceptable solder flow results. In this situation, the following can be observed when inspecting the solder flow:

(i) Contour of the solder preform is visible.

(ii) Melt-able inserts have not flowed.

(iii) Contour of braid and/or lead is obscured by solder.

(4) When there is too much heat used during the soldering process, unacceptable solder flow results. In this situation, the following can be observed when inspecting the solder flow:

(i) Joint area is not visible because of severe darkening of the outer sleeve.

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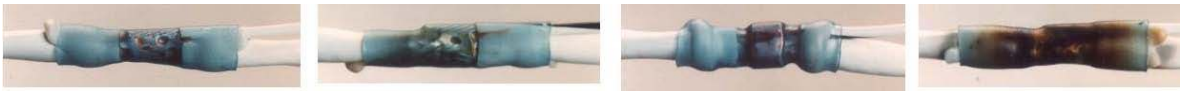
(ii) Solder fillet is not visible along lead and shield interface.

(iii) Wire insulation damaged outside of the sleeve.



Note: The red indicator band identifies the solder sleeve as a low-temperature solder sleeve (max 150 ℃).

FIGURE 27-1. Low-temperature shield termination solder sleeve.



Acceptable
Minimum Solder Flow

Acceptable
Maximum Solder Flow

Unacceptable
Insufficient heat. Solder band is not melted.

Unacceptable
Overheated

FIGURE 27-2. Examples of solder sleeves that have acceptable and unacceptable solder flow.

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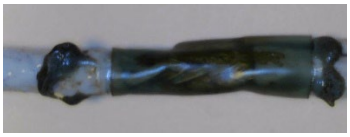


FIGURE 27-3. High-temperature shield termination solder sleeve.









Acceptable Minimum Solder Flow	**Acceptable** Maximum Solder Flow	**Unacceptable** Solder band not melted. Solder indicator wire not melted (top image).	**Unacceptable** Overheated

FIGURE 27-4. Examples of high-temperature solder sleeves that have acceptable and unacceptable solder flow.

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GUIDELINE 28

BONDING STRAP/JUMPER INSPECTION

1. Purpose. This guideline gives criteria for inspecting bonding/jumper straps used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 017 00	Bonding and Grounding

3. General inspection guidelines. When inspecting bonding straps and jumpers used in the EWIS, the following should be examined (see figure 28-1):

a. If there is evidence of electrical arcing, check for intermittent electrical contact between conducting surfaces that may become a part of a ground plane or a current path.

b. Confirm that bond connections are secure and free from corrosion.

c. Verify that bonding jumpers are installed in such a manner as not to interfere in any way with the operation of movable components of the aircraft.

d. Inspect bonding jumper condition and verify that jumpers are not frayed or kinked. Replace if more than one-third of the jumper wire strands are broken.

e. Confirm that self-tapping screws are not being used for bonding purposes. Only standard threaded screws or bolts of appropriate size should be used.

f. Confirm that bonds are attached directly to the basic aircraft structure rather than through other bonded parts.

g. Verify that the resistance across a bonding or grounding jumper is 0.1 ohm or less. This test is made after the mechanical connection is completed and consists of a milliohm-meter reading of the resistance between the cleaned areas of the object and the structure. Refer to the platform/aircraft manual for specific installation and test requirements (see figure 28-2).

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Acceptable
Proper bonding strap applied.



Unacceptable
Frayed and broken bonding strap.



Unacceptable
Wrong orientation and frayed bonding strap.

FIGURE 28-1. Examples of bonding straps in acceptable and unacceptable condition.

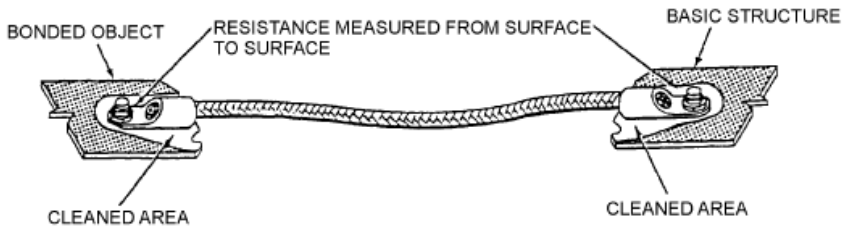


FIGURE 28-2. Bonding resistance test setup.

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GUIDELINE 29

HEAT-APPLIED SPLICE INSPECTION

1. Purpose. This guideline gives criteria for inspecting heat-applied splices (Environmental SAE AS81824/1 splices, Multi-splice SAE AS81824/6 thru /11), and SAE AS81824/13 stub splices used in the EWIS. For cold-applied splices, see guideline 30 (SAE AS81824/12) and guideline 31 (SAE AS81824/14).

2. Applicable documents. The documents listed below are those applicable to this guideline.

SAE AS81824	Splice, Electric, Crimp, Copper, Environment Resistant
SAE AS81824/1	Splice, In-Line, Electric, Crimp, SN/CU, Environmental, Heat-Shrinkable
SAE AS81824/6	Splice, Electric, Permanent, Crimp, Nickel Plated Insulated Wires, Environmental, Class 1, 175 ˚C
SAE AS81824/7	Splice, In-Line, Electric, Crimp, SN/CU, Environmental, Heat-Shrinkable Sleeve (150 ˚C) 1 X 3 Sealant Opening
SAE AS81824/8	Splice, In-Line, Electric, Crimp, NI/CU, Environmental, Heat-Shrinkable Sleeve (175 ˚C) 1 X 3 Sealant Opening
SAE AS81824/9	Splice, In-Line, Electric, Crimp, SN/CU, Environmental, Heat-Shrinkable Sleeve (150 ˚C) 3 X 3 Sealant Opening
SAE AS81824/10	Splice, In-Line, Electric, Crimp, NI/CU, Environmental, Heat-Shrinkable Sleeve (175 ˚C) 3 X 3 Sealant Opening
SAE AS81824/11	Splice, Electric, Permanent, Crimp Style NI/CU, Insulated, Environment Resistant, Class 1, 200 ˚C Max 1 X 1 Sealant Opening
SAE AS81824/12	Splice, Electric, Permanent, Crimp Style, Tin-Coated Copper, Insulated Environment Resistant, Class 1, 150 ˚C, Heatless Sealing
SAE AS81824/13	Splice, Stub, Electric, Permanent, Crimp Style, Nickel/Copper Insulated, Environment Resistant, 175 ˚C Max
SAE AS81824/14	Splice, Electric, Permanent, Crimp Style, Nickel-Coated Copper Insulated, Environment Resistant, Class 1, 175 ˚C, Heatless Sealing
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 014 00	Wire and Cable Splicing and Repair

NOTE

Splice restrictions outlined in NAVAIR 01-1A-505-1 WP 014 00 apply to all EWIS conductor splicing regardless of the type of splice is installed.

NOTE

This guideline applies to environmentally sealed splices other than cold-applied splices meeting M81824/12; for M81824/12 splices see guideline 30 and for M81824/14 splices see guideline 31.

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3. General inspection guidelines. When inspecting splices used in the EWIS, the following should be verified:

a. If multiple splices are installed in a single harness, verify that splices are staggered. If wires are shielded, stagger shield ferrules. See guideline 27 for shield termination inspection.

b. Confirm that splices are not installed in a fuel tank or within 12 inches entering or exiting a fuel tank.

c. Verify that no splices have been installed under support clamps.

d. Confirm that splices are not installed in an area of high flexibility in the harness.

e. Verify that the in-line splice and stub splices are crimped properly:

(1) Conductor should be centered in the crimp nest.

(2) Ensure that the crimp indent is on the side containing the inspection hole.

(3) Verify that no wire strands are protruding from the end of the crimp barrel.

(4) Ensure that the wire insulation gap is 1/32″ maximum against the end of the crimp barrel. This ensures that no insulation is inside the crimp barrel.

(5) Verify that the stripped conductors are butted up flush against the wire stop inside each end of the crimp barrel. Verify that conductors are visible in the crimp barrel inspection window.

(6) For stub splices (SAE AS81824/13) ensure that the conductors are visible and trimmed flush with the end of the crimp barrel (see figure 29-2).

f. Verify that the environmental insulating sleeve for in-line splice and stub splices are properly recovered/shrunk (see figures 29-1 and 29-2):

(1) Ensure that the environmentally sealed sleeve is tightly shrunk/recovered over the metal crimp barrel and wires. No bare wire is exposed beyond the sleeve.

NOTE

For a splice to be environmentally sealed, the hot melt sealant must flow out past the ends of the sleeve.

(2) Ensure that the hot melt sealant from inside the sealing sleeve has flowed out past the ends of the sleeve.

(3) Verify that there is no burning or splitting of the environmental sleeve after heating.

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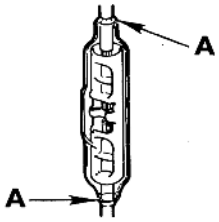
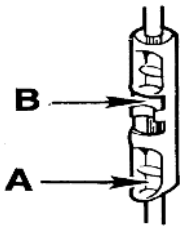
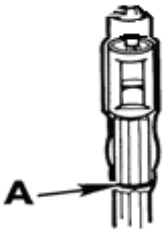
		
Acceptable	**Acceptable**	**Unacceptable**
Wire insulation should have 1/32″ maximum gap at each end. The conductor should be butted against the splice center stop. Crimp indent of both wires is on the side containing the inspection hole. (Insulation sleeve is not shown for clarity.)	Wire conductor must be visible through insulation sleeving. (A) Melted sealing bands have flowed out of both ends of the sleeve.	(A) Crimp is not centered on splice barrel. (B) Top wire conductor is not visible or butted against splice center stop. (Insulation sleeve is not shown for clarity).

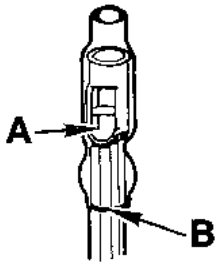
FIGURE 29-1. Examples of acceptable and unacceptable in-line splices.

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Acceptable

Crimp is centered on barrel, the bare wire is visible, and cut flush at the end of the barrel.
(A) Sealant has flowed out of end of sleeve.



Not Acceptable

(A) Crimp is not centered on barrel because of improper positioning in tool. (B) Sealant did not flow out of end of sleeve.

FIGURE 29-2. Examples of acceptable and unacceptable stub splices.

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GUIDELINE 30

COLD-APPLIED SPLICE (SAE AS81824/12) INSPECTION

1. Purpose. This guideline provides the criteria for inspecting repairs using the cold splice (SAE AS81824/12) when used in the EWIS. For the SAE AS81824/1 heat applied splice, see guideline 29. For the SAE AS81824/14 heatless splice, see guideline 31.

2. Applicable documents. The documents listed below are those applicable to this guideline.

SAE AS81824	Splices, Electric, Permanent, Crimp Style, Copper, Insulated, Environment Resistant
SAE AS81824/1	Splice, Electric, Permanent, Crimp Style Copper, Insulated, Environment Resistant, Class 1
SAE AS81824/12	Splice, Electric, Permanent, Crimp Style, Tin-Coated Copper, Insulated Environment Resistant, Class 1, 150 ˚C, Heatless Sealing
SAE AS81824/14	Splice, Electric, Permanent, Crimp Style. Nickel-Coated Copper, Insulated, Environment Resistant, Class 1, 175 ˚C, Heatless Sealing
NAVAIR 01-1A-505-1 Work Package 014 00	Aircraft Electric and Electronic Wiring Wire and Cable Splicing and Repair

NOTE

Splice restrictions outlined in NAVAIR 01-1A-505-1 WP 014 00 apply to all EWIS conductor splicing regardless of the type of splice is installed.

3. General inspection guidelines. When inspecting cold-applied splices meeting SAE AS81824/12 used in the EWIS, the following should be verified. Examples of acceptable and unacceptable cold-applied splices are depicted on figures 30-1 and 30-2, respectively.

a. Verify that the cold-applied splice is installed as follows:

(1) Confirm that the crimp indent is on both sides of the barrel and is parallel with the inspection window (see figure 30-1).

(2) Ensure that the crimp barrel and insulation are not cracked.

(3) Check for adequate crimp of splice by tugging on wire ends. Verify that wire cannot be pulled out of the splice with average hand pull force.

(4) Verify that all conductor strands are inserted in the crimp barrel.

(5) Ensure that no bubbles are visible in gel (both sides) (see figure 30-2).

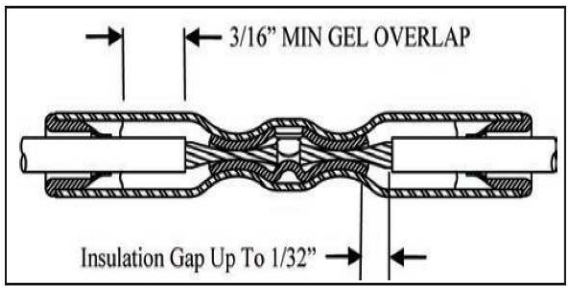
(6) Verify that both wire ends are visible in the inspection window.

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(7) Ensure that only one wire is inserted in each end of the cold applied splice.

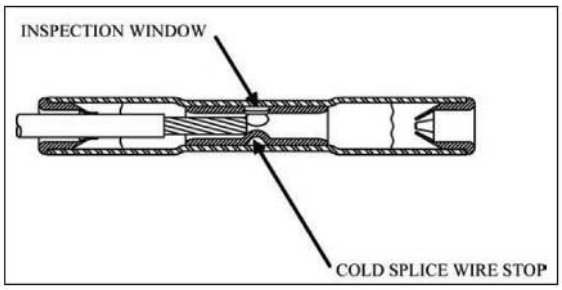
(8) Verify that the gel overlaps the wire insulation at least 3/16 inch.

(9) Ensure that a visible gap of up to 1/32 inch exists between the crimp barrel and the wire insulation.



Acceptable

Before crimping, crimp placement is centered and parallel inspection window; no air bubbles in gel; no cracks in the sleeve, adequate gel coverage on the wire in excess of 3/16″; there is 1/32″ insulation gap visible.

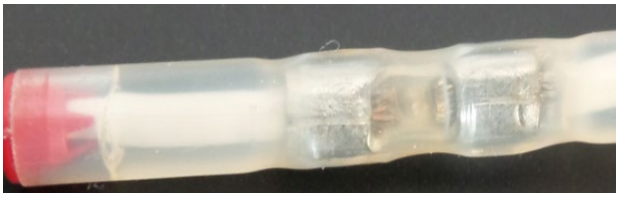


Acceptable

Conductors are butted up against the splice wire stop and visible in the inspection window.

FIGURE 30-1. Examples of acceptable cold-applied splice (SAE AS81824/12).

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Acceptable

No Air bubbles visible in the gel



Unacceptable

Air bubbles visible in the gel

FIGURE 30-2. Example of an acceptable and an unacceptable cold-applied splice (SAE AS81824/12).

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GUIDELINE 31

HEATLESS SPLICE (SAE AS81824/14) INSPECTION

1. Purpose. This guideline provides the criteria for inspecting repairs using the heatless splice (SAE AS81824/14) when used in EWIS. For the SAE AS81824/1 heat applied splice, see guideline 29. For the SAE AS81824/12 cold-applied splice, see guideline 30.

2. Applicable documents. The documents listed below are those applicable to this guideline.

SAE AS81824	Splices, Electric, Permanent, Crimp Style, Copper, Insulated, Environment Resistant
SAE AS81824/1	Splice, Electric, Permanent, Crimp Style Copper, Insulated, Environment Resistant, Class 1
SAE AS81824/14	Splice, Electric, Permanent, Crimp Style. Nickel-Coated Copper, Insulated, Environment Resistant, Class 1, 175 ˚C, Heatless Sealing
SAE AS85485	Cable, Electric, Filter Line, Radio Frequency Absorptive
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 014 00	Wire and Cable Splicing and Repair

NOTE

Splice restrictions outlined in NAVAIR 01-1A-505-1 WP 014 00 apply to all EWIS conductor splicing regardless of type splice is installed.

3. General inspection guidelines. When inspecting heatless splices meeting SAE AS81824/14 used in the EWIS, the following should be verified. Examples of acceptable and unacceptable heatless splices are depicted on figures 31-1 and 31-2, respectively.

CAUTION

SAE AS81824/14 splices are not authorized for use on filter line wire/cable (SAE AS85485) as shorting from inner conductor to conductive cable layer may occur, resulting in component failure or system degradation.

a. Ensure that the SAE AS81824/14 splice is not installed on Filter line wire/cable (SAE AS85485), typically black or purple in color with part number starting with ”M85485”.

b. Verify that the heatless splice is installed as follows:

(1) Ensure that the insulation sleeve is locked and will not slide off with average hand pull force.

(2) Ensure that the insulation sealing sleeve is not cracked.

(3) Rotate the insulation sleeve to view the inspection window. Verify that both conductor ends are visible in the inspection window.

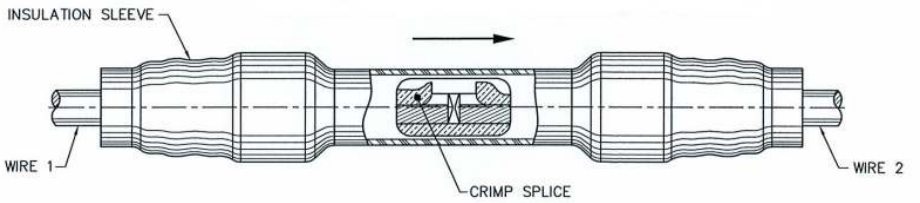
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(4) Ensure that only one wire is inserted in each end of the heatless splice.

(5) Ensure that the ends of the sealing sleeve contact the outside diameter of the wire insulation to form a seal.

(6) Ensure that the inspection window doesn't show evidence of fluids or corrosion inside the splice (see figure 31-2).

(7) Ensure that conductors are butted up against the wire stop and visual in the inspection window (see figure 31-1 upper detail).

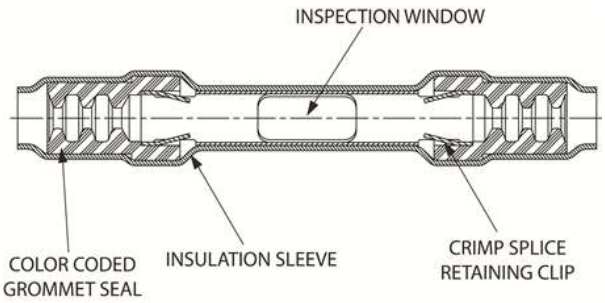


Acceptable

Conductors are butted up against the splice wire stop and visible in the inspection window.



Acceptable



Acceptable

FIGURE 31-1. Examples of acceptable heatless splices (SAE AS81824/14).

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Unacceptable
Only one stripped conductor end is visible in the inspection window. The second (top) stripped conductor is not visible in the inspection window.



Unacceptable
Evidence of corrosion inside sleeve. Multiple wires installed on both ends.

FIGURE 31-2. Examples of unacceptable heatless splices (SAE AS81824/14).

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GUIDELINE 32

WRAP-AROUND/SIDE-ENTRY WIRE INSULATION REPAIR (C-WRAP) INSPECTION

1. Purpose. This guideline provides the criteria for inspecting insulation repairs using the C-Wrap insulation sleeve when used in the EWIS. C-Wrap meets SAE AS6224/1 specification requirements.

2. Applicable documents. The documents listed below are those applicable to this guideline.

SAE AS6224/1	Repair Sleeve, Primary Wire Insulation Damage, Permanent, Environmental Resistant, 150 Degrees C Maximum Operating Temperature
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 014 02	Heat Shrink and Repair
Work Package 029 01	Basic Fault Isolation Methods

3. General inspection guidelines. When using C-Wrap wire insulation repair sleeves (D-150-C-XX and SAE AS6224/1-XX) in the EWIS, the following procedure should be followed:

a. Inspect for the following:

(1) Ensure there is no loose/damaged/protruding portion of the insulation or wire strands. Verify that the damaged wire jacket is completely covered by the repair sleeve.

(2) Confirm that the installed sleeve is fully shrunk onto the wire (see figure 32-1).

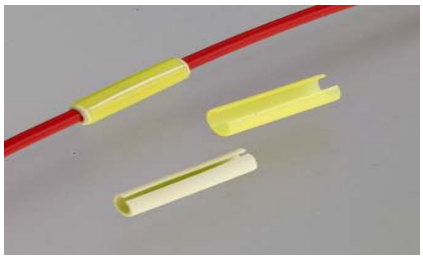
(3) Verify that the adhesive is melted, flowed, and filled the slit of the insulation sleeve.

(4) Ensure that the installed repair sleeve or wire insulation does not show any evidence of overheating (burning, browning, or severe darkening) or damage (see figure 32-2).

(5) Up to three C-Wrap sleeve repairs can be installed per foot of wire.

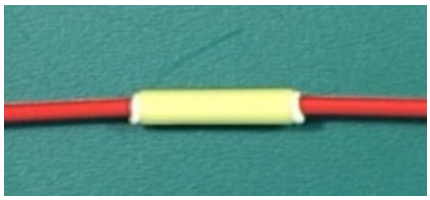
(6) No more than ten sleeve repairs on a wire segment or section.

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Acceptable

Two-piece construction; before and after installation.



Acceptable

Fully melted inner adhesive; no overheat/damage to wire insulation.



Acceptable

Seam fully closed; no exposed conductor.

FIGURE 32-1. Acceptable wrap-around/side-entry wire insulation (C-wrap) repair.

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Unacceptable
Exposed conductor; under-heated or under-sized.



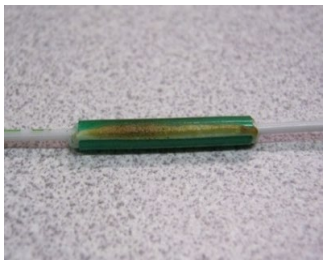
Unacceptable
Misalignment adhesive and outer sleeve.



Unacceptable
Adhesive/insulation mismatch.



Unacceptable
Over-heated installation.



Unacceptable
Mismatched adhesive and sleeve as well as over-heated installation.



Unacceptable
Under-heated/not melted.

FIGURE 32-2. Unacceptable wrap-around/side-entry wire insulation (C-wrap).

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GUIDELINE 33

CONNECTOR INSPECTION

1. Purpose. This guideline gives criteria for inspecting connectors used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 020 00	Military Standard Circular Connectors
Work Package 029 01	Basic Fault Isolation Methods
MS27488	Plug, End Seal, Electrical Connector
SAE AS23053	Insulation Sleeving, Electrical, Heat Shrinkable, General Specification for
SAE AS23053/5	Insulation Sleeving, Electrical, Heat Shrinkable, Polyolefin, Flexible, Crosslinked
SAE AS85049	Connector Accessories, Electrical, General Specification for

3. General inspection guidelines. When inspecting connectors used in the EWIS, the following items should be examined:

a. Inspect the insulator insert for damage and to ensure that there are no bent or splayed contacts. Contacts should be properly seated and locked and not pushed back (recessed or uneven) into the insert (see figure 33-1). Slightly angling the connector during inspection allows for easier identification of recessed contacts, especially socket type contacts.

CAUTION

Do not install the knob end of the sealing plug into connector cavities without having first inserted un-wired contacts. Failure to follow this procedure damages the connector, as the sealing plug is trapped inside the connector by the contact locking mechanism.

b. Ensure that all un-wired contact cavities on the grommet surface area populated with environmental sealing plugs (MS27488); (see figure 33-2).

c. Ensure that all installed sealing plugs are installed with the large end (head) inside the connector grommet (see figure 33-2).

d. Inspect the connector shell plating and finish for corrosion or flaking. Any flaking of plating on the nose of the plug or around the keyways is acceptable (see figure 34-1).

e. Inspect for serviceable condition of attaching mechanism:

(1) For threaded connectors, inspect thread surfaces for wear and condition.

(2) For bayonet connectors, inspect locking pins and mating surface and holes for wear and condition. Ensure that all three locking pins are present and secure.

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f. Confirm that connector receptacles with mounting flanges contain proper mounting hardware in accordance with applicable maintenance manuals.

g. Verify that every connector has a strain relief or backshell installed. For circular connectors, the backshell is part number M85049/XX (or equivalent). Refer to applicable maintenance manuals for backshell and strain relief part numbers. See guideline 36 for information on connector backshell inspection and guideline 37 for connector strain relief inspection.

h. When inspecting environment resistant connectors with grommet seals, the following should be examined (see figures 33-3 and 33-4):

(1) Verify that there are no chips, gouges, or other damage in or extending from chamfered area to the base of the chamfer.

(2) If the wire outside diameter is undersized, the wire may be built up with heat-shrinkable sleeving (SAE AS23053/5) to obtain proper environmental seal and prevent fluid entry in the connector. Applied heat shrink should not cover the crimp contact and should extend 1/2 to 3/4 inch beyond the connector grommet (see figure 33-6). Minimum wire outside diameter tolerances are defined in the applicable connector specification. If heat shrink is installed on wires, verify that the heat shrink is positioned so that it extends 1/2 to 3/4 inch beyond the connector grommet.

(3) Verify that the wires entering the connector's grommet seal do not distort the seal as to create a gap between wire and seal. All wires should extend straight out from the connector, as shown so that a gap is not created (see figures 33-5 and 37-1).

CAUTION

Fiber optic connectors should not be unmated for inspection. Disassemble fiber optic connectors for maintenance or troubleshooting only.

i. Fiber optic connector inspections should always be done before mating. Existing pollutants and damage of an unmated connector (plug or receptacle) can easily be transferred to the mating connector. Prior to mating, verify that the fiber optic ferrule end faces are clean and free of damage.

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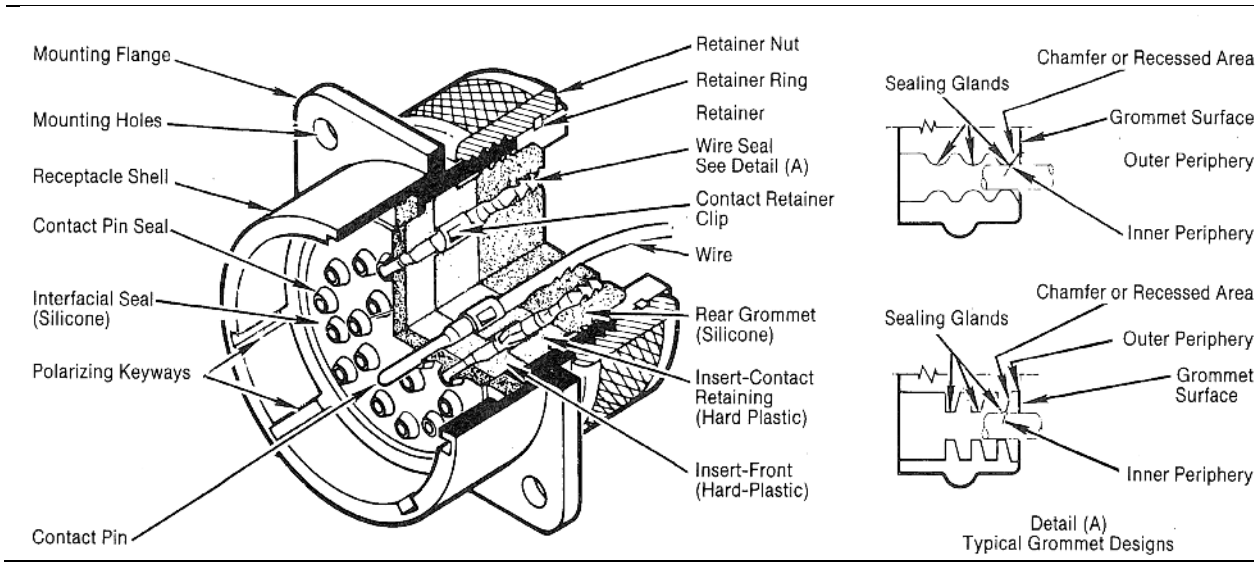


FIGURE 33-1. Illustration of typical connector components.

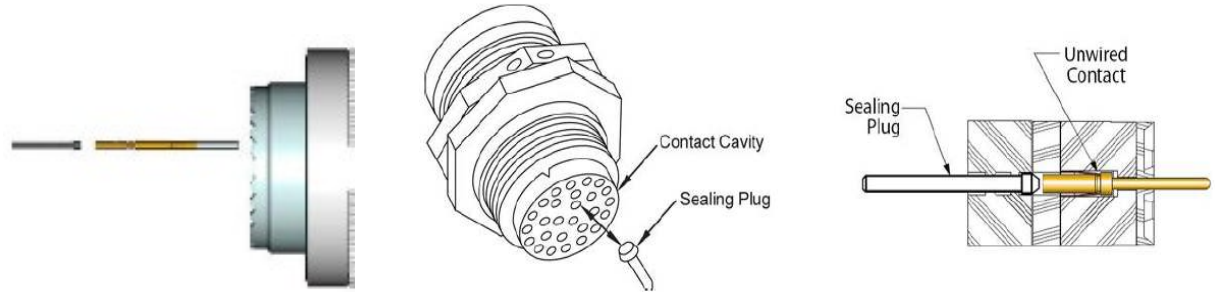


FIGURE 33-2. Sealing plug and contact installation for required connector sealing.

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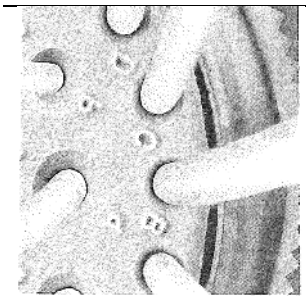
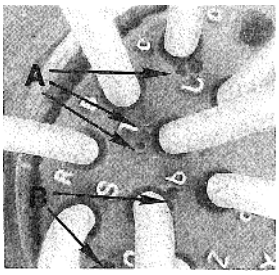
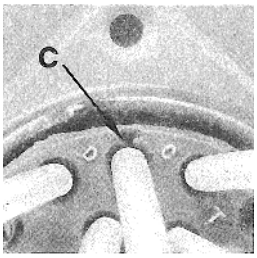
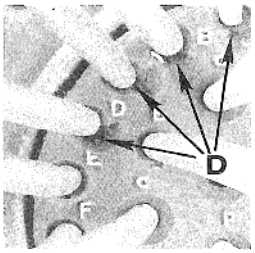
			
Acceptable The connector grommet has no surface gouges that extend to the wire or below the bottom of the chamfer (first sealing gland).	**Acceptable** Although gouges appear on the grommet surface (A) and in the chamfer area (B), none extend to the base of the chamfer.	**Unacceptable** The gouge (C) extends through the chamfered area and comes in contact with the wire. The sealing ability of the grommet is reduced.	**Unacceptable** The large deep gouges (D) extend below the bottom of the chamfer, and destroy the sealing capability of the grommet.

FIGURE 33-3. Examples of connector grommets in acceptable and unacceptable condition.

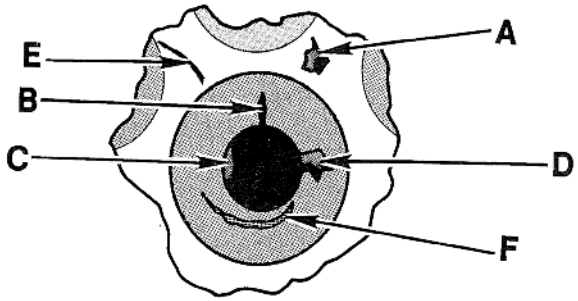
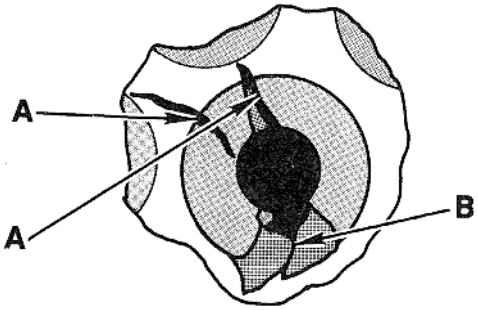
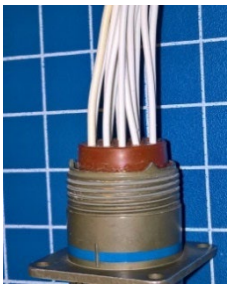
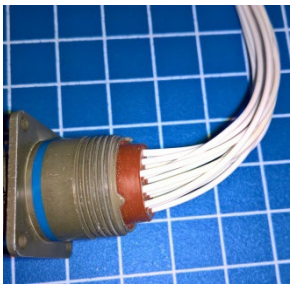
	
Acceptable Any ONE imperfection: (A) Chip, nick, or gouge, (B) Split crack through the sealing gland but not extending out of the recessed or chamfered area, (C) Mold Flash, (D) Chip, nick, or gouge through the sealing gland but not out of the recessed or chamfered area, (E) Split or crack, (F) Chip, nick or gouge not through the sealing gland.	**Unacceptable** (A) Split or crack extending out of the recessed or chamfered area, (B) Chip, nick, or gouge extending out of the recessed or chamfered area. The sealing grommet may be repairable using standard repair procedures.

FIGURE 33-4. Examples of sealing grommet with different levels of damage.

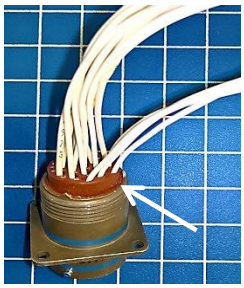
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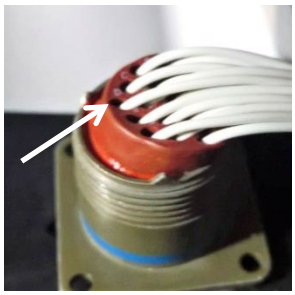
Acceptable
Wires are entering the connector; grommet seals do not distort the seal. All wires extend straight out from the connector.



Acceptable
Normal flexing of wires may exert pressure against the grommet, but a slight distortion of the seal is not enough to create a gap between the seal and wire.



Unacceptable
Wire exits sealing glands normal to the grommet for ¼″ prior to any wire bend. Although some of the wires are routed at a proper angle, four of the wires are short and have been pulled back sharply. A large gap and wire strain (arrow) is the result.



Unacceptable
Wires do not come straight out of the grommet and have distorted the seal. The result is large gaps (arrow). Contaminants and moisture can enter the terminal block.

FIGURE 33-5. Examples of acceptable and unacceptable wire installation in connectors.

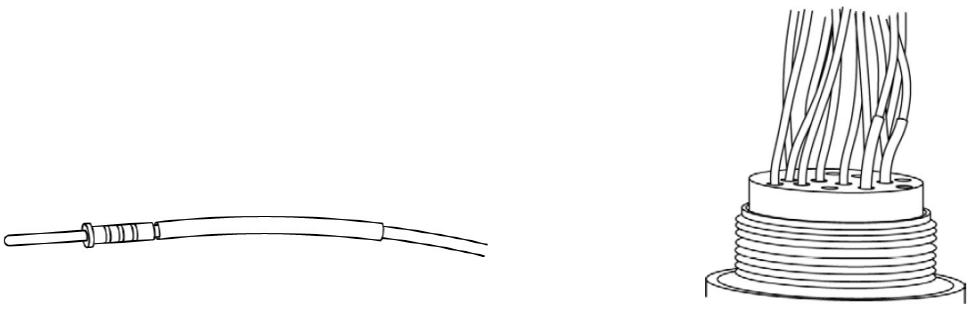


FIGURE 33-6. Wire insulation built up with heat shrink installed in the connector.

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GUIDELINE 34

COMPOSITE CONNECTOR INSPECTION

1. Purpose. This guideline provides the criteria for inspecting composite connectors in the EWIS. Refer to Guideline 33 for all other circular connector inspection criteria.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 004 01	Aircraft Wiring System Inspection
Work Package 020 00	Military Standard Circular Connectors

3. General inspection guidelines. Electrical connectors are manufactured from various metallic and non-metallic (composite) materials. The following guideline applies to only the inspection of composite (non-metallic) connectors (plug and receptacle). Inspect as follows:

NOTE

Plating on composite connectors provides electrical conductivity and EMI protection. Excessive degradation of the plating affects system EMI performance.

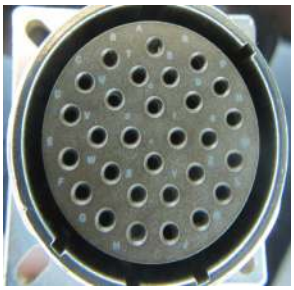
a. On Electroless Nickel (EN) plated composite connectors; inspect for any flaking of nickel plating on the nose of the plug and/or around the keyways. Limited nickel flaking is acceptable. Replace the connector if there is excessive plating wear (large sections missing EN plating inside or out).

CAUTION

Damage to composite connectors adversely affects their structural and electrical properties. Failure to replace damaged composite connectors may lead to system degradation or failure.

b. Inspect the composite connector for any chipped composite material (usually tan or black in color). Chipped composite material anywhere on the plug or receptacle is not acceptable, and the connector requires replacement.

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Acceptable
Serviceable



Acceptable
Limited, plating worn



Unacceptable
Excessive wear/ Connector nickel Flaking



Unacceptable
Cracked connector, flaking

FIGURE 34-1. Acceptable and unacceptable composite connector damage.

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GUIDELINE 35

CONNECTOR EMI GROUNDING RING INSPECTION

1. Purpose. This guideline provides the criteria for inspecting connector EMI grounding rings (finger seals) in EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 004 01	Aircraft Wiring System Inspection
Work Package 020 00	Military Standard Circular Connectors
MIL-DTL-38999	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded, or Breech Coupling), Environment Resistant with Crimp Removable Contacts or Hermetically Sealed with Fixed, Solderable Contacts, General Specification for

3. General inspection guidelines. The following guidelines should be used to perform an inspection on connectors containing EMI grounding rings (finger seal). EMI connectors should be inspected as follows:

WARNING

Ensure that any broken pieces of the EMI finger seal are removed. Failure to remove the loose pieces may cause FOD and/or electrical shock hazards and degrade system performance.

NOTE

The metallic EMI seal is made of spring fingers in the connector to allow shell-to-shell grounding before contacts mate and after they separate. They provide the required EMI protection and help to meet the shield effectiveness requirements of the application (see figure 35-1).

a. If any broken fingers (segments) on the EMI spring finger seal are found on the plugs/receptacles, remove all loose segments from inside of plug and/or mating receptacle (see figure 35-2). Contact Cognizant Engineering Authority for further guidance on EMI system-level requirements for the particular damaged connector.

b. For MIL-DTL-38999 circular connectors, series II plugs with EMI spring finger seal only, spring fingers are allowed to be missing at bayonet pins, provided that the spring continues to be retained about the shell periphery (see figure 35-3).

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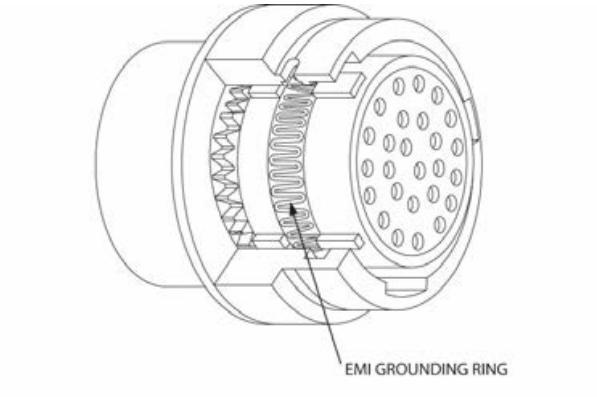


FIGURE 35-1. Connector plug with EMI finger seal.



Acceptable:
EMI finger seal is installed, not damaged and continuous.



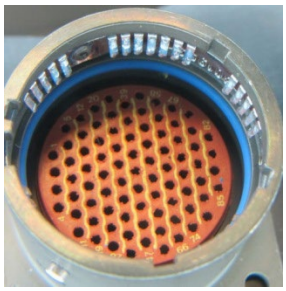
Unacceptable:
Multiple broken EMI seal fingers.



Unacceptable:
Missing and broken EMI seal fingers.

FIGURE 35-2. Connector EMI finger seal inspection.

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Acceptable
MIL-DTL-38999 Series II connector with
EMI fingers missing at bayonet pins

FIGURE 35-3. Acceptable EMI connector finger damage.

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GUIDELINE 36

CONNECTOR BACKSHELL INSPECTION

1. Purpose. This guideline gives criteria for inspecting the electrical connector backshells used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

A-A-59163	Insulation Tape, Electrical, Self-Adhering Silicone
MIL-I-22444	Insulation Tape, Electrical, Self-Bonding, Silicone Rubber Treated Bias Weave or Sinusoidal Weave Glass, Cable Splicing, Naval Shipboard.
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 024 00	Connector Accessories

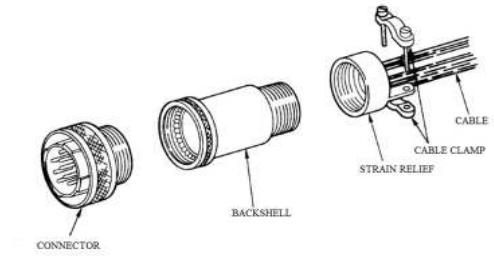
3. General inspection guidelines. Figure 36-1 illustrates components associated with non-environmental, environmental, and EMI backshells. When inspecting backshells used in the EWIS, the following steps should be taken:

a. Confirm that all components are present, are serviceable, and installed in the correct order (See figure 36-1). Ensure that harness orientation meets minimum bend radius requirements. See guideline 16.

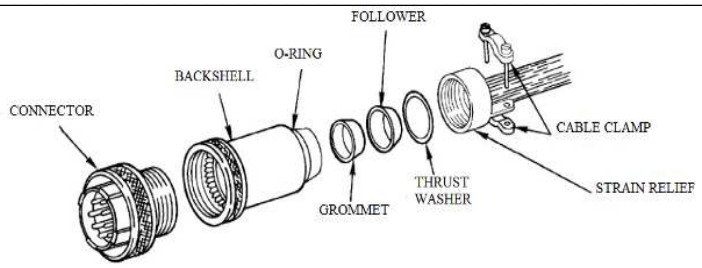
b. Verify that the backshell is tightened by applying a clockwise force as viewed from the connector rear. Refer to NAVAIR 01-1A-505-1 WP 024 00 for backshell torque values.

c. Confirm that silicone tape (A-A-59163 or MIL-I-22444) or a reusable side entry bushing (CS949X-000) to build cable diameter is applied under strain relief if the harness is smaller than the strain relief opening where the rubber grommet is not installed (see figure 36-2).

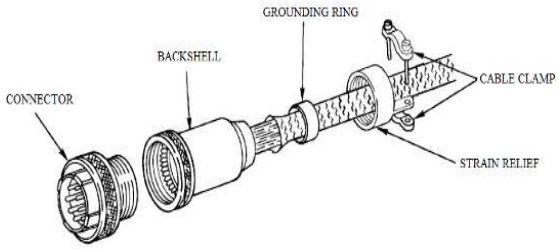
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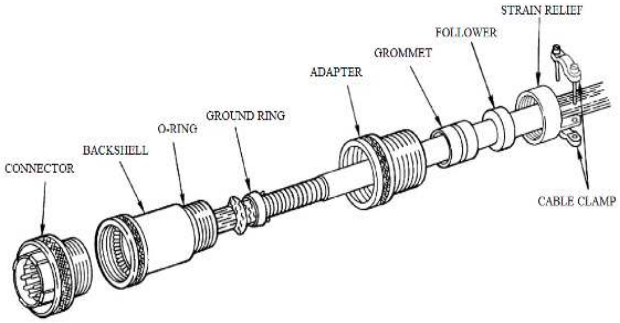
Typical non-environmental backshell



Typical environmental backshell



Typical non-environmental EMI/RFI backshell



Typical environmental EMI/RFI backshell

FIGURE 36-1. Illustrations of components associated with non-environmental, environmental and EMI/RFI backshells.

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Acceptable
Wires are secured in backshell strain relief clamps using silicone tape. Screws are oriented so that the backshell can be rotated clockwise while applying pressure on the screw heads. More than two threads are protruding from screws. The composite backshell is not cracked, and it is tightly threaded onto the connector.



Unacceptable
The composite backshell is cracked, and it is loose on the connector. Note the longitudinal crack in the coupling nut.



Unacceptable
The composite backshell is cracked, and it is loose on the connector. Note the radial crack in the coupling nut.

FIGURE 36-2. Examples of acceptable and unacceptable composite backshell accessories.

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GUIDELINE 37

CONNECTOR STRAIN RELIEF INSPECTION

1. Purpose. This guideline gives criteria for inspecting the connector stress relief in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

A-A-59163	Insulation Tape, Electrical, Self-Adhering Silicone
CS949X-000	Bushing, Side Entry, Reusable
MIL-I-22444	Insulation Tape, Electrical, Self-Bonding, Silicone Rubber Treated Bias Weave or Sinusoidal Weave Glass, Cable Splicing, Naval Shipboard. (Inactive for New Design)
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 020 00	Standard Military Circular Connectors
Work Package 024 00	Connector Accessories

3. General inspection guidelines. Figure 37-1 gives examples of acceptable and unacceptable connector strain relief. When inspecting connector strain relief, the following steps should be used:

a. For angled applications (90 or 45-degree strain reliefs), ensure that the wire exits the connector sealing glands perpendicular to the grommet for 3/8″ prior to any wire bend.

b. Check wire for proper contour and that it is not being pulled tight, inducing stress on sealing glands, distorting the grommet, and prohibiting an environmental seal (see figures 37-1 and 33-5).

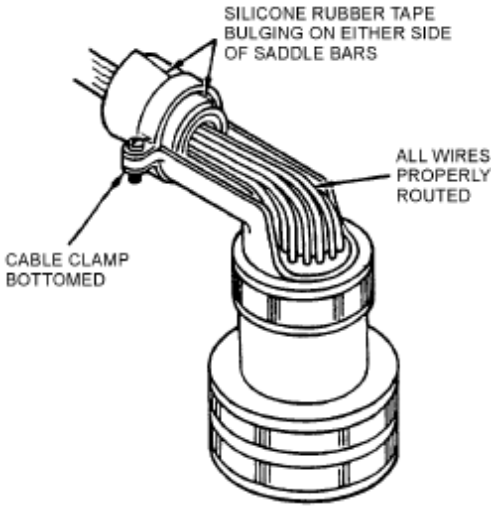
c. Ensure that there is a minimum of two wraps of silicone rubber tape cushion (A-A-59163 or MIL-I-22444) centered securely under the cable clamp or a reusable side entry bushing (CS949X-000) to build cable diameter where necessary.

d. Confirm that cable clamp and attaching hardware are installed.

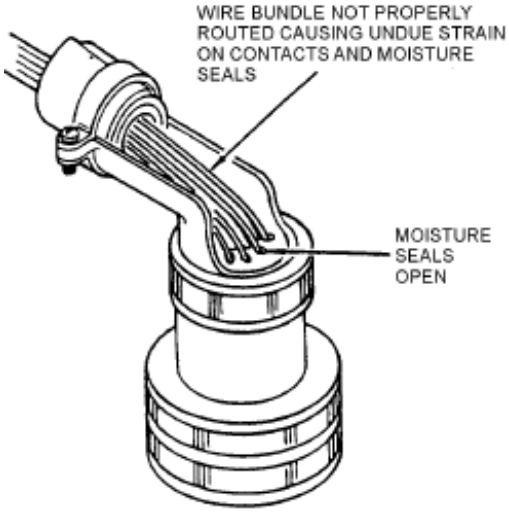
e. Ensure that the bending of cabling exiting a backshell occurs at the strain relief so that it doesn't elongate the sealing grommet. See figures 33-5 and 37-1.

f. For 90 or 45-degree backshells or strain reliefs, ensure that the orientation/angle of the backshell will not contact or interfere with adjacent components. Re-clock or re-position backshell to ensure that there is suitable clearance (see NAVAIR 01-1A-505-1, WP 024 00).

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Acceptable
Wire exits sealing glands perpendicular to the grommet for 3/8″ prior to any wire bend. Does not induce stress on sealing glands. Provides environmental seal around the wire.



Unacceptable
The short wires in the back of the connector (arrow) produce excess stress on the contact crimp joint and sealing glands. Distorts sealing gland, prohibiting environmental seal.

FIGURE 37-1. Examples of acceptable and unacceptable connector strain relief.

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GUIDELINE 38

LOCKWIRE/SAFETY CABLE INSPECTION

1. Purpose. This guideline gives criteria for inspecting lockwire (SAE AS4536) used in the EWIS. For shearwire inspection, see guideline 39.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NASM20995	Wire, Safety, or Lock
SAE AS4536	Safety Cable Kit Procurement Specification and Requirements for Use
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 018 00	Lockwiring, Shear Wiring, and Safety Cables

(Copies of NASM20995 are available from www.aia-aerospace.org.)

3. General inspection guidelines. When lockwire and safety cable used in the EWIS is inspected, the following should be examined:

NOTE

Always refer to the platform/weapon system-specific manual governing maintenance practices to ensure that the use of safety cable is not limited or prohibited from use.

a. Lockwire inspection:

(1) Verify that lockwire is not used for any shear or breakaway applications (see guideline 39 for shearwire applications).

(2) Verify that lockwire is twisted in a clockwise direction, with approximately 8 to 10 twists per inch. The cut end (pigtail) of the lockwire should have 4 to 6 complete turns (¼ to ½ inch long) bent under or back to prevent injury. Lockwire or safety cable should be installed so that the applicable shell of the connector being secured is pulled toward the tightening direction (see figure 38-1).

(3) Confirm that lockwire is in accordance with application requirements. Select approved lockwire from NAVAIR 01-1A-505-1, WP 018 00, Table 1. Ensure that the installed lockwire is routed in the most direct way to the tightening position, is taut, and shows no evidence of nicks, kinks, or breaks (see figure 38-2).

(4) Confirm that the lockwire has not been over twisted and that there are no broken wires (see figure 38-2).

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b. Safety cable installation

(1) Verify that the safety cable is not used for any shear or breakaway applications (see guideline 39 for shearwire applications).

(2) Ensure that the safety cable crimping ferrule is securely installed and cable is tensioned (see figure 38-3). For flex limits of safety cable, refer to NAVAIR-01-505-1, WP 018 00.

(3) Ensure that the safety cable routing is to the tightening position of the device being secured.

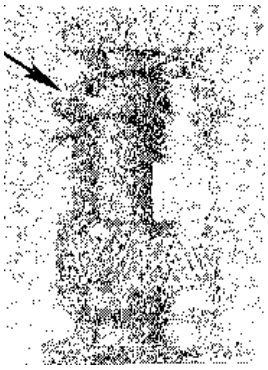
(4) Ensure that the safety cable diameter meets application requirements.



Acceptable
Lockwire is twisted clockwise, as shown. A pigtail of 4 to 6 twists is left after the completion of safety wiring.



Unacceptable
Lockwire has been over twisted. Excessive twists may cause a wire break.



Unacceptable
Excessive twists have resulted in a broken wire (arrow). This is caused by the extreme stress of over-twisting wires.

FIGURE 38-1. Examples of acceptable and unacceptable installation of lockwire.

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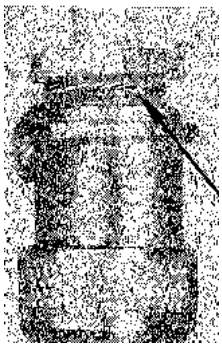
			
Acceptable	**Unacceptable**	**Unacceptable**	**Unacceptable**
Lockwire is routed in the most direct way to the tightening position, is taut, and shows no evidence of nicks, kinks, or breaks.	The kinked lockwire (arrow) should be removed and replaced. A kinked strand transfers the stress load to the remaining strand and may result in lockwire failure.	The wire is over twisted and shows evidence of nicks (arrow) which may fracture and result in loss of lockwire tension.	The broken wire (arrow) is the result of over-twisting during application.

FIGURE 38-2. Examples of acceptable use of lockwire, lockwire that is directly routed and unacceptable installation of lockwire due to kinks and over twisting.

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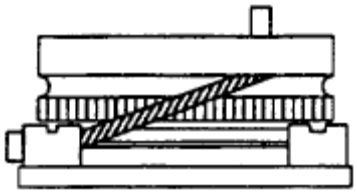
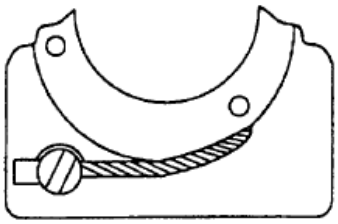
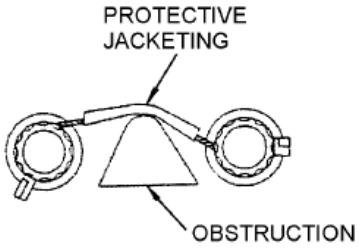
		
Acceptable Safety cable may be used as a substitute for lockwire on electrical equipment and connectors in aircraft to prevent accidental loosening caused by vibration. Safety cable may be used on threaded parts, such as connector coupling mechanisms, backshells, strain relief components, relays, other electrical components, and equipment covers/panels.	**Acceptable** Safety cable installed on a single jam nut receptacle. Safety cable may be installed from a fastener as shown above, or it may require the use of a self-looping safety cable.	**Acceptable** A tubular jacket is installed where it may come into contact with other surfaces that may damage the safety cable or may be damaged by the safety cable.

FIGURE 38-3. Examples of acceptable safety cable installations.

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GUIDELINE 39

SHEARWIRE INSPECTION

1. Purpose. This guideline provides the criteria for inspecting shearwire used in EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1 Aircraft Electric and Electronic Wiring
Work Package 018 00 Lockwiring, Shearwiring, and Safety Cables

3. General inspection guidelines. Shearwire should be examined as follows:

WARNING

Select the correct wire for the application. Ensure that only MS20995-CY-20 shearwire (Copper color) is used for shear wire applications. Shearwire is designed to shear when the switch/guard is actuated, as opposed to lockwire which is supposed to lock safely and secure the device to prevent it from loosening. Loss of life may occur when lockwire is used instead of shearwire.

CAUTION

Shearwire is intended to secure emergency equipment to prevent accidental actuation. Inadvertent actuation of a switch is possible if the switch guard is shearwired to the switch toggle lever.

NOTE

MS20995-CY20 shearwire may be yellow or reddish (copper) colored.

a. Verify that the emergency devices are shearwired with only copper wire, part number MS20995-CY20.

b. Confirm that all applications are using the single wire method so that the shearwire may be easily broken in an emergency (see figure 39-1).

c. Ensure that wire ends are terminated (twisted) in a pigtail and out of the way to protect against injury (see figure 39-2).

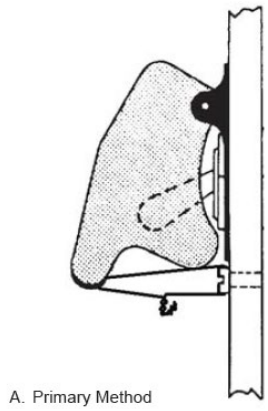
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FIGURE 39-1. Acceptable methods of shearwiring emergency devices.



Unacceptable
Shearwire not installed (right) and broken (left)



Acceptable
Shearwire installed using the primary method

FIGURE 39-2. Unacceptable and acceptable shearwiring of emergency devices.

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GUIDELINE 40

PRESERVATION OF CONNECTOR/COMPONENT INSPECTION

1. Purpose. This guideline gives criteria for inspecting connector/component preservations used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

A-A-52083	Tape, Lacing and Tying, Glass
A-A-52084	Tape, Lacing and Tying, Aramid
MIL-PRF-8516	Sealing Compound, Synthetic Rubber, Electric Connectors and Electric Systems, Chemically Cured
MIL-A-46146	Adhesives-Sealants, Silicone, RTV, Noncorrosive (For Use with Sensitive Metals and Equipment)
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 025 00	Potting and Sealing Connectors, Electrical Cable Assemblies, and Electrical Components
Work Package 026 00	Connector Cleaning and Preservation
NAVAIR 01-1A-509-1	Corrosion Program and Corrosion Theory
NAVAIR 01-1A-509-3	Avionic Cleaning and Corrosion Prevention/Control

3. General inspection guidelines. Figure 40-1 gives examples of connectors in acceptable and unacceptable conditions. When inspecting for connector corrosion, connector sealing methods, and electrical connection potting, the following apply:

WARNING

Ensure that all electrical power is disconnected from the aircraft and all affected systems in the aircraft are de-activated. Working on EWIS with energized systems may result in equipment damage and/or injury to personnel.

NOTE

Ensure that connectors or components to be inspected are first cleaned of contaminants or debris to facilitate inspection.

a. Metallic or plated connectors and backshells should be inspected for signs of corrosion (see figure 40-1). Examples of corrosion and corrosion damage include:

(1) Corrosion deposits/powder, flaking, or loosening of outer plating material. Refer to NAVAIR 01-1A-505-1 WP 026 00, Table 1 for identification of various corrosion types and their appearance.

(2) Inspect environmentally sealed connectors, which employ heat shrink boots. If signs of corrosion are visible on the ends of the boots, sealing boot may have to be removed for inspection (see figure 40-2).

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(3) Identify corrosion as light, moderate, or severe (see figures 40-3 and 40-4). Degrees of corrosion are classified as follows:

(i) **Light**: A mild pale partial tarnish with faded green appearance, some minimal white spotting to the surface.

(ii) **Moderate**: A light powdery surface populated with some spots or grainy white dimples/deposits; roots of threads or partially concealed areas exhibiting higher degrees of damage. Appears like a salt has been dried on to the surface.

(iii) **Severe**: Extensive corrosion, some flaking of the plating, powdery residue on surface, substantial pitting, base metal exposed or corroded.

(4) Clean light or moderate corrosion on connector external surfaces per NA 01-1A-505-1, WP 026 00.

(5) Severe corrosion on connector external surfaces requires affected component replacement and/or documentation in aircraft forms/log.

NOTE

If the connector was sealed using a dual wrap kit (AD28500-36-36-8), inspect for the following:

b. Connector sealed using the dual wrap kit AD28500-36-36-8 (contains the pink-colored Stretch Seal (inner layer), the gray-colored Self-Fusing Silicone tape (outer layer), and HT3326-5FR50 sealing compound (see figure 40-5) should be inspected as follows:

(1) Ensure that there are no voids and 50 percent overlap of gray self-fusing outer layer tape is uniformly applied.

(2) Ensure that the outer layer is secured with lacing and tying tape (e.g., A-A-52083 or A-A-52084).

(3) Ensure that none of the inner, pink stretch seal tape is exposed. Only the gray outer layer tape should be exposed.

c. If connectors are sealed/potted using compounds such as MIL-PRF-8516, MIL-A-46146, and TG2010FR-50 (see figures 40-5 and 40-6) they should be inspected as follows:

(1) Ensure that the potting boot or ring is firmly seated against the back of the connector.

(2) Ensure that the potting boot or ring is completely filled with potting compound.

d. Electrical connection potting, as may apply to exposed relays, contactors, and ground terminals (see figures 40-6 and 40-7), should be inspected as follows:

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(1) Ensure that all exposed metallic surfaces are uniformly potted with a sealing compound.



Acceptable
Serviceable connector.



Unacceptable
Connector corrosion with connector degraded until broken.



Unacceptable
Another example of corrosion on a connector.

FIGURE 40-1. Examples of connectors that are in acceptable and unacceptable condition.



FIGURE 40-2. Examples of corrosion under environmentally sealed connector boot.

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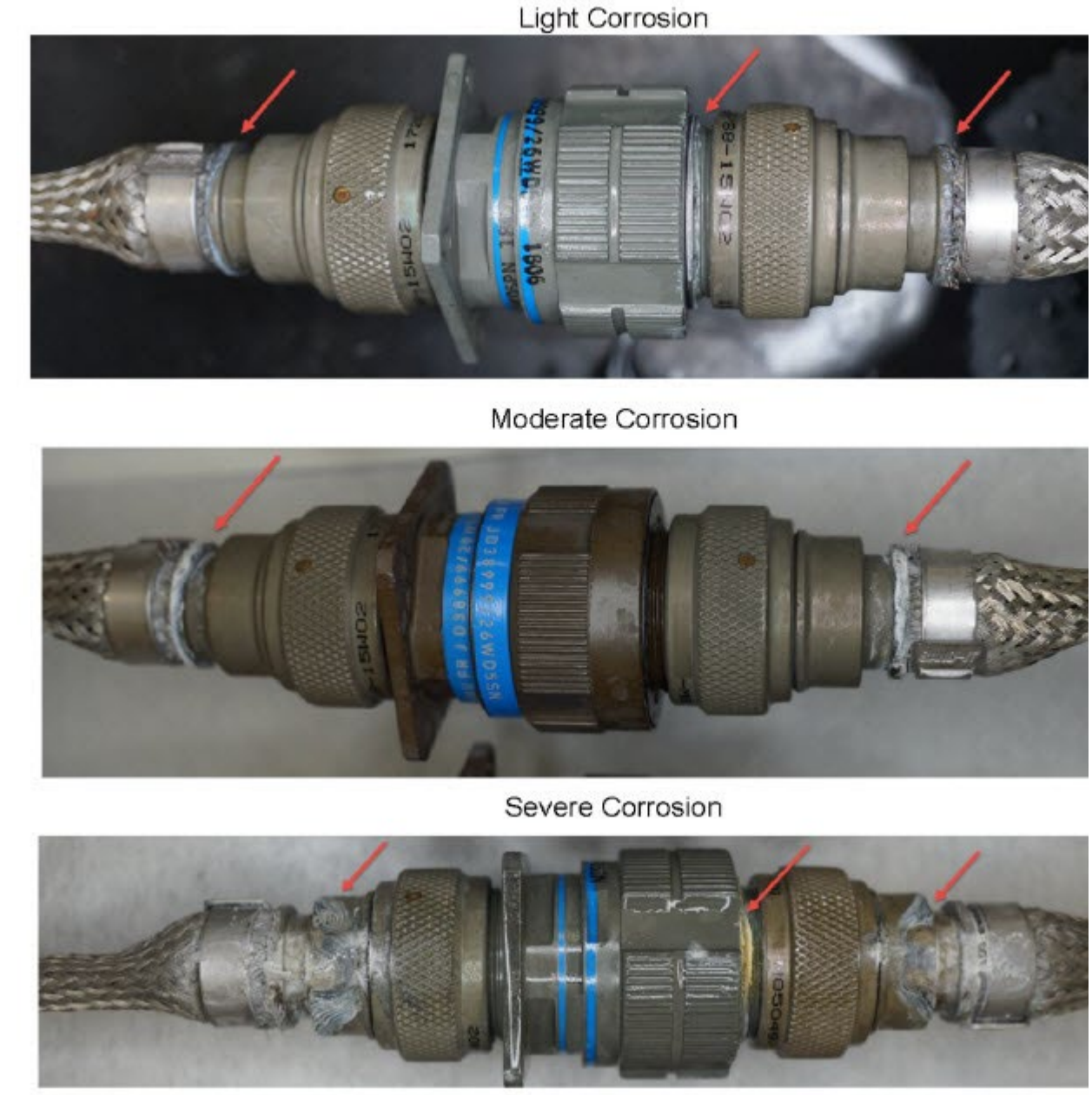


FIGURE 40-3. Corrosion severity examples on cadmium plated connector.

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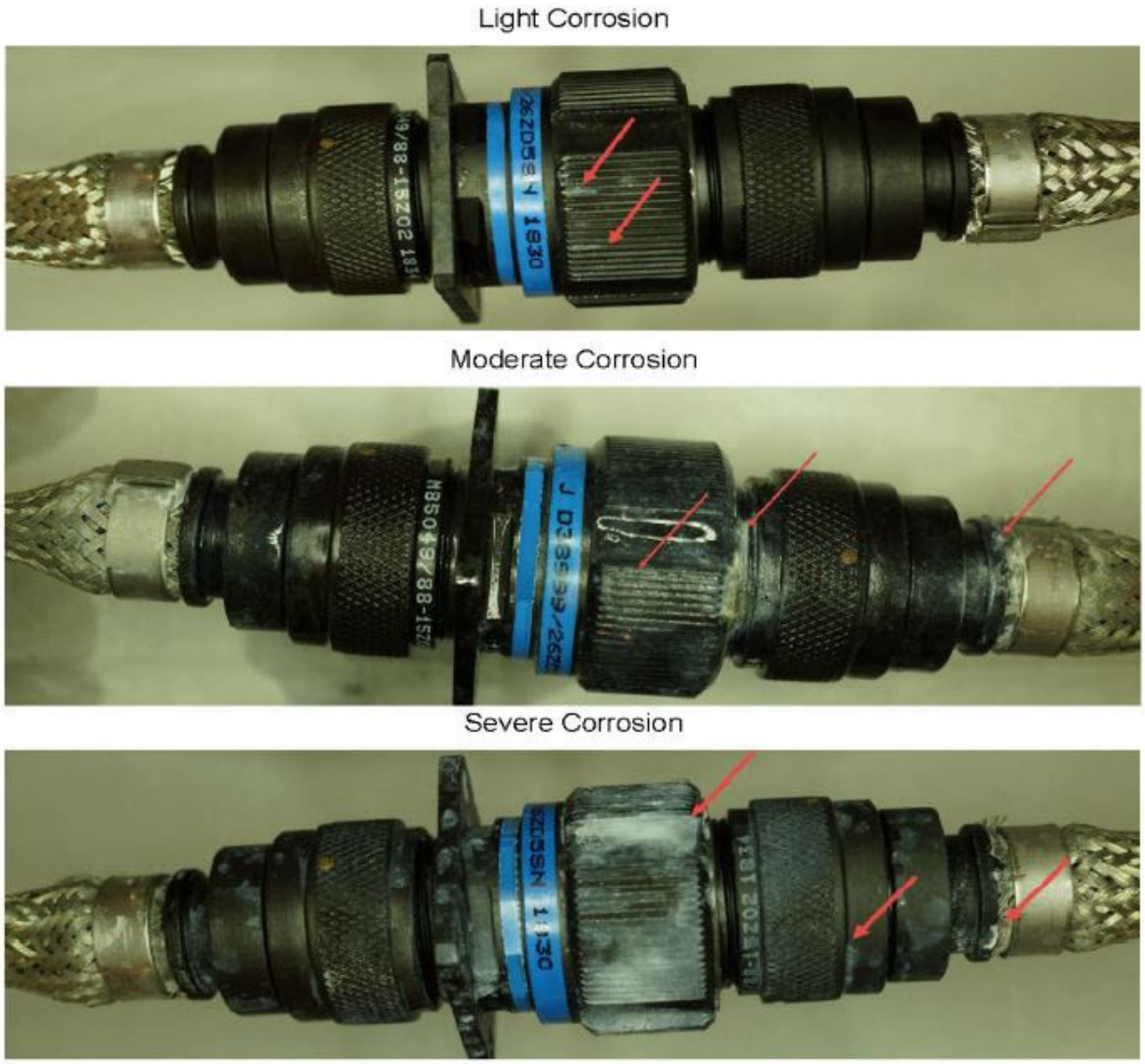


FIGURE 40-4. Corrosion severity examples on zinc-nickel plated connector.

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Acceptable
Dual wrap kit installed properly using self-leveling green, stretch seal, and self-fusing silicone tape, secured with lacing tape.

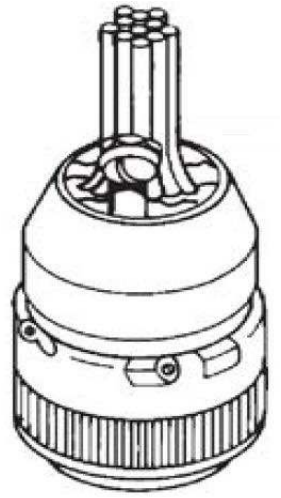


Acceptable
Heat shrinkable sealing boot installed onto environmental resistant connector accessory.

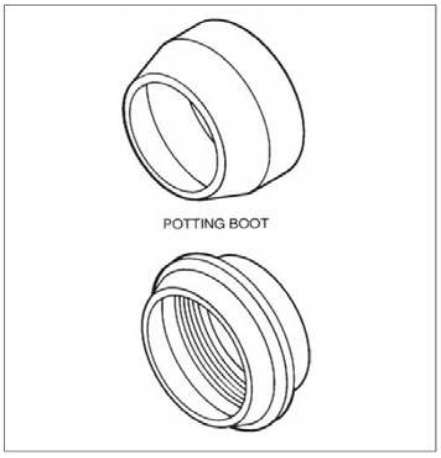


Unacceptable
Self-fusing silicone tape not applied and not secured with authorized lacing tape.

FIGURE 40-5. Examples of acceptable and unacceptable connector sealing.



Acceptable
Potting ring and sealant correctly installed.



Acceptable
Potting boot and ring before installation.



Acceptable
Potted connector accessory on molded /sealed wire harness.

FIGURE 40-6. Examples of potted or sealed connectors.

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Acceptable Sealant uniformly applied and no exposed metallic surfaces.	**Acceptable** Sealant uniformly installed and exposed. Metallic surfaces covered.	**Unacceptable** The potted/sealed terminal post is not fully covered and has voids in the potting compound.

FIGURE 40-7. Examples of acceptable and unacceptable potted or sealed connectors/components.

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GUIDELINE 41

CORROSION PREVENTION COMPOUND (CPC) APPLICATION INSPECTION

1. Purpose. This guideline provides the criteria for inspecting corrosion prevention compounds used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

MIL-PRF-81309	Corrosion Preventive Compounds, Water Displacing, Ultra-Thin Film
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 026 00	Connector Cleaning and Preservation
NAVAIR 01-1A-509-3	Avionic Cleaning and Prevention/Control
NAVAIR 01-1A-509-1	Corrosion Program and Corrosion Theory

3. General inspection guidelines. CPC (such as MIL-PRF-81309, Type III) application should be examined as follows:

WARNING

Internal connector surface CPC application can result in serious damage to equipment, possibly resulting in system failure or fire, and personnel injury may occur.

a. Verify that there is no evidence of CPC build-up on internal sections of connectors and receptacles (see figure 41-1).

b. Verify that there is no evidence of CPC build-up on multiple termini/contact connectors containing fiber optic termini or on fiber optic single ferrule connectors. Any seepage ingress into the connector housing is considered a contaminant and may degrade the system transmission performance or cause failure.

c. Verify that CPC is only applied to the external sections of the plugs and receptacles (see figure 41-2).

d. Verify that there is no evidence of CPC build-up on the insulation of any wire. This includes all types and grades of CPC's (e.g., structural or avionics grade).

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Unacceptable

Evidence of CPC build-up in the connector and on contacts.

FIGURE 41-1. Unacceptable CPC build-up examples.

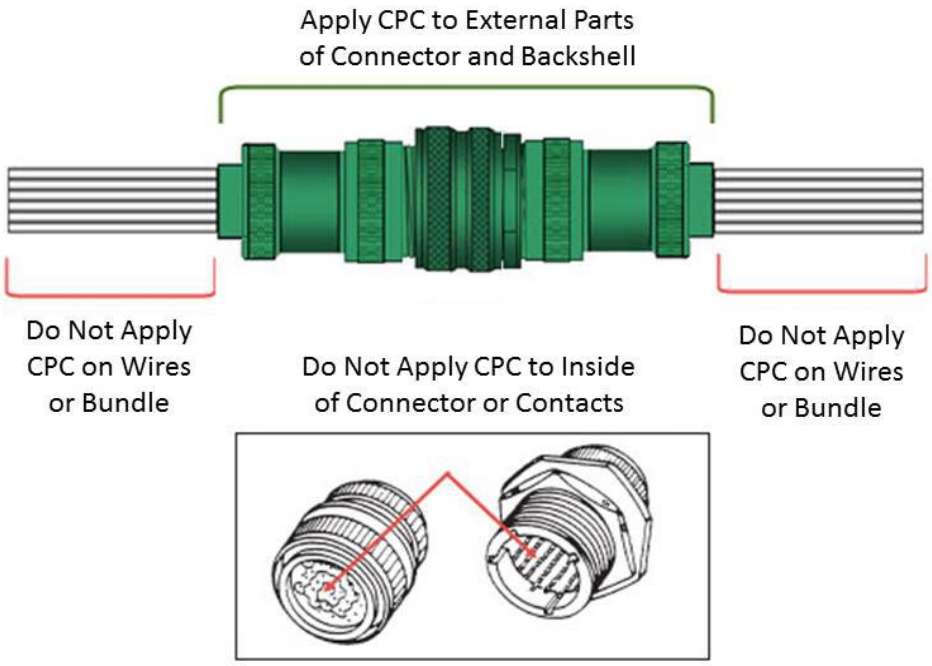


FIGURE 41-2. CPC Application for typical connectors.

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GUIDELINE 42

TERMINAL BOARD AND GROUND STUD INSPECTION

1. Purpose. This guideline gives criteria for terminal board and ground stud inspections.

2. Applicable documents. The documents listed below are those applicable to this guideline.

MIL-PRF-8516	Sealing Compound, Synthetic Rubber, Electric Connectors and Electric Systems, Chemically Cured
MIL-A-46146	Adhesives-Sealants, Silicone, RTV, Noncorrosive (For Use with Sensitive Metals and Equipment)
SAE AS7928	Terminals, Lug: Splices, Conductor: Crimp Style, Copper, General Specification for
SAE AS21919	Clamp, Loop Type, Cushioned Support
SAE AS23190	Wiring, Positioning, and Support Accessories
NASM25440	Washers, for Use with Aircraft Aluminum Terminals
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 013 00	Contacts and Terminals
Work Package 017 00	Bonding and Grounding
Work Package 019 00	Bus Bar and Terminal Board
Work Package 026 00	Connector Cleaning and Preservation

(Copies of NASM documents are available at www.aia-aerospace.org.)

3. General inspection guidelines. When inspecting the terminal board and ground stud used in the EWIS, the following items should be examined:

a. Terminal board mounting and connections should be inspected as follows:

(1) Confirm that no more than four terminal lugs, or three-terminal lugs, and one bus are connected to one terminal stud. Verify that no more than four wires are in a terminal lug, resulting in a maximum of sixteen wires per stud.

(2) Verify that terminal lugs with various diameters are stacked with the largest outer diameter on the bottom and the smallest on top.

(3) Verify that terminal lugs are not over tightened or worn so that the terminal lug or stud is not deformed or damaged.

(4) Check the wires connected to the terminal board to confirm that the wires have a minimum of ¾″ ±¼″ wire length plus required bend radius allowance (see figure 42-1).

(5) Confirm that all wires exit straight from terminal lugs (see figure 42-1).

(6) Confirm that a marking sleeve is installed.

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(7) Verify that terminal lugs are positioned so that bending is not required to remove the fastening screw or nut (see figure 42-2).

(8) Confirm that terminal lugs are positioned so that movement will tighten the nut.

(9) Verify that copper terminal lugs do not have spacers or washers between the tongues of terminal lugs (see figure 42-3).

(10) Confirm that aluminum terminal lugs have the tongue or the total number of tongues are sandwiched between two NASM25440 flat washers. Spacers or other washers are not permitted between the tongues (see figure 42-4).

(11) When two lugs are attached to one side of a stud, verify that the lugs are installed back-to-back (see figure 42-5).

(12) When the maximum of three lugs is installed on one side, confirm that a space washer is placed between back-to-back lugs and the third lug (see figure 42-5).

(13) Verify that stud hardware is stacked as shown on figure 42-3 or figure 42-4, depending on application requirements.

WARNING

Ensure that the cover is securely installed over terminal boards. Failure to install the cover is an electrical shock hazard and may cause injury or death to personnel.

(14) Verify that the terminal board cover is securely installed on the terminal board.

WARNING

Under-torqued or loose terminal lugs may create excessive voltage drops across the connection resulting in degraded system operation and/or overheating.

(15) Ensure that the terminal lugs are not loose and do not rotate around the terminal board stud mount. Torque limits provided in NAVAIR 01-1A-505-1, WP 019 00, Table 1 based on stud size.

b. Ground stud mounting and connections should be as follows:

(1) Confirm that no more than four terminal lugs, or three-terminal lugs, and one bus are connected to one terminal stud. Verify that no more than four wires in a terminal lug, resulting in a maximum of sixteen wires per stud.

(2) Verify that terminal lugs with various diameters are stacked with the largest outer diameter on the bottom and the smallest on top.

(3) Verify that terminal lugs are not over-tightened or worn and the terminal lug or stud is not deformed or damaged.

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(4) Confirm that all wires exit straight from terminal lugs.

(5) Verify that terminal lugs are positioned so that bending is not required to remove the fastening screw or nut (see figure 42-2).

(6) Confirm that the total number of aluminum terminal lugs is sandwiched (enclosed) between two NASM25440 flat washers. Additional spacers or other types of washers are not permitted between the terminal lugs (see figure 42-4).

(7) When two terminal lugs are attached to one side of a stud, verify that the lugs are installed back-to-back (see figure 42-5).

(8) When the maximum of three-terminal lugs is installed on one side, confirm that a space washer is placed between back-to-back lugs and the third lug (see figure 42-5).

(9) Verify that stud hardware is stacked in accordance with figure 42-3 or figure 42-4 depending on application requirements.

(10) Examine the ground stud and attaching hardware for corrosion. Refer to NAVAIR 01-1A-505-1, WP 026 00, Table 1 for appearance of specific corrosion types. Clean or replace as required.

(11) Verify that mounting hardware of the terminal strip is potted or sealed with a suitable potting compound (e.g., MIL-A-46146, TG2010FR-50, or MIL-PRF-8516).

(12) Verify that cushioned clamps (e.g., SAE AS21919 or SAE AS23190) employed as primary support and are not used for any bonding or grounding connection.

WARNING

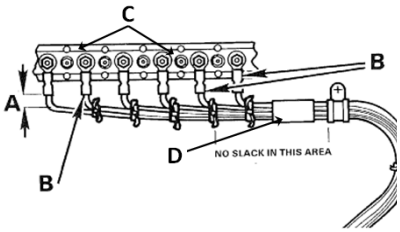
Under-torqued or loose terminal lugs may create excessive voltage drops across the connection resulting in degraded system operation and/or overheating.

(13) Ensure that the terminal lugs are not loose and do not rotate around the stud mount. Torque limits provided in NAVAIR 01-1A-505-1, WP 019 00, Table 1 based on stud size.

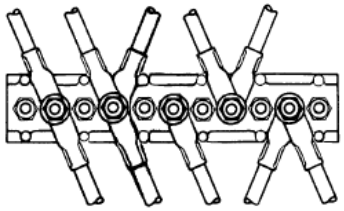
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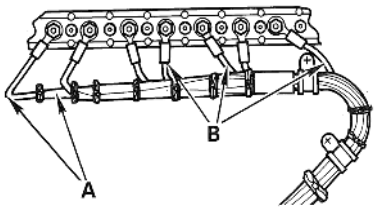
.



Acceptable
(A) Wire length should be ¾″ ±¼″ maximum plus bend radius. (B) All wires exit straight from the terminal lug. (C) Exposed mounting hardware of the terminal strip is potted or sealed with a suitable potting compound. (D) Marking sleeve is attached after the bundle is formed

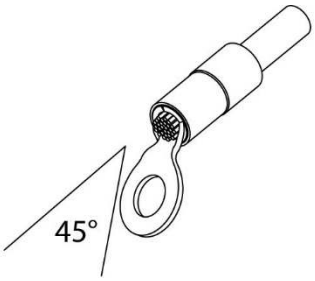


Acceptable
All terminals should be placed so that movement will tighten the nut as shown.

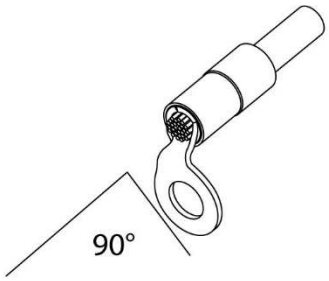


Unacceptable
(A) Excessive wire length and bend radius.
(B) No stress relief at termination, insufficient wire length.

FIGURE 42-1. Examples of acceptable and unacceptable wire length, bend radius, and terminal placement.



For copper terminals size 12-10,
straight terminals only,
45° maximum bend away from barrel.



For copper terminals size 26 - 14,
straight terminals only,
90° maximum bend away from barrel.

FIGURE 42-2. Maximum allowable copper terminal lug bend.

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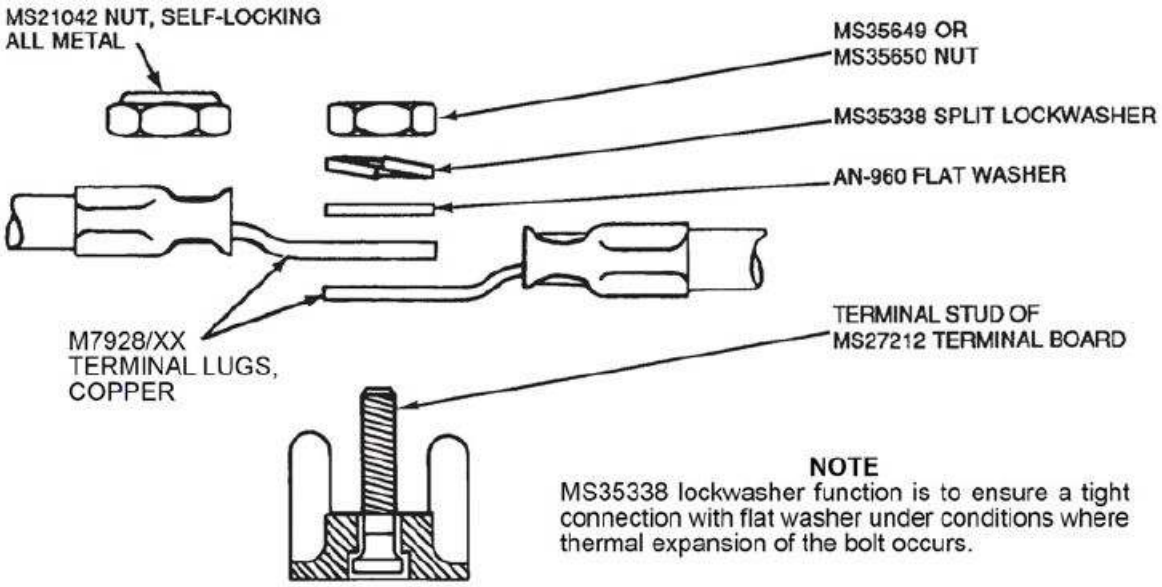


FIGURE 42-3. Hardware for wiring terminal boards with copper terminals.

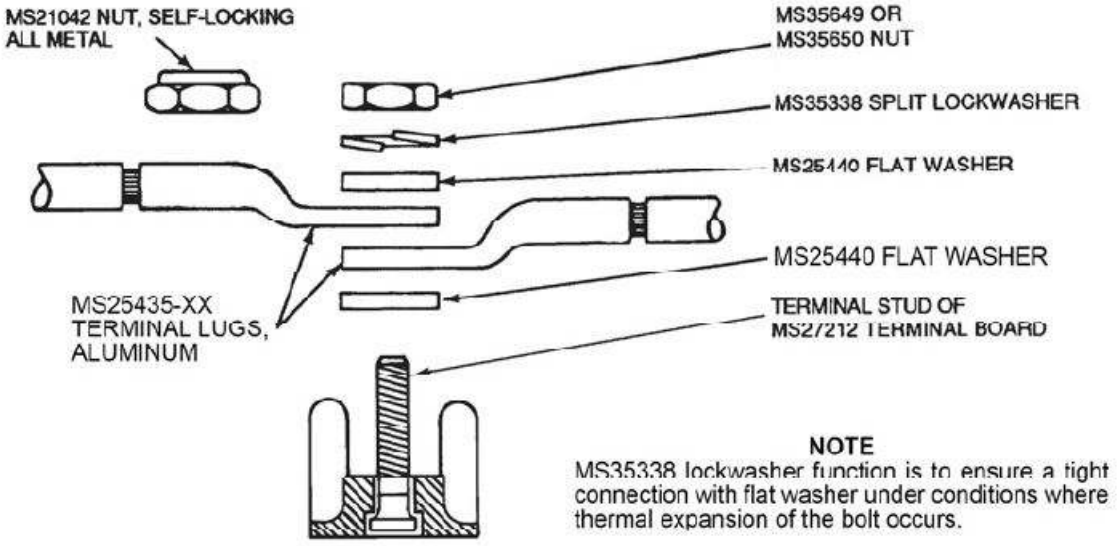


FIGURE 42-4. Hardware for wiring terminal boards with aluminum terminals.

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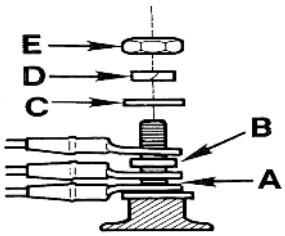
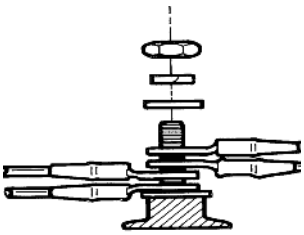
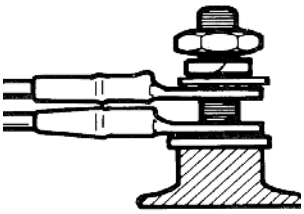
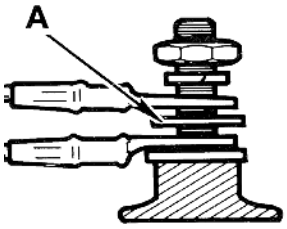
			
Acceptable (A) Two lugs are attached to one side of a stud; the lugs are installed back-to-back. Three lugs are on one side; a space washer is placed between (A) back-to-back lugs and (B) the third lug. Stud hardware is stacked in the following order (C) flat washer, (D) lock washer, (E) nut.	**Acceptable** When four lugs are installed, they will be grouped in pairs, back-to-back, on opposite sides of the studs.	**Unacceptable** The lugs are not back-to-back as required. This places stress on the ring tongue creating cracks and fractures.	**Unacceptable** Improper hardware stackup. (A) A flat washer should be on top of the top lug. No washer required between just two back-to-back lugs.

FIGURE 42-5. Examples of acceptable and unacceptable installation of terminal lugs and associated hardware.

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GUIDELINE 43

CIRCUIT BREAKER INSPECTION

1. Purpose. This guideline gives criteria for inspecting circuit breakers used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 028 00	Protective Devices Installation

3. General inspection guidelines. Perform the following inspection on thermal and arc-fault circuit breakers of single or multi-phase configurations (see figure 43-1):

WARNING

Verify that installed circuit breakers are in the deactivated (pulled) position and external electrical power and batteries is disconnected before proceeding with any of the following instructions or routine maintenance. Failure to do so can result in damage to the equipment and severe injury or death to personnel.

a. Inspect push-button for cracks or deterioration and inspect case for cracks, deterioration, discoloration, and burn marks.

b. Verify that there are no foreign objects present that could cause physical damage or electrical shorts.

c. Check leads of wires for burn marks and physical damage. Also, check for broken wire strands at wire terminations.

d. Examine the circuit breakers for burn marks on the insulating barrier material of three-phase circuit breakers.

e. Inspect circuit breakers and around the circuit breaker for corrosion, discoloration, and hot spots on all metal parts, including bus bars.

f. Ensure that all circuit breaker connecting hardware is tight and secure and verify the correct line and load connection.

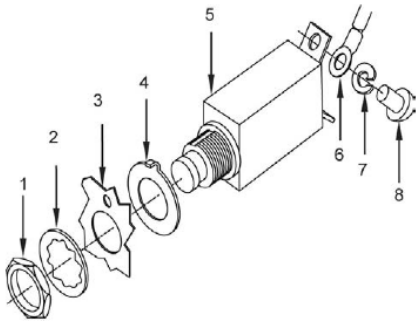
g. Ensure that no more than two terminal lugs or one terminal lug and a bus bar are installed on each circuit breaker terminal.

h. Confirm that the circuit breaker does not have a history of tripping. A tripped breaker maybe faulty, maybe in a faulty circuit, or maybe improperly applied. Fault isolate affected circuit and replace circuit breaker as necessary.

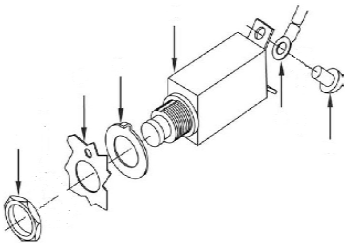
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i. Inspect the circuit breaker boot for splits or deterioration. Do not remove the boot except for inspection.

j. For deactivated (pulled) circuit breakers, ensure that only approved circuit breaker collars/devices are employed (plastic zip ties, string tie, etc., are NOT approved).



Acceptable
1. Nut
2. Lock washer
3. Panel
4. Key washer
5. Circuit breaker
6. Terminal
7. Lock washer
8. Screw



Unacceptable
Missing two lock washers (item 2 and 7)



Unacceptable
Screw too long and is causing damage to circuit breaker housing.

FIGURE 43-1. Examples of circuit breakers in acceptable and unacceptable condition.

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GUIDELINE 44

SOLDER INSPECTION

1. Purpose. This guideline gives criteria for inspecting solder contacts and joints in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 016 00	Soldering
NAVAIR 01-1A-23	Standard Maintenance Practices Miniature/Microminiature (2M) Electronic Assembly Repair
Work Package 020 00	Introduction to Lead-Free Solder

3. General inspection guidelines.

NOTE

The following guideline applies to conventional, tin-lead solder applications. For lead-free solder applications, refer to NAVAIR 01-1A-23, Work Package 020 00.

a. When inspecting solder joints within the EWIS, a good solder joint has the following characteristics (see figure 44-1):

(1) A bright silvery appearance, with smooth fillets and feathered, not sharp, edges.

(2) The entire joint will be covered with a smooth, even coat of solder, and the contour of the joint will be visible.

(3) The insulation is properly cut and free of solder.

b. When inspecting solder joints within the EWIS, any of the following indicates a poor solder joint and are cause for rejection (see figure 44-1):

(1) Dull gray, chalky, or granular appearance (evidence of a cold joint).

(2) Hairline cracks or irregular surfaces (evidence of a disturbed joint).

(3) Grayish, wrinkled appearance (evidence of excessive heat).

(4) Partially exposed joint (evidence of insufficient solder).

(5) Scorched wire insulation or burned connector inserts are present.

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(6) Globules, drips, or tails of solder are present.

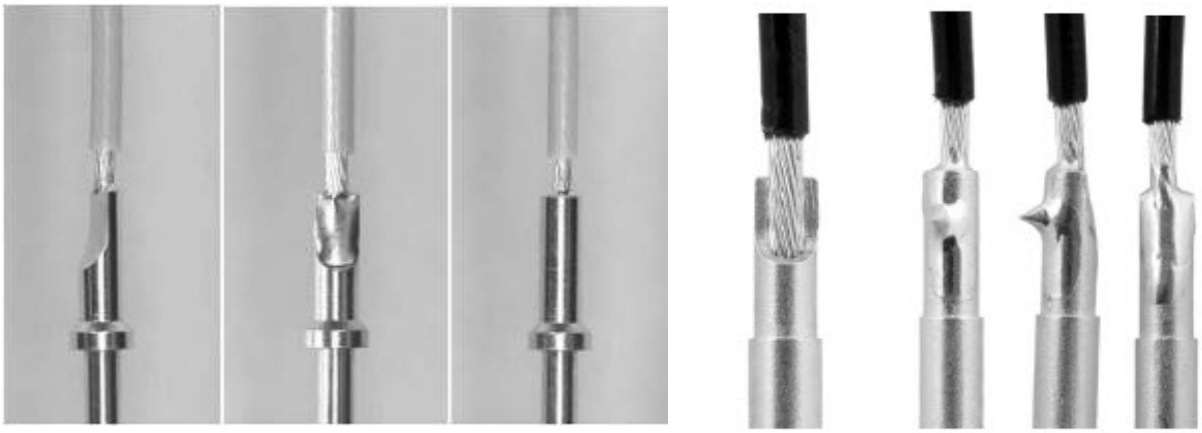
c. When inspecting wire insulation at solder joints, if any of these conditions are present, they are cause for rejection:

(1) Insulation is charred, burned, or blistered (evidence of overheating).

(2) Insulation has frayed or has an uneven appearance.

(3) Solder present on insulation.

(4) Ensure that the wire insulation clearance is equivalent to one wire diameter (including insulation) between the end of the insulation and the outside edge of the terminal or solder contact barrel. Exposed conductor should not exceed two wire diameters (see figure 44-1).



Acceptable
Good solder quantity with no spillage. Wire insulation clearance is equivalent to one wire diameter (including insulation) between the end of the insulation and the outside edge of the terminal or solder contact barrel. The exposed conductor does not exceed two wire diameters

Unacceptable
Insufficient solder quantity with uncovered wire strands.

Unacceptable
Excess solder flow with protrusions.

FIGURE 44-1. Examples of acceptable and unacceptable soldering.

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GUIDELINE 45

GROMMET INSPECTION

1. Purpose. This guideline provides the criteria for inspecting grommets on airframes and structures used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

MS35489	Grommets, Synthetic and Silicone Rubber, Hot-Oil and Coolant Resistant
SAE AS21919	Clamp, Loop Type, Cushioned Support
NASM21266	Grommet, Plastic, Edging
NASM22529/1	Grommet, Composite, Edging
NASM22529/2	Grommet, Cushion Composite, Edging
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 010 00	Harness Installation

(Copies of NASM22529/1 are available at www.aia-aerospace.org.)

NOTE

NASM22529/1 flat edge grommet type does not incorporate a cushion and offers no wire chafe protection. NASM2259/1 is not approved EWIS protection applications.

3. General inspection guidelines. When inspecting donut (MS35489) or caterpillar grommets (NASM22529/2, NASM21266), the following procedure should be followed:

CAUTION

Grommets are designed for incidental contact only and should not be used as a primary or secondary means of chafe protection.

a. Ensure that grommets are used when the physical separation between wiring and equipment or structure edges where a minimum of 3/8 inch clearance cannot be maintained. A minimum 3/8-inch clearance is acceptable for lightening hole applications where a grommet is installed (see figures 45-1 and 45-4). For holes through the structure (i.e., lightening holes, feed-through holes, or cutouts), the wire harness/cable is required to be supported with a primary support cushioned clamp adjacent to the hole in the structure so that a minimum clearance (grommet to closest wire) is no less than 1/8-inch.

b. Verify that the grommet is not damaged. If damaged, replace as necessary.

c. Verify that grommet installation is secured to an airframe and is not a potential Foreign Object Damage (FOD) hazard.

d. Caterpillar grommets (NASM22529/2) are limited to 2″ diameter holes or larger, with flat edge (not beveled or curved) applications. Ensure that the grommet cushion layer is secured,

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does not separate, or is not missing; (see figure 45-2). The use of donut grommet (MS35489) is intended for 2″ and smaller diameter holes; (see figure 45-1).

CAUTION

NASM21266 grommets require sealant/adhesive for securing to structure. Failure to secure this grommet type to structure may result in FOD and/or equipment damage.

e. Ensure that NASM21266 grommet is installed with sealant/adhesive. Failure to secure this grommet to the edges of the airframe hole or bulkhead may result in FOD (see figure 45-3).

f. Caterpillar grommets (NASM21266 and NASM22529/2) are cut between the castles at 90° or 45° angle depending on grommet configuration. This is to assure the absence of deformation of adjacent castles close to the cut-off, thus leaving no sharp, protruding castle edges (see figures 45-2 and 45-3).

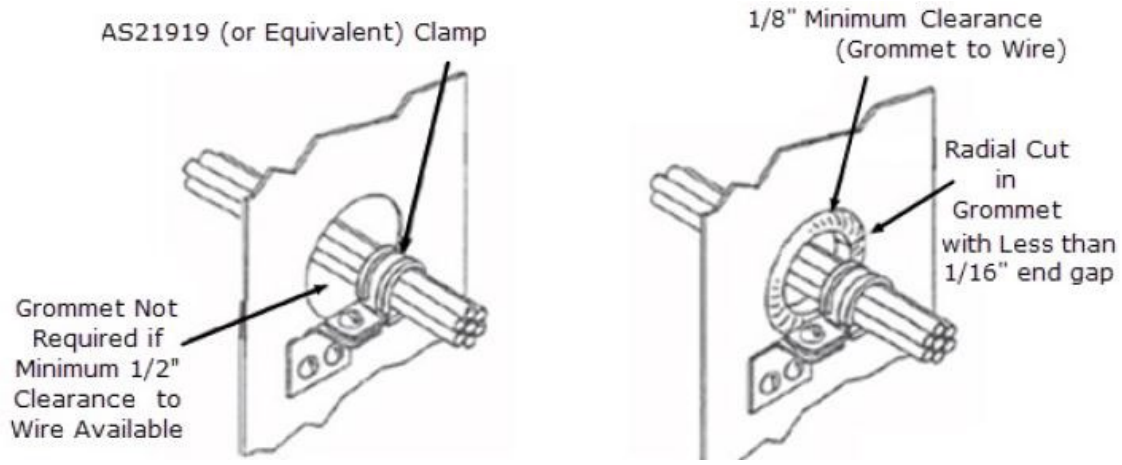


FIGURE 45-1. Acceptable donut (MS35489) grommet installation.

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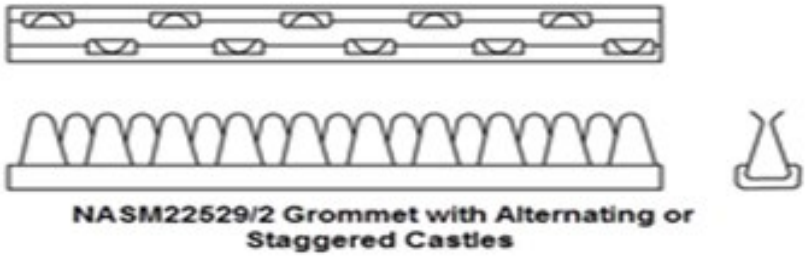


FIGURE 45-2. Typical caterpillar grommet (NASM22529/2) configuration.

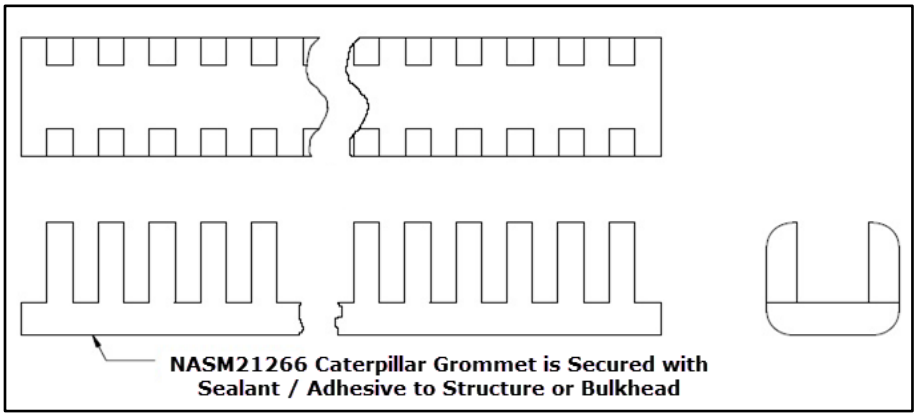
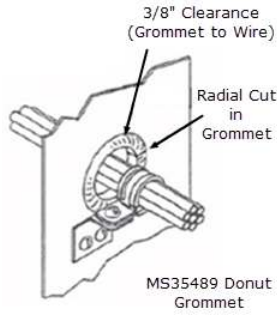


FIGURE 45-3. Caterpillar grommet (NASM21266) configuration.

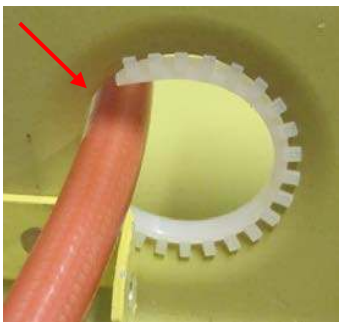
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Acceptable
MS35489 donut grommet securely installed, 1/8 inch gap to the wire, radial cut at 90 degrees.



Acceptable
NASM22529/2 grommet maintains the minimum 3/8 inch clearance to the wire. The grommet is securely fit to lightening hole in the airframe structure. Note that the airframe hole is flat, not beveled or flanged.



Unacceptable
The grommet does not cover the full circumference of the lightning hole. Minimum 3/8 inch clearance grommet to cable not maintained. NASM21266 is not secured with adhesive/sealant (FOD potential).



Unacceptable
Cable is pinched in MS35489 donut grommet gap. Grommet gap over .30 inch wide. Minimum 1/8 inch clearance grommet to wire not maintained.

FIGURE 45-4. Acceptable and unacceptable grommet installation.

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GUIDELINE 46

LARGE GAUGE TERMINAL LUG INSPECTION

1. Purpose. This guideline gives criteria for inspecting large gauge terminal lugs (SAE AS7928 and SAE AS20659) used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 013 00	Contacts and Terminals
SAE AS5259	Crimping Tool and Accessories, Electrical, Size 8 to 700 KCMIL, General Purpose Use, Die and Dieless
SAE AS7928	Terminals, Lug: Splices, Conductor: Crimp Style, Copper, General Specification For
SAE AS20659	Lugs, Terminal, Crimp Style. Copper, Uninsulated, Ring Tongue, Type 1, Class 1, for 175°C or 260°C Total Conductor Temperature

3. General inspection guidelines. Improper crimping techniques may degrade the mechanical (tensile strength) and electrical performance of large gauge terminal lugs. Large terminal lugs are regarded as those that can accommodate wire/cable bigger than 12 American Wire Gauge (AWG). To verify that large terminal lugs are crimped properly, they should be examined as follows:

WARNING

Ensure that terminal lug is longitudinally centered in the crimp die or against the die stop (if available). If crimp indent is located at either end, it may result in a joint with high electrical resistance and low mechanical pull strength. Overheating and fire may result.

a. Verify that the crimp indent is positioned in the center of the terminal.

NOTE

Large gauge terminal lug crimp dies meeting SAE AS5259 incorporate the marking of the wire size and terminal lug type, such that after crimping, it leaves the mark embossed on the terminal lug. Not all in-service-approved and qualified crimp tools and dies have this feature.

b. Verify that the crimp die wire gauge number (i.e., “2/0” is for 2/0 wire) and type (i.e., “CU” means Copper terminal lug) is embossed on the completed crimp. These markings need to align with the “AIR AWG” rating of the terminal lug wire gauge (see figure 46-1). The 2/0 gauge should also match the wire size/gauge the terminal lug was crimped on. The size or gauge of the wire is identified in the wire part number printed (typically) every 12 inches.

c. Check that the wire insulation is inserted in the support area of the terminal barrel (see figure 46-2).

d. Confirm that the stripped wire conductor is pushed fully into the lug.

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e. Verify that the stripped wire ends are flush to terminal stop with not more than 1/32 inch protruding (see figure 46-2).

f. Verify that no wire insulation is present in the wire crimp barrel.

g. Confirm that conductor insulation has been inserted into the insulation support area of the insulated terminal lug barrel (see figure 46-2).

h. Verify that the amount of insulation stripped from the wire is not excessive, such that conductors extend into the hardware mounting area (see figure 46-2).

NOTE

Not all terminal lug configurations employ the "AIR Gauge" marking convention. This marking signifies that the wire employed is for aviation applications and employs smaller wire strand with a dedicated *crimp setting. If the "AIR" and "AWG" values differ on the terminal lug marking, select the "AIR" as they align with aerospace applications. Refer to the terminal lug specification for requirements or the Cognizant Engineering Authority.*

i. Verify that the correct "AIR Gauge" rating (if so marked) on the terminal lug matches the appropriate wire crimp indent and type (i.e., Copper or Aluminum) (see detail B of figure 46-1)



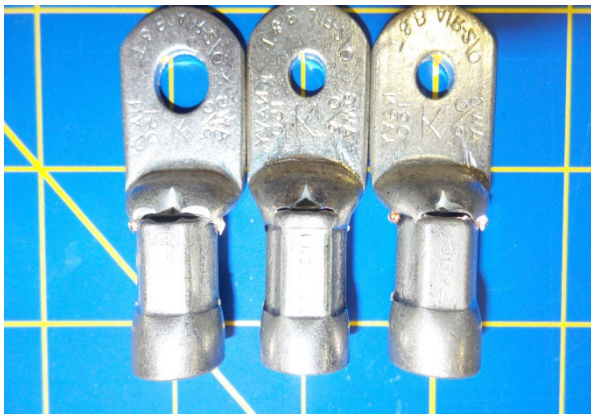
(A) Die index number embossment on completed crimp. (B) AIR Gauge marking.

FIGURE 46-1. Terminal lug markings.

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Acceptable
Lug has been properly positioned in the tool. The crimp mark is centered on the crimp barrel. Stripped wire ends are flush to terminal stop (not more than 1/32 inch protruding). No wire insulation is inside of the crimp barrel.



Unacceptable
Lug improperly positioned in tool resulted in crimp indent not centered on barrel.

FIGURE 46-2. Examples of acceptable and unacceptable crimping of terminal lugs.

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GUIDELINE 47

SOLDER CONTACT (TWINAX/SOLDER TACT) INSPECTION

1. Purpose. This guideline gives criteria for inspecting solder contact terminations (e.g., SAE AS39029/73, /74) used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 013 00	Contacts and Terminals.
SAE AS39029/73	Contacts, Electrical Connectors, Socket, Solder, Removable, Shielded, Size 12 (for MIL-DTL-83723 Series III, MIL-DTL-26482, Series 2, MIL-DTL-83733, and MIL-DTL-83527 Connectors)
SAE AS39029/74	Contacts, Electrical Connectors, Pin, Solder, Removable, Shielded, Size 12 (for MIL-DTL-83723 Series III, MIL-DTL-26482, Series 2, MIL-DTL-83733, and MIL-DTL-83527 Connectors)

3. General inspection guidelines. Solder contact configurations vary, but the typical design incorporates features resembling those on figure 47-1. When inspecting solder contacts, the minimum and maximum solder flow should be verified (see figure 47-2):

a. For acceptable minimum solder flow, the following should be present:

(1) Both solder preforms have melted and flowed.

(2) Slight flare of sleeve ends is acceptable.

b. For acceptable maximum solder flow, the following should be present:

(1) Both solder preforms have melted and flowed.

(2) Joint areas are visible despite browning of the sleeve.

c. When there is insufficient heat during the soldering process, unacceptable solder flow results. In this situation, the following can be observed when inspecting the solder flow:

(1) Solder preforms not fully melted; original form is still visible.

(2) Tubing has not fully recovered.

d. When there is too much heat used during the soldering process, unacceptable solder flow results. In this situation, the following can be observed when inspecting the solder flow:

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(1) Joint areas are not visible because of severe darkening of the sleeves.

(2) Wire insulation damaged outside of the sleeve.

(3) Solder “wicked” away from termination areas; no fillets visible.

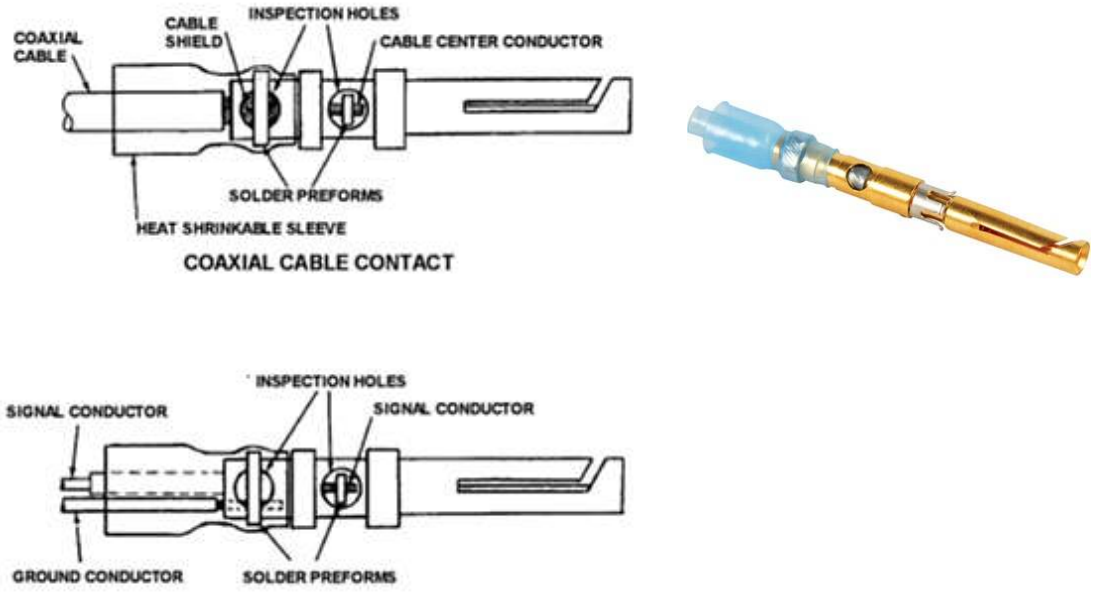
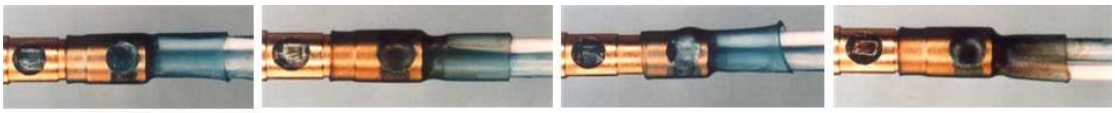


FIGURE 47-1. Examples of solder contacts SAE AS39029/73 and /74 typical configuration.

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Acceptable Minimum Solder Flow	**Acceptable** Maximum Solder Flow	**Unacceptable** Insufficient heat. Solder band not melted.	**Unacceptable** Overheated

FIGURE 47-2. Examples of solder contacts that have acceptable and unacceptable solder flow.

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GUIDELINE 48

CONDUIT INSPECTION

1. Purpose. This guideline gives criteria for inspecting conduit for proper installation and damage (e.g., SAE AS6136, SAE AS81914) used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 010 00	Harness Installation.
SAE AS6136	Conduit, Electrical, Flexible, Shielded, Aluminum Alloy for Aircraft Installations
SAE AS81914	Tubing, Plastic, Flexible, Convoluted, Conduit, General Specification for

3. General inspection guidelines. When inspecting EWIS conduit (e.g., SAE AS6136, SAE AS81914), the following should be verified (see figure 48-1):

CAUTION

No ties or splices are permitted inside a conduit. Collateral damage and insulation chafing of wires can occur due to the relative motion of unsupported wires in the conduit.

NOTE

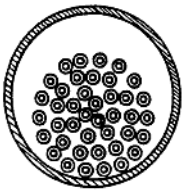
Wiring is not to be tight/stretched, nor to have excessive slack inside conduits.

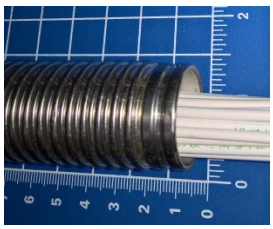
a. Ensure that the wire bundle diameter fill does not exceed 80% of the internal diameter of the conduit (see figure 48-1).

b. Verify that the conduit is not deformed, dented, or compressed beyond 20 percent of the outer tube diameter (see figure 48-1). Replace damaged conduit with the same type, size, and configuration.

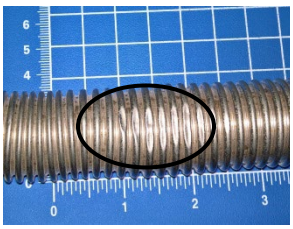
c. Verify that the conduit has no open cuts, holes, perforations, or cracks exposing the wires within (see figure 48-1). Replace damaged conduit with the same type, size, and configuration.

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Acceptable
Less than 80 percent filled with wires. No wire ties inside conduit. No deformation, cuts, or dents in conduit.



Acceptable
Dent/deformation is rounded and is less than 20 percent of the diameter of the conduit.



Unacceptable
Dent/deformation greater than 20 percent of diameter.



Unacceptable
Dent and cut are exposing wires. Sharp edges damaging wires.

FIGURE 48-1. Examples of EWIS conduit that have acceptable and unacceptable installation and/or damage.

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GUIDELINE 49

TERMINAL JUNCTION INSPECTION

1. Purpose. This guideline gives criteria for inspecting terminal junctions and terminal junction blocks used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

MS27488	Plug, End Seal, Electrical Connector
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 027 00	Terminal Junction System

3. General inspection guidelines. Figures 49-1 through 49-3 show the different parts that make up typical electrical application terminal junctions and brackets. When inspecting terminal junctions used in the EWIS, the following items should be examined:

a. Ensure that the terminal junction(s) are secured in the bracket/rail assembly (see figure 49-1).

b. Inspect the terminal junction insulator insert for damage and to ensure that there are no bent or splayed contacts. Contacts should be properly seated and locked (not protruding) from the terminal junction (see figure 49-1 and 49-3).

NOTE

Sealing plugs (e.g., MS27488) inserted in terminal junctions are installed without the unwired contacts. The large end (head) of the sealing plug is inserted first into the terminal junction.

c. Ensure that all un-wired contact cavities on the grommet surface are populated with environmental sealing plugs (MS27488) (see figure 49-3).

d. Ensure that all installed sealing plugs are installed with the large end (head) inside the terminal junction grommet (see figure 49-3).

e. When inspecting terminal junction grommet seals, verify that there are no chips, gouges, or other damage in or extending from chamfered area to the base of the chamfer; (see figures 49-4 and 49-5).

f. Verify that wires entering terminal junctions/blocks' grommet seals do not distort the seal so as to create a gap between the wire and the seal. All wires should extend straight out from the terminal junction, as shown so that a gap is not created (see figure 49-6).

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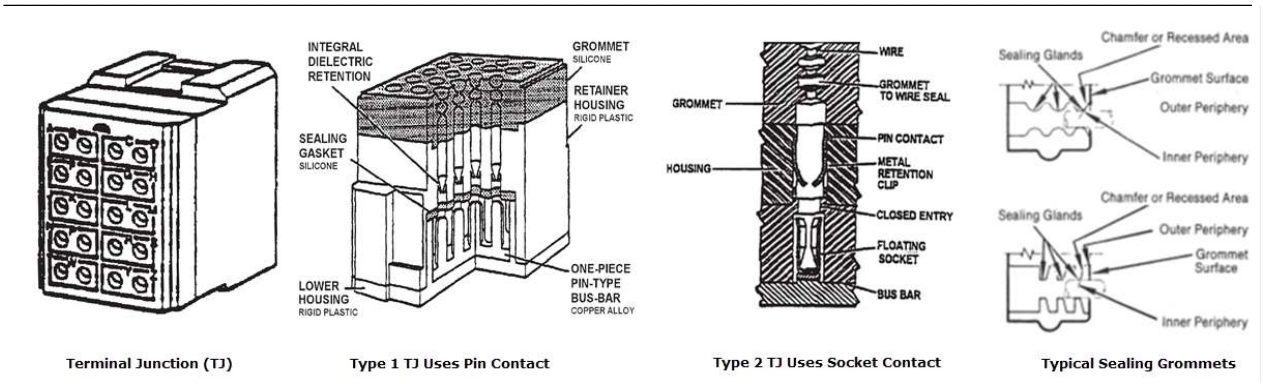


FIGURE 49-1. Illustration of typical terminal junction components.

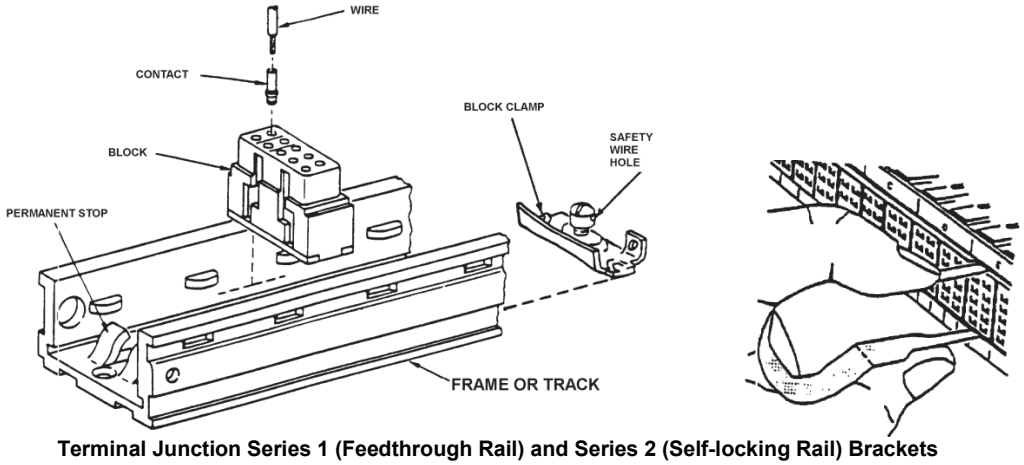


FIGURE 49-2. Illustration of typical terminal junction bracket/rail components.

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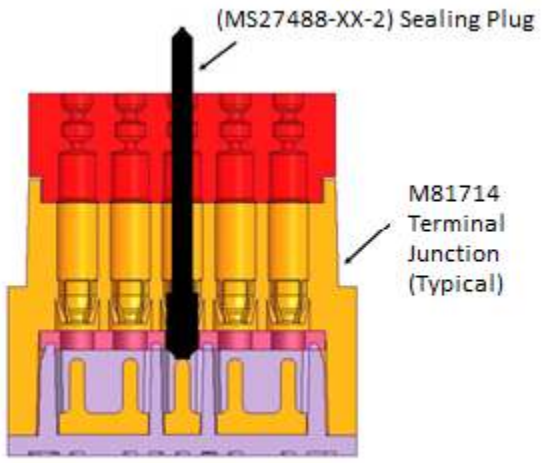
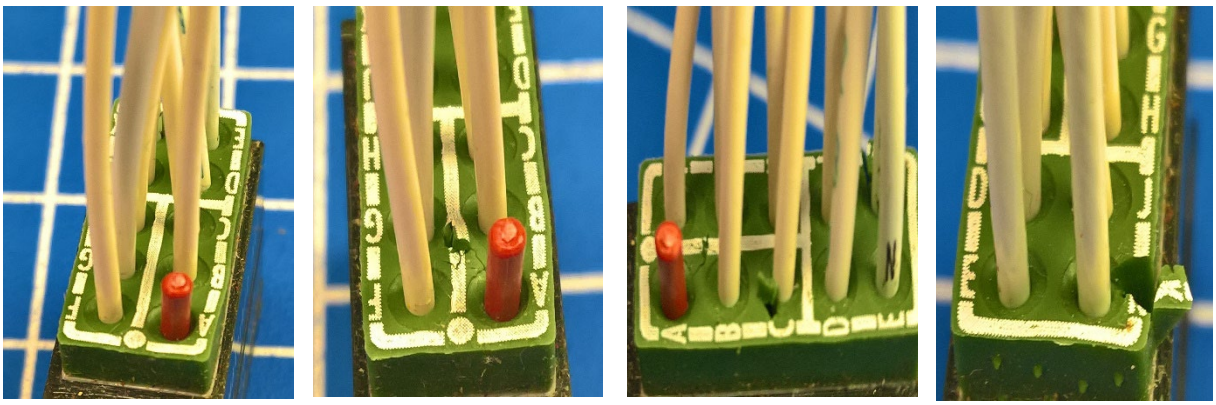


FIGURE 49-3. Sealing plug and wired contact installation for terminal junction sealing.



Acceptable
The connector grommet has no surface gouges that extend to the wire or below the bottom of the chamfer (first sealing gland).

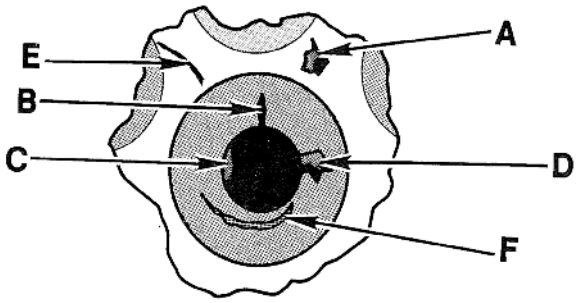
Acceptable
Although a gouge appears on the grommet surface and in the chamfer area, it does not extend to the base of the chamfer.

Unacceptable
The gouge extends through the chamfered area and comes in contact with the wire. The sealing ability of the grommet is reduced.

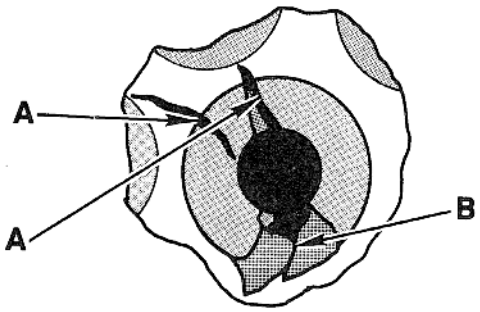
Unacceptable
The large deep gouge extends below the bottom of the chamfer, and destroys the sealing capability of the grommet.

FIGURE 49-4. Examples of connector sealing grommets in acceptable and unacceptable condition.

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Acceptable
Any ONE imperfection: (A) Chip, nick, or gouge, (B) Split crack through the sealing gland but not extending out of the recessed or chamfered area, (C) Mold Flash, (D) Chip, nick, or gouge through the sealing gland but not out of the recessed or chamfered area, (E) Split or crack, (F) Chip, nick, or gouge not through the sealing gland.



Unacceptable
(A) Split or crack extending out of the recessed or chamfered area, (B) Chip, nick, or gouge extending out of the recessed or chamfered area. May be repairable using standard repair procedures.

FIGURE 49-5. Examples of sealing grommet with different levels of damage.

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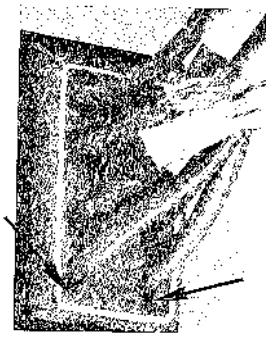
			
Acceptable Wires are entering connectors and terminal blocks; grommet seals do not distort the seal. All wires extend straight out from the connector.	**Acceptable** Normal flexing of wires may exert pressure against the grommet (arrows), but a slight distortion of the seal is not enough to create a gap between the seal and wire.	**Unacceptable** Although the wires are routed at a proper angle, one wire has been pulled back sharply during spot tie installation. A very large gap (arrow) is the result.	**Unacceptable** Wires do not come straight out of the grommet and have distorted the seal. The result is large gaps (arrows). Contaminants and moisture can enter the terminal block.

FIGURE 49-6. Examples of acceptable and unacceptable wire installation in terminal junction blocks.

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GUIDELINE 50

SAE AS34591 CONNECTOR MATED INSPECTION

1. Purpose. This guideline gives criteria for inspecting part number MS3459 connectors. These connectors meet SAE AS34591 specification requirements and are used in the EWIS.

2. Applicable documents. The documents listed below are those applicable to this guideline.

SAE AS34591	Connector, Plug, Electrical, Self-Locking, Coupling Nut, Rear Release, Crimp Contact, AN Type
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 020 00	Military Standard Electrical Connectors

3. General inspection guidelines. Figures 50-1 and 50-2 show typical part number MS3459 Class K, L, or W environmental resisting electrical connectors. The part number MS3459 series connectors are not lock-wired. The part number MS3459 connector has a self-locking coupling device, which is built into the coupling ring. Some manufacturers' coupling ring designs use an internal locking device which is not inspectable (see figure 50-1). Other manufacturer's designs include an orange color band indicator identifying an externally inspectable locking mechanism (see figure 50-2). When the plug is mated, pressure is applied, which ratchets the device against the connector shell. When inspecting part number MS3459 connectors with an orange band indicator, the following items should be examined:

a. Ensure that the part number MS3459 plug is fully mated by the locking indicator being in a position just beyond the orange color band (see figure 50-3).

CAUTION

If the coupling nut is exposed to excessive torque and the locking indicator is forced past the illustrated position, damage to the retaining ring may result.

b. Inspect the part number MS3459 plug is fully mated by ensuring no visible threads on the mating receptacle (see figure 50-4).

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FIGURE 50-1. Illustration of typical part number MS3459 plug with internal locking mechanism.

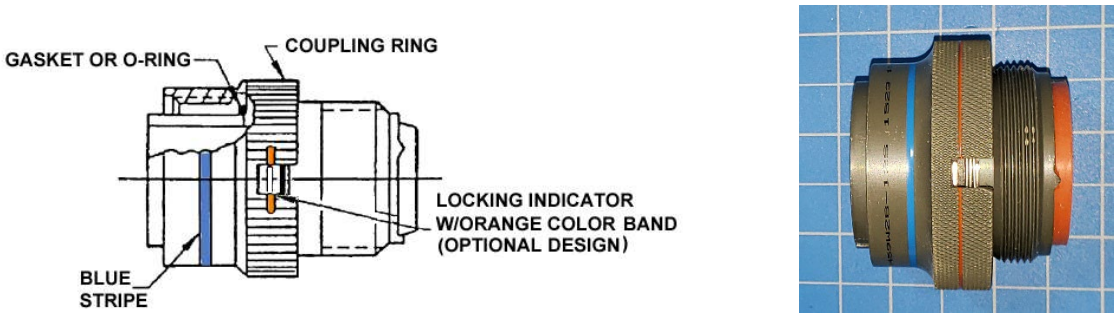


FIGURE 50-2. Illustration of typical part number MS3459 plug with an orange color band and external inspectable locking mechanism.

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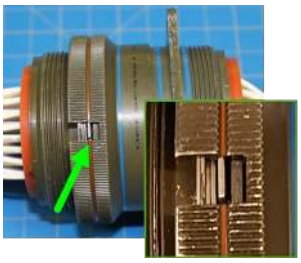
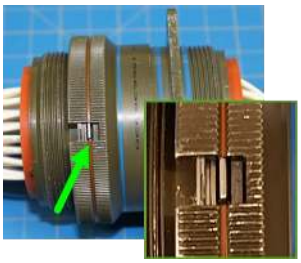
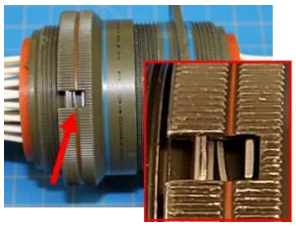
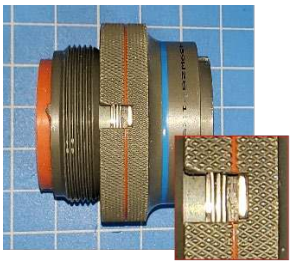
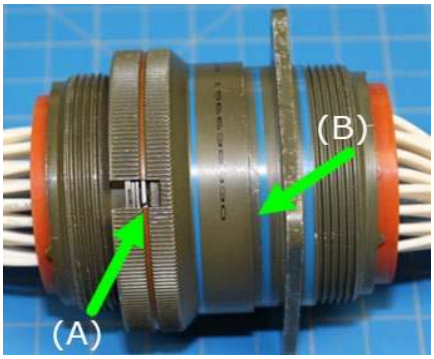
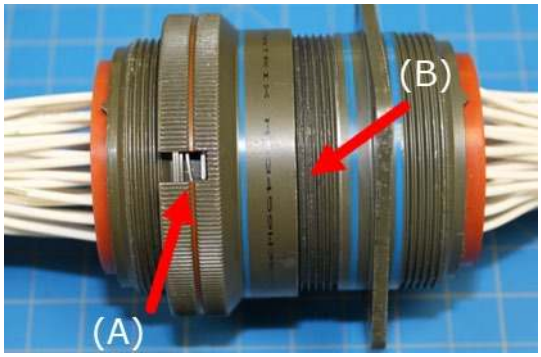
			
Acceptable The locking indicator is just beyond the orange band indicator. Full coupling ring engagement.	**Acceptable** The locking indicator is even with the orange band minimum coupling ring engagement.	**Unacceptable** The locking indicator is not aligned or to the left (beyond) of the orange band.	**Unacceptable** The locking indicator is not aligned or to the left (beyond) of the orange band. The indicator is to the far right with plug de-mated.

FIGURE 50-3. Examples of part number MS3459 connector with the orange color band and external inspectable locking mechanism.



Acceptable
Both required fully mated connector conditions are met.

(A) The locking indicator is beyond the orange band.
(B) No exposed threads beyond the coupling ring/nut.



Unacceptable
Neither of the two required mated connector conditions are met.

(A) The locking indicator is not aligned or beyond the orange band.
(B) Several exposed threads beyond the coupling ring/nut.

FIGURE 50-4. Examples of part number MS3459 connector with orange band and thread engagement.

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GUIDELINE 51

CABLE LACING FASTENER (CLF) INSPECTION

1. Purpose. This guideline provides the criteria for inspecting part number M32554 Cable Lacing Fasteners. These Cable Lacing Fasteners (CLF) are employed as secondary support devices for wire bundles in the EWIS and meet MIL-DTL-32554 specification requirements. As an EWIS secondary support device, the CLF is deemed a suitable substitute for the hand-installed and tied Lacing Tape (e.g., A-A-52084).

2. Applicable documents. The documents listed below are those applicable to this guideline.

A-A-52084	Tape, Lacing and Tying, Aramid
MIL-DTL-32554	Straps, Tie-Down, Adjustable, Non-Metallic, Cable Bundling, General Specification for
MIL-DTL-32555/1	Tool, Installation, For Adjustable Tie -Down Straps
NAVAIR 01-1A-505-1 Work Package 010 00	Aircraft Electric and Electronic Wiring Harness Installation

3. General inspection guidelines. Figure 51-1 shows typical part number M32554 Cable Lacing Fasteners (CLF) and components. Part number M32554 CLF length variations allow for multiple wraps, one to three, to be applied around different size bundles prior to termination (see figure 51-2). Multiple wraps are used for increased bundle strength, to provide support in vibration environments, and to provide extreme resistance to lateral or radial motion. Installation of CLF is limited to wrapping around bundle no more than three times. CLF is available in various colors of lacing tape. The part number M32554 CLF is installed using the designated installation tool part number M32555/01-01. This tool tensions the CLF around the bundle and cuts off the excess strap. When inspecting part number M32554 Cable Lacing Fastener installations, the following items should be examined:

a. Ensure that the part number M32554 CLF is installed so that it is snug around the wire bundle and does not easily able to slide laterally or rotate around the bundle (see figure 51-3).

b. Ensure that the strap is not twisted.

c. The installed CLF fastener head should be free of cracks or signs of damage.

d. No more than three wraps around the bundle.

e. CLF positioned so that the fastener head is exposed for inspection capability.

f. Locking pin should have angled feature pointing in the opposite direction as the cut end of lace; confirmation that the fastener head and strap is locked.

g. Cable Lacing Tape should be cut cleanly; about 0.5 in. +/-0.25 in. from fastener head.

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If cut strap end is not cleanly cut; tool blade replacement may be necessary.

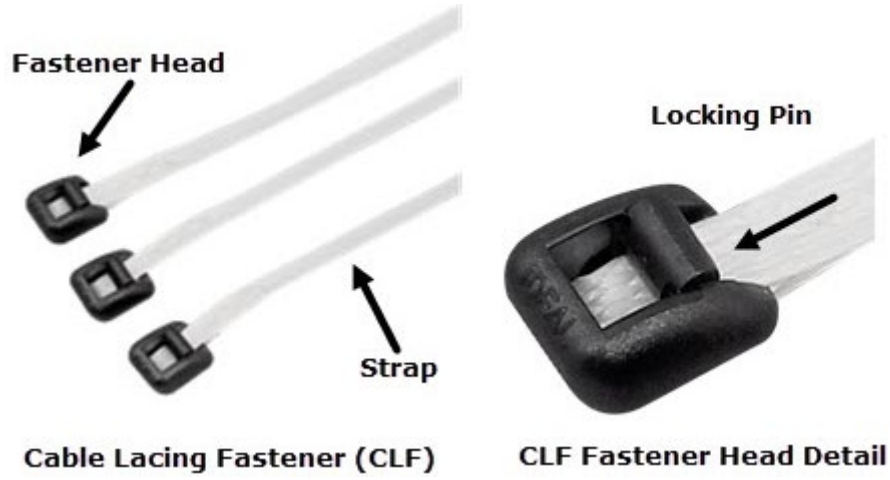


FIGURE 51-1. Illustration of typical part number M32554 CLF.



FIGURE 51-2. Illustration of typical part number M32554 CLF installation configurations.

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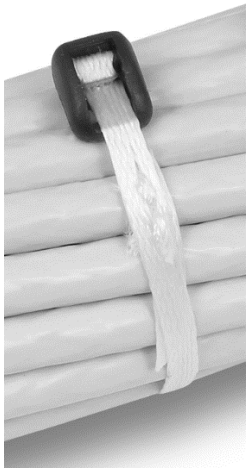
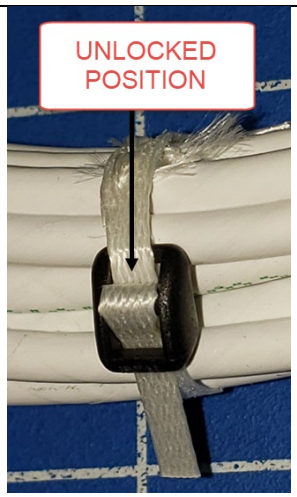
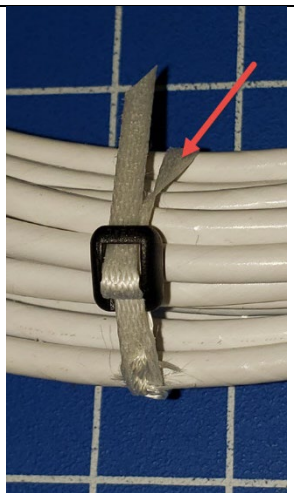
			
Acceptable The part number M32554 CLF inspection should show: - Strap is not twisted - Head is exposed and not cracked or broken - Tight fit on the wire bundle - Strap end cut cleanly 3/4 to 1/4 inch long - Strap is not twisted - Maximum 3 wraps	**Acceptable** The part number M32554 CLF The strap is locked with angled pin orientation against buckle stop. The strap is taught and tight-fitting, securing the wire bundle. The fastener head is not cracked or broken.	**Unacceptable** The part number M32554 CLF is not locked, and the strap is loose. Angled Pin orientation shows unlocked.	**Unacceptable** The part number M32554 CLF is not locked, and the strap is twisted.

FIGURE 51-3. Examples of part number M32554 cable lacing fastener with securely locked installation.

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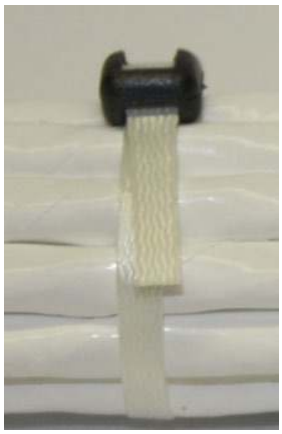
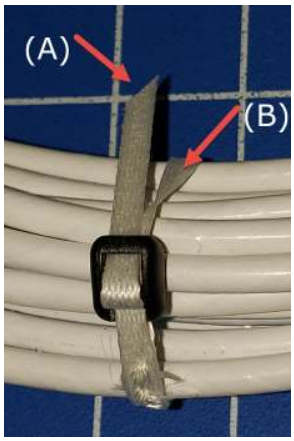
			
Acceptable The part number M32554 CLF fits tightly on wire bundle, locked on, and with a strap end cut cleanly 3/4 to 1/4 inch in length.	**Acceptable** The part number M32554 CLF-fits tightly on wire bundle, locked on, and with a strap end cut cleanly 3/4 to 1/4 inch long.	**Unacceptable** The part number M32554 CLF strap is loosely fitting on the bundle, twisted, and not locked.	**Unacceptable** The part number M32554 CLF strap is not locked and shows: (A) Strap over 3/4 inch long and not square. (B) Strap is twisted.

FIGURE 51-4. Examples of the part number M32554 cable lacing fastener with correct and incorrect fit and strap end length.

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GUIDELINE 52

SPRING BAND SHIELD TERMINATION INSPECTION

1. Purpose. This guideline provides the criteria for inspecting part number M32628/01-XX Spring Band Shield Terminations. The part number M32628/01 Spring Bands are employed as metallic overbraid shield terminations for EMI/RFI shielded wire bundles in EWIS. The spring bands meet MIL-DTL-32628 and MIL-DTL-32628/1 specification requirements.

2. Applicable documents. The documents listed below are those applicable to this guideline.

MIL-DTL-32628	Backshell, Electrical Connector, Spring Band, Accessory General Specification for
MIL-DTL-32628/1	Backshell, Electrical Connector, Tri-Dent Spring Band
SAE AS85049/82	Connector Accessories, Electrical, Backshell, Straight, Self-Locking, Shield Band Termination (RFI/EMI), Shrink Sleeve Accommodation, Category 3B (for MIL-C-5015 Crimp, MIL-C-26482 Series 2, MIL-C-81703 Series 3, and MIL-C-83723 Series III Connectors)
SAE AS85049/83	Connector Accessories, Electrical, Backshell, 45 Degrees, Self-Locking, Shield Band Termination (RFI/EMI), Shrink Sleeve Accommodation, Category 3B (for MIL-C-5015 Crimp, MIL-C-26482 Series 2, MIL-C-81703 Series 3, and MIL-C-83723 Series III Connectors)
SAE AS85049/84	Connector Accessories, Electrical, Backshell, 90 Degrees, Self-Locking, Shield Band Termination (RFI/EMI), Shrink Sleeve Accommodation, Category 3B (for MIL-C-5015 Crimp, MIL-C-26482 Series 2, MIL-C-81703 Series 3, and MIL-C-83723 Series III Connectors)
SAE AS85049/85	Connector Accessories, Electrical, Backshell, Straight, Self-Locking, Shield Band Termination (RFI/EMI), Shrink Boot Accommodation, Category 3B (for MIL-C-38999 Series I and II Connectors)
SAE AS85049/86	Connector Accessories, Electrical, Backshell, 45 Degrees, Self-Locking, Shield Band Termination (RFI/EMI), Shrink Sleeve Accommodation, Category 3B (for MIL-C-38999 Series I and II Connectors)
SAE AS85049/87	Connector Accessories, Electrical, Backshell, 90 Degrees, Self-Locking, Shield Band Termination (RFI/EMI), Shrink Sleeve Accommodation, Category 3B (for MIL-C-38999 Series I and II Connectors)
SAE AS85049/88	Connector Accessories, Electrical, Backshell, Straight, Self-Locking, Shield Band Termination, (RFI/EMI) Shrink Boot Accommodation, Category 3B (for MIL-C-38999 Series III and IV Connectors)

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SAE AS85049/89	Connector Accessories, Electrical Backshell, 45 Degrees, Self-Locking, Shield Band Termination, (RFI/EMI) Shrink Sleeve Accommodation, Category 3B (for MIL-C-38999 Series III and IV Connectors)
SAE AS85049/90	Connector Accessories, Electrical, Backshell, 90 Degrees, Self-Locking, Shield Band, Termination, (RFI/EMI) Shrink Sleeve Accommodation, Category 3B (for MIL-C-38999 Series III and IV Connectors)
SAE AS85049/93	Connector Accessories, Electrical, Termination, Shield Split Support Ring, Composite, Nonenvironmental, Straight, Category 7
NAVAIR 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 011 01	Protected Wire Harness Fabrication and Repair

3. General inspection guidelines. Figure 52-1 shows a typical part number M32628/01 Spring Band and features. The part number M32628/01 Spring Band is available in four diameter sizes for various size wire bundles/backshell cable entry sizes. As an EWIS shield termination device, the reusable part number M32628/01 spring band is for use with or on part number M85049/82 through /90 connector backshells for EMI/RFI applications. The part number M32628/01 spring band is not approved for use on or with the part number M85049/93 shield splice due to the splice geometry limitations. The part number M32628/01 spring band is installed and removed by hand, requiring no special tools. When inspecting part number M32628/01 spring band installations, the following items should be examined:

CAUTION

Proper material, marking, and installation are critical to the performance of part number M32628/01 spring bands. Failure to ensure that the proper part number/size and installation may lead to premature failure or compromised EMI shielding performance.

a. Ensure that the “tri-dent” feature and the part number (“M32628/01-1X”) are exposed. This guidance is not applicable to the part number M32628/01-10, which the smallest size of these spring band shield terminations because this part might only show a partial part number, (see figure 52-1).

b. Ensure that a close, tight, secure fit around the circumference of the EMI shield as installed on the banding platform of the part number M85049/82 through /90 backshell accessory (see figure 52-2).

c. Spring band shows no gaps, bends, or kinks.

d. The part number M32628/01-1X Spring band is installed on connector accessories (part number M85049/82 through /90). The part number M32628/01-1X spring band is not for shield splice part number M85049/93 applications.

e. If any items (4.a through 4.d above) are not met, replace with a new spring band per the application requirement.

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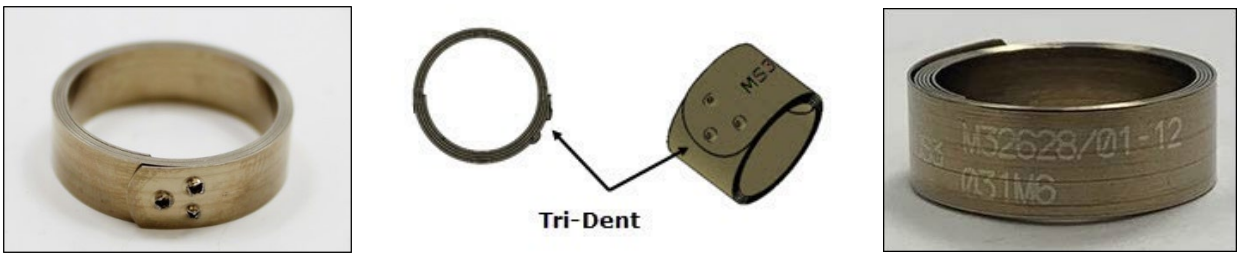


FIGURE 52-1. Illustration of typical part number M32628/01 shield termination spring band.



FIGURE 52-2. Illustration of typical part number M32628/01 shield termination spring band installation.

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Acceptable The part number M32628/01 spring band inspection should show: - Band fits tightly on the backshell. - “tri-dent” feature is exposed. - “M32628/01-1X” part number is exposed. - Band is installed on part number M85049/82 through /90 backshells. - No bends, gaps, or kinks.	**Acceptable** The spring band is shown with the “tri-dent” feature exposed and the correct part number is legible.	**Unacceptable** The spring band has no “tri-dent” feature exposed. Incorrect band configuration installed.	**Unacceptable** The part number M32628/01 spring band has no “M32628/01-1X” part number exposed. Incorrect band configuration installed.

FIGURE 52-3. Examples of part number M32628/01 spring band of correct configuration.

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Acceptable The part number M32628/01 spring band inspection should show: - Band fits tightly on the backshell. - “tri-dent” feature is exposed. - “M32628/01-1X” part number is exposed. - Band is installed on part number M85049/82 through /90 backshells. - No bends, gaps, or kinks.	**Unacceptable** Spring band of unidentified configuration installed on part number part number M85049/88-X backshell. Spring band type is wider and incompatible with the backshell it is installed on.	**Unacceptable** The spring band installed inside out and “tri-dent” feature is not exposed and visible.	**Unacceptable** The spring band size too large for this backshell. It is loose, not a tight fit, and installed inside out. Spring band is not installed in the banding platform of the backshell but on top of the aft edge, not compressing the shield material.

FIGURE 52-4. Examples of part number M32628/01spring band with correct and incorrect fit and installation.

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GUIDELINE 53

EWIS DAMAGE REPORTING AND EWIS DAMAGE INVESTIGATION

1. Purpose. This guideline provides criteria for EWIS damage reporting and EWIS damage investigations.

2. Applicable documents. The documents listed below are those applicable to this guideline.

NA 01-1A-505-1	Aircraft Electric and Electronic Wiring
Work Package 004 01	Aircraft Wiring System Inspection

WARNING

For EWIS damage that is beyond that of standard repair practices and/or approved routing configurations, the CEA must be contacted for determining the repair disposition. Failure to comply with reporting requirements may result in further damage to aircraft components and severe injury or death to personnel.

3. General inspection guidelines. The following procedures should be followed when reporting damage to the EWIS or determining if an Engineering Investigation is needed.

a. Based on severity, some discrepancies may need to be corrected prior to operational use of the aircraft. An EWIS Inspection procedure should be performed prior to repairing EWIS discrepancies. The EWIS Damage Investigation (paragraph d below) will aid in determining if Cognizant Engineering Authority (CEA) (e.g., Fleet Support Team (FST) for Navy/Marine Corps) guidance is required.

b. EWIS damage reporting. If determined that a CEA repair disposition is required, report incident to platform CEA as an HMR/EI via JDRS following the applicable service reporting requirements. Use the EWIS damage investigation process (see the next paragraph) to aid in determining if CEA repair guidance is required.

NOTE

The EWIS damage investigation is to be performed using the findings from the EWIS inspection (see guideline 1). Additional inspections may be required to accurately complete this investigation.

c. EWIS damage investigation. The purpose of the EWIS damage investigation is to aid in determining if the damage is repairable using standard repair practices, or if CEA (e.g., FST for Navy/Marine Corps) repair guidance is required prior to returning the aircraft or component to an operational status. This investigation is to be used in conjunction with in-service HMR/EI reporting requirements outlined in the applicable service's deficiency reporting program (for Navy/Marine Corps, see the NAMP).

(1) If the answer is "YES" to any of the following six questions, report the extent of EWIS damage to platform CEA per in-service HMR/EI process.

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(i) If repaired using standard repair procedures or reinstalled per approved routing, is there a reason to believe the damage will reoccur or worsen, or that personnel or flight safety will be compromised?

(ii) Does the level of repair exceed that detailed in the aircraft platform technical manuals, component SM&R code and/or this series manual?

(iii) Does the EWIS damage exceed the material or technical abilities of the assigned technicians to the repair?

(iv) Did the EWIS damage result in an un-commanded weapon/store release, firing of ordnance or cartridge actuating device (CADs) or affect the flight controllability of the aircraft?

(v) For electrical short incidents, did the circuit breaker or other protective device fail to respond/trip?

(vi) Has there been a trend or other reported instances of similar EWIS damage occurring?

(2) If the answer is “YES” to any of the following four (4) questions and the root cause of the EWIS damage cannot be determined or the root cause cannot be corrected using standard repair procedures and/or approved routing, report EWIS damage to platform CEA following in-service HMR/EI requirements:

(i) Are there signs of electrical arcing or burning?

(ii) Were wires chafing on components, or tubing for systems containing flammable fluids, gasses, or emergency systems (e.g. fuel, oil, oxygen, ejection system)?

(iii) Were wires chafing on airframe/structure, flight controls or other components?

(iv) As the result of a single event, where six (6) or more wire/cable conductors are damaged?

(3). If the answer was “NO” to the questions in steps (1) and (2) perform an EWIS repair (NA 01-1A-505-1 or platform-specific manual).

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85
DLA - CC

Review activities:
Air Force - 11, 20

Preparing activity:
Navy - AS

(Project 6145-2022-012)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above by using the ASSIST Online database at https://assist.dla.mil.

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